

BAR ISLAND DAM RECONSTRUCTION

BUREAU OF LAND MANAGEMENT

PHILLIPS COUNTY, MONTANA

NID – MT023923

TECHNICAL SPECIFICATIONS

PREPARED FOR



BUREAU OF LAND MANAGEMENT

MALTA FIELD OFFICE

&

MONTANA STATE OFFICE

PREPARED BY



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BAR ISLAND DAM RECONSTRUCTION TECHNICAL SPECIFICATIONS

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DIVISION 1

GENERAL REQUIREMENTS

PART 1 - GENERAL

1.01 WORK INCLUDED:

This project will reconstruct Bar Island Dam located in Phillips County, Montana, approximately 59 miles southwest of Malta, MT. Major items of work at this dam include:

1. Dewater reservoir and construction area.
2. Excavate a portion of the existing dam embankment and foundation to remove the existing outlet works pipe and allow construction of the new outlet works.
3. Reconstruct the dam embankment, including
 - a. Placement of compacted fill from onsite borrow sources,.
 - b. Placement of lime-amended backfill.
 - c. Placement of imported rock for riprap slope protection within designated areas on the upstream slope.
 - d. Placement and compaction of imported sand for internal filter and drain features.
 - e. Construction of a toe drain.
4. Construct a precast concrete outlet works riser
5. Construct concrete encased, precast reinforced concrete pipe outlet works conduit.
6. Construct grouted riprap outlet works stilling basin.
7. Excavation, fill, and grading of the spillway.
8. Place gravel road surfacing within specified limits on the crest of the dam.
9. Reclaim all disturbed areas including staging areas and temporary roads. Reclamation consists of preparation of soil and seeding.

1.02 INVASIVE SPECIES:

Wash and clean all vehicles and equipment prior to entering the project area, etc. to prevent introduction of invasive plants and seeds.

1.03 CONTRACTOR USE OF PREMISES:

Confine operations to the site of the proposed Work or within the right-of-way or construction easements provided and shown on the drawings. Construction camps and staging areas outside the construction area must be approved by the GOVERNMENT. Obtain advance approval from the GOVERNMENT of all staging areas, excavated soil stockpile and waste areas and camp areas.

It shall be understood that the responsibility for protection and safekeeping of equipment and materials on or near the site will be entirely that of the CONTRACTOR and that no claim shall be made against the GOVERNMENT by reason of any act of an employee or trespasser whether GOVERNMENT has paid CONTRACTOR for equipment or materials in storage or not. It shall be

further understood should any occasion arise necessitating access by the GOVERNMENT to the sites occupied by these stored materials and equipment, the CONTRACTOR owning or responsible for the stored materials or equipment shall immediately remove same. No materials or equipment may be placed upon the property of the GOVERNMENT until the GOVERNMENT has agreed to the location contemplated by the CONTRACTOR to be used for storage. CONTRACTOR is solely responsible for obtaining and shall pay all costs in connection with any additional work area, storage sites, access to the site, or temporary right-of-way, which may be required for proper completion of the Work.

1.04 PROTECTION OF EXISTING UTILITIES:

Utility service connections are generally shown on the Drawings, but the CONTRACTOR is solely responsible for locating all existing underground installations, including service connections, in advance of excavating or trenching, by contacting 811 (in-state) or 1-800-424-5555 (outside of Montana) or the owners thereof. The CONTRACTOR shall use his own information and shall not rely solely upon information shown on the Drawings concerning existing underground installations. Repair all damage to existing utilities or property at the CONTRACTOR's expense.

1.05 PROTECTION OF EXISTING STRUCTURES:

Where excavation will be required adjacent to existing structures, the CONTRACTOR is solely responsible to maintain the structural integrity of the existing structures. Take whatever means necessary to ensure the existing structures are not damaged. Repair all damage to the existing structures at CONTRACTOR's expense. Repair any fences destroyed during construction to the satisfaction of the Government or property owner. Unless otherwise noted on the Construction Drawings, restore all existing ditches disturbed by construction to their original size, line, and grade.

Structures located downstream of the dam shall also be protected during reservoir dewatering activities. Restore or repair to original condition roads, culverts, bridges, cattle guards and fences or any other structures damaged by the CONTRACTOR's activities and access to the sites at no additional cost to the project.

1.06 FIELD CHECK OF EXISTING STRUCTURES:

Dimensions and elevations of existing structures and locations of existing fences, pipelines, conduits, cables, and equipment shown on the Drawings were taken for the most part from available records and survey data and are not guaranteed for accuracy.

Check all dimensions and elevations of existing structures, pipelines, conduits, cables, equipment, or other existing items, both above and below ground, affected by or affecting the Work under this contract, prior to the start of construction or ordering materials and equipment affected thereby.

1.07 SITE ACCESS:

Determine the adequacy of all roads, culverts, cattle guards, and bridges used in moving or gaining access for equipment and materials to the construction site. Provide alternative methods of access, such as a temporary crossing for any equipment that exceeds the structural limits of existing facilities. Improvements for road access including fill, culverts, and fencing to remain after project completion

as approved by GOVERNMENT.

1.08 LIVESTOCK:

The CONTRACTOR shall provide temporary fencing for excluding livestock from construction areas, excavations, trenches, etc.

CONTRACTOR shall be solely responsible for any livestock injuries or losses as a result of CONTRACTOR's activities.

1.09 FIELD OFFICES AND SHEDS:

The CONTRACTOR is responsible for providing their own temporary field offices and storage sheds required for completion of the work. Locate these temporary structures in areas approved by the GOVERNMENT. All expenses for telephone, utilities, security and waste disposal are the responsibility of the CONTRACTOR.

1.10 REQUIRED SUBMITTALS

A. Land Use Plan

1. Show anticipated use of construction area and extent of impact. Uses include, but are not limited to, the following
 - a. Proposed campsite(s)
 - b. Temporary field offices and storage facilities
 - c. Access and haul roads
 - d. Stockpile, material processing, and waste areas
 - e. Water storage pond(s)

1.11 REFERENCES:

References to documents, such as ASTM, shall be for the current printing.

1.12 ABBREVIATIONS:

AASHTO - American Association of State Highway and Transportation Officials
ACI - American Concrete Institute
ASTM - American Society for Testing and Materials
AWWA – American Water Works Association
GOVERNMENT - Contracting Officer
OSHA - Occupational Safety and Health Act
PVC - Polyvinyl Chloride
RCP – Reinforced concrete pipe
CMP – Corrugated Metal Pipe

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION – NONE

Not Used.

END OF SECTION

PART 1 - GENERAL

1.01 WORK INCLUDED:

The CONTRACTOR is responsible for providing construction staking for horizontal and vertical alignment of the control lines, grading, and all appurtenant features of the work, including offset lines if necessary. CONTRACTOR is responsible for performing measurements and maintaining controls to complete the construction to the lines and grades shown in the Contract Documents.

GOVERNMENT has established primary control with control points adjacent to the Work as shown in the Contract Documents.

1.02 SUBMITTALS:

A. Survey Plan:

1. Describe work layout and survey methods.
2. Include survey schedule.

B. Resume

1. Engineer or surveyor responsible for supervising and directing survey work.

C. Control Survey Verification:

1. Accuracy check of existing primary control.

D. Quantity Survey Notes and Computations

1. Required for progress payments. Include notes and computations as supporting information for relevant progress payment quantities.

1.03 PRIMARY CONTROL:

A. As shown on the Drawings, existing control points are available for establishing work lines and grades.

B. The CONTRACTOR shall confirm the horizontal and vertical accuracy of the existing control. If discrepancies are identified, resolve discrepancies with the GOVERNMENT before beginning work.

C. Preserve and maintain primary control points unless otherwise authorized by the GOVERNMENT.

1.04 QUALIFICATIONS:

A. Surveyors: Experienced construction surveyors with a minimum of two years of experience in charge of construction surveys.

PART 2 - PRODUCTS

2.01 SURVEY MATERIALS AND EQUIPMENT:

- A. Provide required materials and equipment for surveying work including, but not limited to, instruments, stakes, spikes, steel pins, templates, platforms, and tools.
- B. Except as required to be incorporated in work or left in place, survey materials and equipment are, and will remain, property of the CONTRACTOR.

PART 3 - EXECUTION

3.01 LAYOUT OF WORK SURVEYS:

- A. Establish lines, grades, and work layout from the primary control points.
- B. Provide stakes, markers, and other survey controls necessary to control, check, and guide construction.

3.02 QUANTITY SURVEYS:

- A. Where required for a progress payment, perform surveys and computations to determine quantities of work performed or placed during each payment period.
- B. Unless specifically waived, perform quantity surveys in presence of the GOVERNMENT. Notify the GOVERNMENT at least 24 hours before performing a quantity survey.

3.03 SURVEY REQUIREMENTS:

- A. Minimum Alignment Staking Intervals: 50 feet on tangent and 25 feet on curves.
- B. Minimum Slope Staking Intervals: 50 feet on tangent and 25 feet on curves, and every 10 feet in elevation.
- C. Structures: Stake out structures before and during construction.
- D. Blue Top Intervals for Roads: 50 feet, on tangent and 25 feet on curves.
- E. As-builts: As required for structures and other features of work.

3.04 ACCURACY:

- A. Degree of Accuracy:
 - 1. Alignment of Tangents and Curves: Set within 0.1 foot.
 - 2. Structure Points: Set within 0.01 foot, except where installation or operation considerations require tighter tolerances.
 - 3. Blue Tops: Set within 0.1 foot.
 - 4. Cross-Section Points: Locate within 0.1 foot, horizontally and vertically.
 - 5. Vertical Elevation Surveys: Close within 0.05 foot times the square root of the circuit

length in miles.

3.05 FIELD RECORDS:

- A. Record original field notes, computations, and other survey data in fieldbooks.
- B. Record survey data in accordance with recognized professional survey standards.
 - 1. Notes or data not in accordance with standard formats will be rejected.
 - 2. Illegible notes or erasures on any page of a fieldbook will be sufficient cause for rejection of part or all of fieldbook.
 - 3. Corrections by ruling or lining out errors will be permitted.
 - 4. Rejection of part or all of a fieldbook may necessitate resurveying.
- C. Data and notes may be collected on electronic data collection devices.
 - 1. Electronic files: In approved format.
 - 2. Include both electronic files and paper copies of notes in submittals.

END OF SECTION 01050

PART 1 - GENERAL

1.01 WORK INCLUDED:

The CONTRACTOR shall be fully responsible for preparing, obtaining, and maintaining the Storm Water Discharge Permit/Storm Water Pollution Prevention Plan as required by the Montana Department of Environmental Quality. The CONTRACTOR will pay all fees associated with obtaining the permit. Throughout the construction of the reservoir, the CONTRACTOR is responsible for all maintenance of the pollution prevention plan. Upon completion of the project and receipt of final payment, the CONTRACTOR shall continue to maintain the pollution prevention plan until all criteria have been met. The CONTRACTOR is also responsible for filing the Notice of Termination for the permit.

Except as otherwise noted, identify, obtain, maintain, and proceed in conformity with any and all additional required permits necessary for completion of the Work.

Any costs associated with these Permits are included as part of the Contract Price as incidental costs.

1.02 SUBMITTALS:

A. Stormwater Pollution Prevention Plan (SWPPP)

1. The CONTRACTOR shall prepare a Stormwater Pollution Prevention Plan as required to obtain a Stormwater Discharge Permit.
2. Comply with terms and conditions imposed by the Stormwater Discharge Permit.
3. Provide monitoring and water treatment, if necessary, to achieve compliance with SWPPP discharge requirements.
4. Provide recordkeeping required by the SWPPP.

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION – NONE

Not Used.

END OF SECTION 01060

PART 1 - GENERAL

1.01 SCOPE:

The method of measurement and basis of payment is described in this section.

1.02 GENERAL:

- A. Bid price for each item of the Contract covers all Work shown on the Drawings and required by the specifications and other Contract Documents.
- B. All costs are included in the unit and lump-sum prices bid including all materials, equipment, supplies and appurtenances; all construction plant, equipment, and tools; and all labor and supervision necessary to fully complete the Work.
- C. No item that is required by the Contract Documents, for the proper and successful completion of the Work, will be paid for outside of or in addition to the prices submitted in the Bid.
- D. All work not specifically set forth as a pay item in the Bid Form is considered an incidental obligation of the CONTRACTOR, and all costs in connection therewith shall be included in the unit prices bid.

1.03 ESTIMATED QUANTITIES:

- A. All quantities in the Price Schedule of the Bid Form or other Contract Documents, which are to be paid per field measurement, are approximate only, and are to be used only as:
 - 1. A basis for estimating the probable cost of the Work, and
 - 2. For the purpose of comparing Bids submitted for the Work.
- B. Actual work performed, and materials furnished under the unit-price items for those items to be field measured may differ from the estimated quantities.
- C. Basis of payment for work and materials will be the actual amount of work done and materials furnished to the pay lines defined in this section.
- D. Payment will be paid at the unit price bid for listed items successfully completed according to Contract requirement.
- E. Lump-sum items and those based on neat-line plan dimensions are not field-measured. They will be paid at neat-line or the lump-sum price bid for the items listed.

1.04 AUTHORITY:

- A. GOVERNMENT will perform all measurements and compute quantities for payment.
- B. GOVERNMENT will verify measurements and quantities provided by CONTRACTOR.
- C. CONTRACTOR provides access to work areas for survey measurements, as required.

1.05 MEASURED QUANTITIES:

A. Measurement by Volume: Measured by cubic dimension using mean length, width, and height or thickness.

1. Measurement for payment of excavations upon or against which concrete or cementitious backfill is not to be placed will be limited to the lines and grades shown on the Drawings or to the most practical lines, grades, and dimensions established by the GOVERNMENT, in writing. No measurement will be made of over excavations beyond design lines and grades.
2. Fill quantities will be computed using the average-end-area method or other computation method approved of by the GOVERNMENT.

B. Measurement by Area: Measured by square dimension using mean length and width. Items that are measured by area will be measured parallel to the ground surface.

C. Linear Measurement: Measured by linear dimension at the item centerline or mean chord. Items that are measured by the lineal foot will be measured parallel to the ground surface, unless otherwise specified.

D. Stipulated Sum/Price Items: Measured by weight, volume, area, or linear means, or combination, as appropriate, as completed items or units of the work.

E. Lump-Sum Items: Will not be measured for payment. Interim measurements may be made to monitor work progress.

1.06 SCHEDULE OF VALUES:

A. A schedule of values or “allocation of contract price” is required for all lump sum items.

B. The schedule of values is used only to help assess the intermediate value of work completed for the purpose of making progress payments. The schedule of values is not considered as a commitment to prices in the event of later negotiations.

1.07 MEASUREMENT AND PAYMENT ITEMS:

A. Item 1 – Mobilization & Preparatory Work:

1. This item includes, but is not limited to:
 - a. Preparatory work and operations performed by the CONTRACTOR for work and operations that are performed or costs incurred before beginning work on the various items on the project site.
 - b. Providing required bonds and insurance.
 - c. Preparing and maintaining the project schedule.
 - d. Delivering and removing the contractor's equipment, supplies, and materials; and

- final site clean-up and demobilization.
- e. Preparing and securing approvals of submittals.
 - f. Preparing record drawings.
 - g. This item also includes constructing and maintaining detour road(s) and traffic control devices as necessary to provide public access through the project site on the BLM Lands Road year-round and for the duration of the project.
 - h. Mobilization costs for subcontracted work are also included.
 - i. Providing all other related and necessary labor, equipment, and materials to complete the work that are not accounted for in the other bid items is included in this item.
2. Measurement and payment for mobilization and preparatory work will be made at the Contract lump sum price. Progress payments will be made as follows:
- a. 1. When 5 percent of the original contract amount is earned from other bid items, 50 percent of the amount bid for mobilization or 5 percent of the original contract amount, whichever is lesser, will be paid.
 - b. 2. When 10 percent of the original contract amount is earned from other bid items, 100 percent of the amount bid for mobilization or 10 percent of the original contract amount, whichever is less, will be paid.
 - c. 3. Upon completion of work for the project, payment of the amount bid for mobilization in excess of 10 percent of the original contract amount will be paid.
 - d. 4. The total sum of all payments shall not exceed the original contract amount bid for mobilization, regardless of the fact that the Contractor may have shut down the work on the project or moved equipment away from the project and then back again.
3. Basis of payment: No direct payment for moves between project sites will be made. The Contractor shall include consideration for moves between project sites in the Schedule of Items (SOI) for other items of the contract. Prices and payment will be full compensation for the work described in this section. No modification and adjustments shall be allowed for costs and changes to mobilization despite changes in fuel or transportation costs. Payment will be made at the Contract lump sum price."

B. Item 2 – Site Clearing & Restoration: Consists of stripping, stockpiling and replacing topsoil from all staging, borrow, excavation, embankment, and waste areas. Seeding will be completed by the CONTRACTOR. Topsoil stripping, replacement, and seeding will be measured and paid at the Contract lump sum price.

C. Item 3 – Existing Outlet Works Demolition: Includes removal of existing outlet works pipes, riser, and any other associated features. This item includes removal of pipes, concrete headwalls, concrete cutoff collars, concrete thrust blocks, miscellaneous concrete, wire, rock, debris; protection of existing materials not designated for removal; legal disposal at offsite facilities; removal of loose material and all cleaning necessary or incidental to demolition. Measurement will be based on acceptable demolition, cleanup, and removal. Payment will be made at the Contract lump sum price.

D. Item 4 – Common Excavation: Consists of excavation and stockpiling or wasting as necessary to construct work in accordance with the Contract Documents. Excavation areas include the outlet works area and the spillway area. The quantity measured for payment will be to the defined neat lines or as otherwise directed by the GOVERNMENT. Unauthorized over-excavation will not be included in the measurement for this item. Waste of materials from excavation areas that are unsuitable for embankment placement is included in this item. Payment for this work will be made at the Contract unit price per cubic yard.

E. Item 5 – Muck Excavation: (Field Measured). Consists of excavation and wasting, as necessary, to remove muck (saturated soil) from the area of the existing outletworks riser, outlet, dam foundation and the area of the construction of the riprap outlet basin. Wasting will be in areas as directed by the GOVERNMENT. Payment for this work will be made at the Contract unit price per cubic yard.

F. Item 6 – Compacted Embankment: Consists of loading from stockpiles or excavating from borrow areas, hauling, placing and compacting embankment in accordance with the Contract Documents. Embankment areas include outlet works and spillway areas. Excavation and backfill for examination test pits are considered incidental to the price of this item. The quantity measured for payment will be to the defined neat lines or as otherwise directed by the GOVERNMENT. Embankment placed to compensate for unauthorized over-excavation will not be included in measurement for this item. Payment will be made at the Contract unit price per cubic yard.

G. Item 7 – Lime Amended Embankment: Consists of furnishing hydrated lime, hauling, mixing lime with soil, water conditioning, placing, and compacting the lime treated soil in accordance with the Contract Documents. The quantity measured for payment will be computed using the neat line dimensions shown on the Drawings. Payment for this work will be made at the Contract unit price per cubic yard.

H. Item 8 – Filter Sand: Consists of furnishing and placing filter sand material in accordance with the Contract Documents. The quantity measured for payment will be computed using the neat line dimensions shown on the plans. Payment for this work will be made at the Contract unit price per cubic yard.

I. Item 9 – Drain Gravel: Consists of furnishing and placing drain gravel material in accordance with the Contract Documents. The quantity measured for payment will be computed using the neat line dimensions shown on the Drawings. Payment for this work will be made at the Contract unit price per cubic yard.

J. Item 10 – Road Gravel: Consists of furnishing and placing road gravel material in accordance with the Contract Documents. Supplying and installing geotextile stabilization fabric between the road gravel and subgrade is considered incidental to this item. The quantity measured for payment will be computed using the neat line dimensions shown on the Drawings. Payment for this work will be made at the Contract unit price per ton.

K. Item 11 – 18-Inch CMP Culvert: Consists of furnishing and installing the culvert in accordance with the Contract Document. This item includes excavation, shoring, temporary controls, disposal of excavated material, furnishing and installing pipe, bedding, backfill, compaction, and all other materials, tools, equipment, labor, and work required to complete this item. Payment will be made at the Contract price per linear foot.

L. Item 12 – Outlet Works Riser and Inlet Structure: Consists of providing all materials for

construction of the concrete outlet works tower structure. Item includes precast concrete manhole sections, stainless steel vent pipe, non-shrink grout for vent pipe encasement, concrete slab, concrete bedding around the foundation, concrete floor, and all other items necessary to construct the outlet works in accordance with the Contract Documents. Excavation and backfill will be measured and paid for separately under other items of work. Measurement and payment will be made at the Contract lump sum price.

M. Item 13 – Outlet Works Conduit: Consists of furnishing and installing precast reinforced concrete pipe outlet conduit in accordance with the Contract Documents. This item includes outlet trench excavation through Compacted Embankment, pipe support blocks, reinforced concrete pipe, flap gate, steel reinforcing bars, and providing and placing concrete bedding and concrete cap. Compacted Embankment around the outlet is not included in this item and will be paid separately. Measurement and payment will be made at the Contract lump sum price.

N. Item 14 – Stilling Basin: Consists of furnishing and placing rock riprap material from an offsite source and furnishing and placing grout in accordance with the Contract Documents to construct the grouted riprap basin. PVC and steel pipes for drain weeps, HDPE pipe for dewatering sumps, and geotextile fabric associated with the outlet basin are considered incidental to this item. Excavation and backfill will be measured and paid separately under other items of work. Measurement and payment will be made at the Contract lump sum price.

O. Item 15 – Toe Drain System: Consists of furnishing and installing the toe drain collection system along the downstream toe of the dam and includes the drainpipe, bends, and end treatments. Drain gravel and filter sand will be paid separately. Measurement and payment will be at the Contract lump sum price.

P. Item 16 – New Class 1 Riprap Slope Protection: Consists of supplying from offsite sources and placing riprap on the upstream face in accordance with the Contract Documents. Measurement will be computed using the neat line dimensions shown on the Drawings. Payment for this work will be made at the Contract unit price per ton.

Q. Item 17 – Water Control/Dewatering: Consists of dewatering the reservoir and other water control measures necessary to complete the Work in accordance with the Contract Documents. Includes all design, equipment, facilities (including, but not limited to pumps, hoses, trenches, dikes, wells, sediment control), and operation required to maintain excavations and work areas free from water. This item also includes the removal of dewatering facilities upon completion of construction. The CONTRACTOR is responsible to design and maintain all construction water control features to prevent damage to the Work. Measurement and payment will be made, subject to acceptable construction progress, based on the Contract lump sum price for this item, prorated in equal monthly amounts throughout the construction schedule.

R. Item 18 – Stabilization Fabric: (Field Measured). Consists of furnishing and installing geotextile stabilization fabric in accordance with the Contract Documents in areas directed by the GOVERNMENT. Payment for this work will be made at the Contract unit price per square yard as measured and accepted in-place, excluding overlaps and anchoring.

S. Item 19 – Filter Fabric: (Field Measured). Consists of furnishing and installing geotextile filter fabric in accordance with the Contract Documents in areas directed by the GOVERNMENT. Payment for this work will be made at the Contract unit price per square yard as measured and accepted in-place, excluding overlaps and anchoring.

T. Item 20 – General Construction Operations:

1. This item includes, but is not limited to:
 - a. Furnishing, establishing, maintaining, and removing all construction facilities.
 - b. Obtaining, complying with, and transferring permits.
 - c. Installing and maintaining environmental controls for the duration of the construction Contract.
 - d. Temporary construction utilities such as electricity and sanitary facilities.
 - e. General site maintenance and clean-up for the duration of the construction Contract.
 - f. All CONTRACTOR's costs for construction surveying and staking, including, but not limited to, verifying the primary survey control points, establishing temporary construction survey control, construction staking, quantity surveys, and as-constructed surveys.
 - g. All CONTRACTOR's costs for quality control testing and inspection.
2. Payment will be at the Contract lump sum price.
 - a. Payment will be based on the percent of the total Project Costs completed to date, excluding the price for the General Construction Operations item.

U. Item 21 – Spillway Armoring (Additive Bid Item): Consists of furnishing and installing concrete block mat spillway armoring in accordance with the Contract Documents in areas directed by the GOVERNMENT. All costs are included in the unit and lump-sum prices bid, including all materials, equipment, supplies, and appurtenances; all construction plant, equipment, and tools; and all labor and supervision necessary to fully complete the Work.

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION – NONE

Not Used.

END OF SECTION 01150

PART 1 - GENERAL

1.01 PRE-CONSTRUCTION CONFERENCE:

A. A pre-construction conference will be held prior to the start of work. CONTRACTOR will be notified in advance of the meeting time, date, and place. The purpose will be to review required work, project drawings and specifications, construction schedules, payroll and payments, and administrative provisions of the Contract.

B. CONTRACTOR, subcontractors, and the persons responsible for the coordination of the work are required to be present at the meeting.

C. CONTRACTOR is required to be prepared to summarize and explain procedures planned for the project and to present the pre-construction submittals required.

1.02 PROGRESS MEETINGS:

A. Progress meetings may be requested by either the GOVERNMENT or the CONTRACTOR. The request shall state who should attend and include an agenda.

1.03 SUBSTANTIAL COMPLETION INSPECTION:

A. Notify the GOVERNMENT, in writing, at least five working days prior to the substantial completion date, allowing the GOVERNMENT to schedule an inspection.

1.04 FINAL INSPECTION:

A. Notify the GOVERNMENT, in writing, at least five working days prior to the completion date, allowing the GOVERNMENT to schedule a final inspection.

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION – NONE

Not Used.

END OF SECTION 01200

PART 1 - GENERAL

1.01 DESCRIPTION:

This section includes the procedures for submittals. Items that need to be reviewed by the GOVERNMENT are included.

PART 2 - PRODUCTS – NONE

PART 3 - EXECUTION

3.01 SUBMITTAL PROCEDURE:

A. Coordinate all submittals and review them for accuracy, completeness, and compliance with contract requirements; and indicate CONTRACTOR's approval thereon as evidence of such coordination and review. Items submitted without evidence of the CONTRACTOR's approval will be returned for resubmission.

B. Date and Number: Forward to the GOVERNMENT at least 30 days prior to needing approval or delivery to the site, all items required by the individual sections of the specifications.

C. Cover Letter: Forward all submittals with a cover letter from the CONTRACTOR, identifying the project and the portion of the Project to which it applies. Submittals that are related to or affect each other shall be forwarded simultaneously as a package to facilitate a coordinated review. Uncoordinated submittals will be rejected. The GOVERNMENT reserves the right to require submittals in addition to those called for in individual sections.

D. Electronic Submittals: An internet-based electronic submittal and review system will be available for this project. The use of this system will be organized by the GOVERNMENT at no cost to the CONTRACTOR and will be the means for receiving and reviewing submittals.

3.02 GOVERNMENT'S APPROVAL:

The GOVERNMENT will indicate approval or disapproval of each submittal and, if not approved, will indicate reasons therefor. Any work done prior to approval shall be at the CONTRACTOR's own risk. Any work done, or materials provided, prior to approval will not be eligible for payment until approval is obtained. Approvals do not relieve the CONTRACTOR from responsibility for complying with the requirements of this Contract. If submittals show variations from the Contract requirements, the CONTRACTOR shall describe such variations in writing, in the cover letter at the time of submission. The request for a variation shall include a proposal addressing the net effect on contract price and time.

3.03 REQUIRED SUBMITTALS:

- A. Subcontractors: At the preconstruction conference, provide a list of all suppliers and subcontractors to be used on the project and a completed Form SF1413 (form provided at the end of this section) for each subcontractor.
- B. Certificates: For those items called for in individual sections, furnish certificates from manufacturers, suppliers, or others certifying that materials or equipment being furnished under the Contract comply with the requirements of these specifications.
- C. Shop Drawings: See the individual sections for specific requirements.
- D. Work Schedule and Progress Payment Schedule: Submit a Work Schedule and an estimated Progress Payment Schedule.
- E. SUMMARY OF REQUIRED SUBMITTALS: The following table is a summary of submittals, by section, required by these specifications. Consult individual sections for a more complete description of individual requirements.

List of Submittals*			
Submittal Number	Submittal	Specification Reference	Date Due**
01010-1	Land Use Plan	01010	Preconstruction Conf.
01050-1	Survey Plan	01050	**
01050-2	Surveyor Resume	01050	**
01050-3	Control Survey Verification	01050	**
01050-4	Quantity Survey Notes and Computations	01050	As produced
01060-1	Contractor-Obtained Permits	01060	Preconstruction Conf.
01150-1	Schedule of Values for Lump Sum Items	01150	**
01300-1	List of Subcontractors & Suppliers with completed Form SF1413 for each subcontractor	01300	Preconstruction Conf.
01300-2	Work Schedule	01300	Preconstruction Conf.
01300-3	Estimated Progress Payment Schedule	01300	**
01400-1	Quality Procedures Plan	01400	Preconstruction Conf.
01400-2	Quality Control Test Results		As produced
01570-1	Traffic Control Plan	01570	**
01700-1	Contractor Record Drawings	01700	At project closeout
01700-2	Supplier O&M Instructions	01700	At project closeout
02275-1	Riprap Source Data and Test Results	02275	**
02300-1	Off-site Material Test Results	02300	**
02300-2	Filter Sand and Drain Gravel Quality Control Tests	02300	As produced
02300-3	Foundation Treatment and Protection Plan	02300	**
02300-4	Embankment Construction Plan	02300	**
02300-5	Lime Treatment Plan	02300	**
02370-1	Geotextile Filter Fabric Product Data & Samples	02370	**
02370-2	Geotextile Stabilization Fabric Product Data & Samples	02370	**
02380-1	Concrete Block Mat Data	02380	**
2610-1	Product Data	2610	**
2610-2	Installation & Testing Plan	2610	**
2610-3	Survey Reports	2610	**
2610-4	Inspection Video	2610	**

List of Submittals*			
Submittal Number	Submittal	Specification Reference	Date Due**
02724-1	Reinforced Concrete Pipe Materials Shop Drawings, and Product Data	02724	**
02724-2	Pipe Anchor Details to Prevent Flotation	02724	**
02724-3	Waterproofing Products	02724	**
02724-4	Manhole Shop Drawings, Product Data, and Placement Information	02724	**
02724-5	Flap Gate Shop Drawings and Product Data	02724	**
02724-6	Steel and PVC Pipe Shop Drawings and Data	02724	**
02724-7	Vent Pipe Encasement Form, Support, and Placement Plan	02724	**
02800-1	Product Data	02800	**
02910-1	Seeding Plan	02910	**
03300-1	Concrete Mix Design	03300	**
03300-2	Concrete Materials Product Data	03300	**
03300-3	Reinforcement Shop Drawings	03300	**
03300-4	Reinforcement Certified Materials Test Report (CMTR)	03300	**
03300-5	Formwork Design and Assembly Drawings	03300	**
03300-6	Concrete Placement Drawings	03300	**
03300-7	Concrete Placement Schedule	03300	**
03300-8	Concrete Placement Plan(s)	03300	**
03300-9	Cold Weather Placement Plan	03300	**
03300-10	Hot Weather Placement Plan	03300	**
03300-11	Certifications	03300	**
03300-12	Concrete Batch Tickets	03300	As produced
03300-13	Protection Period Temperature Records	03300	As produced
05500-1	Manufacturer Product Data Sheets	05500	**
05500-2	Steel Fabrication Shop Drawings	05500	**

*This list is not intended to be comprehensive; consult individual sections.

**Due date for all submittals is 30 days prior to needing approval or delivery to the site, unless stated otherwise.

STATEMENT AND ACKNOWLEDGMENT					OMB Control Number: 9000-0014 Expiration Date: 1/31/2021									
Paperwork Reduction Act Statement - This information collection meets the requirements of 44 U.S.C. § 3507, as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 9000-0014. We estimate that it will take .05 hours to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate, including suggestions for reducing this burden, or any other aspects of this collection of information to: U.S. General Services Administration, Regulatory Secretariat Division (M1V1CB), 1800 F Street, NW, Washington, DC 20405.														
PART I - STATEMENT OF PRIME CONTRACTOR														
1. PRIME CONTRACT NUMBER		2. DATE SUBCONTRACT AWARDED		3. SUBCONTRACT NUMBER										
4. PRIME CONTRACTOR				5. SUBCONTRACTOR										
a. NAME				a. NAME										
b. STREET ADDRESS				b. STREET ADDRESS										
c. CITY		d. STATE	e. ZIP CODE	c. CITY		d. STATE								
						e. ZIP CODE								
6. The prime contract ☐ does, ☐ does not contain the clause entitled "Contract Work Hours and Safety Standards Act -- Overtime Compensation."														
7. The prime contractor states that under the contract shown in Item 1, a subcontract was awarded on the date shown in Item 2 to the subcontractor identified in item 5 by the following firm:														
a. NAME OF AWARDING FIRM														
b. DESCRIPTION OF WORK BY SUBCONTRACTOR														
8. PROJECT				9. LOCATION										
10a. NAME OF PERSON SIGNING			11. BY (Signature)		12. DATE SIGNED									
10b. TITLE OF PERSON SIGNING														
PART II - ACKNOWLEDGMENT OF SUBCONTRACTOR														
13. The subcontractor acknowledges that the following clauses of the contract shown in Item 1 are included in this subcontract:														
Contract Work Hours and Safety Standards Act - Overtime Compensation (If included in prime contract see Block 6) Payrolls and Basic Records Withholding of Funds Disputes Concerning Labor Standards Compliance with Construction Wage Rate Requirements and Related Regulations Construction Wage Rate Requirements Apprentices and Trainees Compliance with Copeland Act Requirements Subcontracts (Labor Standards) Contract Termination - Debarment Certification of Eligibility														
14. NAME(S) OF ANY INTERMEDIATE SUBCONTRACTORS, IF ANY														
<table border="1" style="width: 100%; border-collapse: collapse;"><tr><td style="width: 5%; padding: 5px;">A</td><td style="width: 45%;"></td><td style="width: 5%; padding: 5px;">C</td><td style="width: 45%;"></td></tr><tr><td style="padding: 5px;">B</td><td></td><td style="padding: 5px;">D</td><td></td></tr></table>							A		C		B		D	
A		C												
B		D												
15a. NAME OF PERSON SIGNING			16. BY (Signature)		17. DATE SIGNED									
15b. TITLE OF PERSON SIGNING														
AUTHORIZED FOR LOCAL REPRODUCTION PREVIOUS EDITION IS NOT USABLE STANDARD FORM 1413 (REV. 4/2013) Prescribed by GSA/FAR (48 CFR) 53.222(e)														

END OF SECTION 01300

PART 1 - GENERAL

1.01 SECTION INCLUDES:

A. This section covers the requirements for Quality Control (QC) on all work performed to ensure it conforms to the drawings and specifications.

1.02 QUALITY SUBMITTALS:

A. Quality Procedures Plan.

1. A plan that identifies the CONTRACTOR's representative responsible for the quality of the completed construction and outlines the quality control program (including construction management controls, quality control testing, test frequency, and test agencies).

B. Quality Control Test Results

A. Submit records of all CONTRACTOR tests to the GOVERNMENT within 24 hours of the completion of testing. The Quality Control laboratory is to notify the CONTRACTOR and the GOVERNMENT promptly of irregularities or deficiencies observed in the Work during performance of the Quality Control Testing.

1.03 DEFINITIONS:

A. Quality Assurance (QA): Inspection and tests performed by the GOVERNMENT to assure compliance with the terms of the contract. QA testing is not used to determine construction procedures or operations (i.e., rolling patterns, lift thickness, etc.). Quality assurance also includes oversight of the CONTRACTOR's quality control, verifying the results of the CONTRACTOR's testing and additional assurance sampling and testing. QA testing and oversight will be provided by the GOVERNMENT at a level/frequency determined necessary by the GOVERNMENT.

B. Quality Control (QC): Activities performed by the CONTRACTOR to ensure work conforms to contract requirements.

C. Quality Control Testing: Tests to be performed by the CONTRACTOR.

1. The GOVERNMENT may use the contractor's QC test results for Quality Assurance.
2. The Contractor may include the GOVERNMENT's QA test records in the QC test report documenting compliance with the construction documents.
3. The tests required by the individual specification sections do not relieve the CONTRACTOR of providing adequate quality control measures to ensure the completed construction complies with the contract documents.

PART 2 - PRODUCTS

Not Used

PART 3 - EXECUTION

3.01 COOPERATION WITH QUALITY ASSURANCE:

A. Assure the GOVERNMENT has access to all work areas at all times work is in progress. Provide any special facilities or equipment to access work areas at CONTRACTOR expense.

Notify the GOVERNMENT when the work is ready for quality assurance/quality control testing. Establish and update the construction schedule to provide the GOVERNMENT with estimated sampling/testing dates and times. The GOVERNMENT may choose to obtain a split and comparison test of any samples collected for quality control testing. Provide adequate notice of testing plans to coordinate collection of samples

END OF SECTION 01400

PART 1 - GENERAL

1.01 DESCRIPTION:

This work is the furnishing of labor, materials and equipment for installing, maintaining and operating traffic control devices to ensure the safety of the general public and project personnel.

1.02 REQUIREMENTS:

Perform work under this section meeting Manual of Uniform Traffic Control Devices (MUTCD), the BLM publication Surface Operating Standards and Guidelines for Oil and Gas Exploration and Development (Gold Book), and contract requirements. CONTRACTOR shall obtain approval of the Traffic Control Plan (TCP) from the GOVERNMENT.

1.03 SUBMITTALS:

A. Conform with Section 01300 – Submittals.

B. Traffic Control Plan – Submit at least 30 days prior to the beginning of construction. In the TCP, name one individual whose purpose shall be to coordinate traffic control and provide a 24-hour-a-day telephone number or numbers to contact this person. Cooperate with the GOVERNMENT and make adjustments to the traffic control plan as required or as necessary to fit specific field conditions.

1.04 SAFETY PROVISIONS:

Safety provisions are the responsibility of CONTRACTOR. CONTRACTOR must meet County, State, and Federal laws, regulations, and ordinances to protect the public.

1.05 DETOURS:

Where required by the authority having jurisdiction that traffic be maintained over any construction work in a public street, road, or highway, and the traffic cannot be maintained on the alignment of the original roadbed or pavement, construct and maintain a detour around the construction work. Include with each detour all necessary barricades, guard rails, approaches, lights, signals, signs, and other devices and precautions necessary for protection of the Work and safety of the public all in accordance with the current edition of the MUTCD.

1.06 BARRICADES AND LIGHTS:

Provide all open trenches and other excavations with suitable barriers, signs, and lights to the extent adequate protection is provided to the public. Provide similar warning signs and lights to indicate obstructions such as material piles and equipment.

Illuminate all barricades and obstructions by means of warning lights from sunset to sunrise. Place materials stored upon or alongside public roads (if any) and conduct the work at all times so as to cause minimum obstruction and inconvenience to the traveling public. Install all barricades, signs, lights and other protective devices, and maintain in conformity with applicable statutory or regulatory requirements and as required by the authority having jurisdiction.

PART 2 - PRODUCTS

1.01 TRAFFIC CONTROL DEVICES:

A. Compliance with Manual on Uniform Traffic Control Devices: Ensure all traffic control devices and their application or usage conform to the current edition of "Manual of Uniform Traffic Control Devices" (MUTCD): <https://mutcd.fhwa.dot.gov>.

B. Reflectorized Signs and Barricades:

1. Ensure all signs and barricades are reflectorized. Ensure all traffic control devices in use are equipped with lighting devices as required in the most current edition of the MUTCD.
2. In particular ensure the channelization spacing, taper rates, standard mounting heights and size of signs are adhered to on this project. Where lighting devices are used at night, ensure they are flashing or constantly on (steady burn) depending upon their use, as defined in the MUTCD.

C. Assure all traffic control devices are clean, legible, reflective for night-time use, and operable.

PART 3 - EXECUTION

1.02 WORK METHODS:

A. Place all traffic control devices as planned before permitting men or equipment on the traveled way.

B. Inspect the work area at least twice each day during construction and maintain records of traffic control devices used and their location.

C. Assure traffic control is appropriate to the work. Assure traffic control devices are appropriate and clean before suspending work for the day.

D. Remove and store all unnecessary traffic control devices away from traffic's view.

END OF SECTION 01570

PART 1 - GENERAL

1.01 COMPLETION OF WORK:

It is understood that, except as otherwise specifically stated in the Contract Documents, the CONTRACTOR shall provide and pay for all materials, labor, tools, equipment, water, light, power, transportation, supervision, temporary construction of any nature, and all other services and facilities of any nature whatsoever necessary to execute, complete, and deliver the Work within the specified time.

1.02 CONTRACTOR RESPONSIBILITY:

Store materials so as to ensure the preservation of their quality and fitness for the Work. Locate stored materials to be incorporated in the Work so as to facilitate prompt inspection by the GOVERNMENT. The CONTRACTOR is responsible for the condition and security of all materials furnished by CONTRACTOR, and shall replace all such material found to be defective or damaged after delivery. This includes the replacement of material that is found to be defective at any time prior to the expiration of the guarantee period.

1.03 MANUFACTURER'S RECOMMENDATIONS:

Apply, install, connect, erect, use, clean, and condition manufactured articles, materials and equipment as directed by the manufacturer.

1.04 SAMPLES:

Provide materials, supplies and equipment in accordance with samples submitted by the CONTRACTOR and approved by the GOVERNMENT.

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION – NONE

Not Used.

END OF SECTION 01600

PART 1 - GENERAL

1.01 DESCRIPTION:

The work covered under this section includes items required in the final closeout of this Project. These include items such as post-construction inspection, acceptance of the work, closeout records, and Project record drawings.

1.02 SUBMITTALS:

Provide closeout submittals for approval prior to final payment. Items to be submitted are:

- A. Project Record Documents - See Item 3.02 for requirements.
- B. Operation and Maintenance Instructions - Furnish electronically complete literature, operating instructions, installation instructions, maintenance instructions, parts lists, and technical data on all items required herein or by the GOVERNMENT.

PART 2 - PRODUCTS

2.01 MISCELLANEOUS PRODUCTS:

Provide all maintenance materials, spare parts, tools, keys, and other items either required in the Contract Documents or furnished by the suppliers of the various products.

PART 3 - EXECUTION

3.01 CLEANING:

Use cleaning materials recommended by the manufacturer of the surface to be cleaned. Follow the instructions on the container.

Remove all tools, equipment, surplus materials, and rubbish. Repair all marred surfaces, and remove grease, dirt, stains and foreign materials from finished surfaces.

3.02 PROJECT RECORD DRAWINGS:

Using red-colored ink, CONTRACTOR shall mark changes on a set of clean prints. Indicate all changes and revisions to the original design that affect the permanent structures and that exist in the completed work. Reference all underground facilities to permanent physical objects. Reference elevation of all existing lines on profile sheets or call out elevations in plan if no profile exists. Keep record drawings current during construction activities. Certification of accuracy and completeness is required on monthly payment requisitions.

3.03 POST-CONSTRUCTION INSPECTION:

Prior to the expiration of the one-year correction period, the GOVERNMENT will inspect the Project to determine whether corrective work is required. The CONTRACTOR will be advised 14 days prior to inspection and notified in writing of all deficiencies. CONTRACTOR shall attend the post-construction inspection.

END OF SECTION 01700

DIVISION 2

SITEWORK

PART 1 - GENERAL

1.01 WORK INCLUDED:

Remove and dispose of existing concrete and metal structures and pipes encountered in completing project excavations. The original Drawings are included in Appendix B.

1.02 PROTECTION AND RESTORATION OF PROPERTY:

The CONTRACTOR is responsible for all methods and operations of demolition. The CONTRACTOR is solely responsible for protection of all public and private property. Promptly repair any damage caused to adjacent facilities or property to the satisfaction of the GOVERNMENT at no additional cost to the GOVERNMENT.

PART 2 - PRODUCTS – NONE

Not Used.

PART 3 - EXECUTION

3.01 SUBGRADE:

A. Remove all pieces of concrete, soil, and rock loosened by demolition unless otherwise directed by GOVERNMENT.

B. Completely fill and compact areas and voids resulting from demolition work. Prepare soil subgrade surfaces as required beneath concrete structures and earth fills in accordance with the Contract Documents.

3.02 DISPOSAL OF MATERIALS REMOVED:

Concrete demolition material, including grout from inside pipes, may be buried on-site in designated waste area as approved by GOVERNMENT. Steel and metal demolition material becomes property of the CONTRACTOR and shall be removed from the site. CONTRACTOR is responsible for lawful transport and disposal of materials.

END OF SECTION 02050

PART 1 - GENERAL

1.01 WORK INCLUDED:

The work within this section consists of removing all trees, root balls, shrubs, brush, tall grass and similar vegetation as necessary to complete the Work.

PART 2 - PRODUCTS

Not Used.

PART 3 - EXECUTION

3.01 REQUIREMENTS:

Remove all trees, shrubs, brush, tall grass, and similar vegetation located on the dam embankment within the limits indicated on the Drawings. Remove all trees, shrubs, brush, tall grass, and similar vegetation located upstream and downstream of the spillway as directed by the GOVERNMENT. Perform all clearing in a safe, prudent and lawful manner and only within the construction limits indicated on the Drawings or as directed by the GOVERNMENT. Dispose of all trees onsite as directed by the GOVERNMENT at no additional cost to the GOVERNMENT. Do not remove trees from the site without approval from the GOVERNMENT.

3.02 TOPSOIL REUSE:

Do not use topsoil removed by stripping for backfill or constructing embankments.

END OF SECTION 02110

PART 1 - GENERAL

1.01 WORK INCLUDED:

Material requirements for upstream slope protection and outlet works stilling basin rock riprap supplied from offsite and riprap placement requirements for the Project.

1.02 SUBMITTALS:

A. Submit source of riprap.

B. Submit laboratory test results that show conformance to the material requirements of this section. Test reports shall not be more than 3 years old at the time of submittal.

1.03 REFERENCES:

The publication(s) listed below form a part of these specifications to the extent referenced. The publications are referred to by the basic designation only.

A. American Society of Testing and Materials (ASTM)

- | | | |
|----|------------|--|
| 1. | ASTM C88 | Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate |
| 2. | ASTM C535 | Test Method for Resistance to Degradation of Large-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |
| 3. | ASTM D4992 | Standard Practice for Evaluation of Rock to be Used for Erosion Control |
| 4. | ASTM D6473 | Test Method for Specific Gravity and Absorption of Rock for Erosion Control |

PART 2 - PRODUCTS

2.01 RIPRAP ROCK:

A. Riprap is to be quarried from an offsite source and shall be dense, angular, reasonably well-graded, sound rock fragments resistant to abrasion, free of cracks, seams, clay, organic material, and defects that would hasten degradation by water and/or frost action.

B. Quarried rock for use as slope protection or in the stilling basin from offsite shall have the following properties:

1. Bulk specific gravity (saturated-surface dry): 2.50 minimum, (apparent specific gravity minimum: 2.60) when tested in accordance with ASTM D6473.
2. Absorption: 2 percent or less, when tested in accordance with ASTM D6473.
3. Sodium sulfate soundness, loss: 10 percent loss or less for five cycles, when tested in

accordance with ASTM C88, 5 cycles.

4. Resistance to abrasion, loss: 10 percent or less after 100 revolutions and 40 percent or less after 500 revolutions, when tested in accordance with ASTM C535.

C. Shape:

1. Angular or semi-angular boulders or cobbles are acceptable.
2. Maximum dimension is not greater than 3 times minimum dimension.

D. Gradation: Furnish stone that is reasonably well-graded, conforming to the gradation listed in Table 02275-1.

TABLE 02275-1
RIPRAP GRADATION

Diameter* (inch)	Percent Smaller by Weight
13	100
10	70 – 90
8	25 – 50
3	0 – 10**

* Diameter is the median dimension when measuring length, depth and breadth.

** Include rock spalls or cobbles in an amount sufficient to fill spaces between larger rocks, but not to excess. Larger rocks should be in point-to-point contact.

PART 3 - EXECUTION

3.01 AREAS TO RECEIVE SLOPE PROTECTION:

A. Place riprap slope protection to the lines and grades as shown on the Drawings.

3.02 PLACING:

A. Subgrade Preparation:

1. Excavate, fill, or grade foundation soils in accordance with Section 02300 so riprap limits are as indicated on the Drawings or as directed by the GOVERNMENT.
2. As indicated on the Drawings, install bedding gravel or non-woven geotextile fabric underlayment on prepared surfaces. For geotextile fabric, ensure required overlaps are maintained and the outer edges of the fabric are buried a minimum of 6 inches below grade.

B. Slope-protection rock need not be hand-placed or compacted but shall be handled and dumped to avoid material segregation and damage or displacement of the underlying material that would impair its functionality.

- C. Place rocks so larger rocks are evenly distributed and small rock fragments fill the spaces.
- D. Dump and smooth by moving rocks into position so material, when in place, is stable.
- E. Begin slope-protection at toe and proceed up the slope.
- F. Leave no large, unfilled spaces within slope protection.

3.03 STILLING BASIN ROCK TO BE GROUTED:

- A. Prior to placing the stilling basin rock, ensure the rock is free of fines or soil, which would affect the grout bond. The requirement for cleaning the stilling basin rock applies to both salvaged and imported rock. Rocks shall be placed so that they are in contact with each other and the spaces between adjacent boulders are as small as the character of the rock will permit but will allow full penetration of the grout between all rocks and down to the subgrade.
- B. Any loose material between rocks shall be removed to ensure complete grout penetration down to subgrade and in all voids between rocks. All rock placed shall be approved by the GOVERNMENT at least 24 hours in advance of the scheduled time for grouting to avoid any unnecessary delays in the grouting operation. Rock surfaces shall be moist at the time of grout placement but shall not contain free water. Vibrators shall be used to make sure all voids are filled between and under rock. The intent is to fill all voids with grout from the subgrade level through the rock layer. In all cases, grout must penetrate to subgrade.
- C. At construction joints between grouting, the grout shall be held down at least 6 inches below the tops of the rock to allow grout placed for the next layer to penetrate down and around the tops of the core rocks.
- D. Immediately after placement of the grout, protect the grout from premature drying and excessive hot or cold temperatures in accordance with Section 03300.

END OF SECTION 02275

PART 1 - GENERAL

1.01 REFERENCES

A. ASTM International (ASTM)

1. ASTM C25	Test Methods for Chemical Analysis of Limestone, Quicklime, and Hydrated Lime
2. ASTM C110	Test Methods for Physical Testing of Quicklime, Hydrated Lime, and Limestone
3. ASTM C997	Specification for Quicklime and Hydrated Lime for Soil Stabilization
4. ASTM C127	Specific Gravity and Absorption of Coarse Aggregate
5. ASTM C136	Sieve Analysis of Fine and Coarse Aggregate
6. ASTM D422	Particle-Size Analysis of Soils
7. ASTM D653	Terminology Relating to Soil, Rock, and Contained Fluids
8. ASTM D698	Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft ³)
9. ASTM D1140	Amount of Material in Soils Finer than the No. 200 (75- μ m) Sieve
10. ASTM D1556	Density and Unit Weight of Soil in Place by Sand-Cone Method
11. ASTM D2216	Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
12. ASTM D2487	Classification of Soils for Engineering Purposes (Unified Soil Classification System)
13. ASTM D4253	Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
14. ASTM D4254	Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density

15. ASTM D4318	Liquid Limit, Plastic Limit, and Plasticity Index of Soils
16. ASTM D4643	Determination of Water (Moisture) Content of Soil by the Microwave Oven Heating
17. ASTM D5080	Rapid Determination of Percent Compaction
18. ASTM D6938	In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods
19. ASTM D382	Maximum Dry Unit Weight and Water Content Range for Effective Compaction of Granular Soils Using a Vibrating Hammer

1.02 DEFINITIONS

A. Use definitions from ASTM D653.

B. Borrow: Material from required site excavations or from designated borrow areas.

C. Coverage: One “coverage” is the result of successive passes by a piece of compaction equipment, which by means of sufficient overlap, ensures all areas have been subjected to one pass of the compaction equipment.

D. Dam Embankment: All portions of the dam embankment including: Previously Placed Material, Compacted Embankment, Lime Amended Fill, Riprap, Riprap Bedding Filter Sand, Drain Gravel, and Gravel Surfacing.

E. Fines: Material passing the No. 200 sieve size (ASTM D1140).

F. Optimum Water Content: The water content that will result from the estimated maximum dry unit weight of soil when subjected to the ASTM D698 compaction test.

G. Percent Compaction: The ratio expressed as a percentage of the in-place dry density to the estimated maximum dry density, in accordance with ASTM D698, at the location of the in-place density test.

H. Percent Relative Compaction: The percent compaction of a cohesionless soil where the laboratory maximum density is determined by Maximum Index Unit Weight test (ASTM D7382).

I. Percent Relative Density: The ratio of, (1) the difference between void ratio of a cohesionless soil in the loosest state and any given void ratio, to (2) the difference between its void ratios in the loosest state and densest state (ASTM D4254).

J. Proof Rolling: Rolling a soil surface with a minimum of 4 passes with compaction equipment for the purpose of detecting loose or soft areas.

K. Suitable Material: Material imported or from required excavations that meets the specification requirements for a designated fill.

L. Topsoil: Native surface soil capable of supporting native plant growth.

M. Unsuitable Materials: Materials that contain waste, debris, roots, organic matter, frozen matter,

stone or rock with any dimension greater than 3-inches, or any other materials determined by the GOVERNMENT that does not meet the requirements for any embankment zone or required fill.

N. Well Graded: A mixture of particle sizes that has no specific concentration, or lack thereof, of one or more sizes. A material that, when compacted, produces a strong, relatively incompressible soil mass with a minimum of voids.

1.03 SUBMITTALS

A. Source Submittals

1. Submit test results from a qualified testing agency demonstrating that off-site sources conform to the requirements of this section. Test reports shall not be more than 2 years old at the time of submittal. Provide submittals a minimum of 30 days prior to delivery of material to the site.
2. Submit quality control gradation test results for filter sand material during production and delivery. Submit a minimum of one test per 500 cubic yards of filter sand delivered.
3. Submit quality control gradation test results for drain gravel material during production and delivery. Submit a minimum of one test per 500 cubic yards of drain gravel delivered.

B. Foundation Treatment and Protection Plan:

1. Describe means and methods for foundation protection and maintaining natural water content at all times the foundation is exposed.
2. Describe the equipment proposed for cleaning, protecting, and treating the foundation. Include equipment catalogs with weight, dimensions, and operating data.

C. Embankment Construction Plan:

1. Describe the equipment proposed for soil processing, placement, compaction, and water (moisture) control. Include equipment catalogs with weight, dimensions, and operating data.
2. Provide a proposed work schedule.
3. Describe proposed methods for protecting work. Include:
 - a. Temporary dewatering, drainage, irrigation
 - b. Water conditioning
 - c. Frost-protection measures.

D. Filter Sand and Drain Gravel Placement Plan

1. Develop and submit a plan for the placement of the filter sand and drain gravel materials.
2. Detail the equipment and methods proposed for the spreading, placement, and compaction of embankment materials that limit the waste of drain materials.

E. Lime Treatment Plan

1. Describe means and methods for applying lime at the specified rate, mixing soil and lime, and moisture control. Include descriptions of the proposed equipment to be used in lime

treatment and the location of the processing area.

2. Detail the means and methods proposed for sealing the lime-treated area.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. When stockpiling material onsite, clear and strip stockpile areas prior to stockpiling.
- B. Handle, load, haul, stockpile, and place materials to minimize segregation, contamination, and breakdown.
- C. All practical means shall be taken by the Contractor to ensure that the filter sand and the drain gravel materials do not become contaminated with fines or other deleterious materials. Examples of sound measures to avoid or prevent contamination include:
 1. Controlling surface runoff, minimizing construction traffic over these materials, maintaining dry excavations, protection during shutdown periods, and keeping these materials higher than the surrounding grade.
- D. At equipment crossings, protect filter sand and drain gravel materials from contamination.
- E. At equipment crossings or after shutdown, remove all temporary materials used to protect the permanent embankment materials before resuming placement of filter sand and/or drain gravel materials. If the permanent embankment materials are contaminated or otherwise harmed, remove and replace the unacceptable embankment materials with no additional payment.
- F. Earthfill material used for temporary protection may be reused for performing other embankment work if the material is suitable for such use.
- G. Do not place and compact soil under following conditions.
 1. When higher than specified soil moisture content is present or the specified moisture content cannot be achieved.
 2. Heat or wind, or both, that dries material below specified moisture conditions.
 3. Ice, snow, or frozen materials are present.

1.05 PROJECT/SITE CONDITIONS

- A. Exploratory investigations are not sufficient to disclose all defects that may exist in the foundation and that will be encountered during construction.
- B. Final excavation lines will be determined by the GOVERNMENT in the field based on the conditions encountered and may not be the same as the estimated excavation lines shown on the Drawings.

PART 2 - PRODUCTS

2.01 GENERAL

- A. Obtain embankment materials from required excavations, existing stockpiles, designated borrow areas, or approved off-site sources. Materials from required excavations, stockpiles and designated borrow areas are comprised of Lean Clay (CL), Fat Clay (CH), and Clayey Sand (SC).

Some regions in the required excavation contain beds, layers and lenses of silt, sand, clay, gravel, and cobbles that may require sorting or blending and processing to be suitable for use.

2.02 COMPACTED EMBANKMENT FILL

A. Obtain embankment soil from required excavations and designated borrow areas. Soil from required excavations and designated borrow areas are generally comprised of fat clay and fat clay with gravel(CH). Some regions in the required excavations and borrow areas will contain beds, layers, and lenses of silt, sand, clay, bentonite, shale, and gravel that will require sorting, blending, processing, and moisture conditioning to be suitable for use as Compacted Embankment Fill. Some soil may not be suitable even after processing, which would require the soil to be wasted. Soil that will be considered suitable for use as embankment fill will have the following characteristics:

1. Maximum particle size of 2 inches,
2. at least 35 percent passing the No. 200 sieve, and
3. a plasticity index of greater than 7.

B. Soils with visible organic materials or salt crystals shall not be used as fill to reconstruct the embankment.

2.03 ROAD GRAVEL

A. Well-graded granular material meeting the following grading limits listed in Table 02300-1 – Road Gravel Surfacing Gradation. The liquid limit for the aggregate fraction passing the No. 40 sieve shall not exceed 25, nor plasticity index exceed 6, as determined by ASTM D4318.

Table 02300-1
Road Gravel Surfacing Gradation
MDT Type “A” Grade 6A

Sieve Size	Percent Smaller By Weight
1 ½ inch	100
¾ inch	74-96
3/8 inch	40-76
No. 4	25-60
No. 40	6-34
No. 200	0-8

2.04 FILTER SAND

A. Provide from a commercial source.

B. Filter sand material shall be a natural, rounded to sub-rounded quartz sand. Manufactured sand

from crusher operations will not be allowed.

C. Sand with material properties conforming to ASTM C 33, fine aggregate. The gradation for filter sand shall generally comply with the gradation for fine aggregate defined in ASTM C 33, except as modified in Table 02300-2 to limit the amount of material passing the No. 200 sieve. Material must meet specified gradation requirements after placement and compaction.

Table 02300-2
Filter Sand Gradation

Sieve Size	Percent Smaller By Weight
3/8 inch	100
No. 4	95-100
No. 8	80-100
No. 16	50-85
No. 30	30-60
No. 50	10-30
No. 100	0-10
No. 200	0-5

2.05 DRAIN GRAVEL

A. Provide from a commercial source.

B. Gravel with material properties conforming to ASTM C 33. The gradation of the drain gravel is generally consistent with the gradation for size 89 coarse aggregate, as in Table 02300-2.

Table 02300-3
Drain Gravel Gradation

Sieve Size	Percent Smaller By Weight
1 inch	100
1/2 inch	100
3/8 inch	90-100
No. 4	20-55
No. 8	5-30
No. 16	0-10

A. Drain gravel shall meet the aggregate durability requirements of ASTM C33.

2.06 HYDRATED LIME

A. Hydrated lime shall meet the following requirements when tested in accordance with ASTM C25:

1. Minimum available lime, reported as $\text{Ca}(\text{OH})_2 = 90\%$.
2. Maximum carbon dioxide = 7%.
3. The physical gradation of hydrated lime when tested in accordance with ASTM C110 shall meet the requirements of ASTM C977.

2.07 PREPARATION/SPREADING/CONDITIONING/PROCESSING/COMPACTING EQUIPMENT

A. General: Maintain equipment in good working condition at all times, and according to manufacturers' specifications.

B. Foundation Preparation/Cleaning Equipment:

1. Hand Tools: Including but not limited to shovels, bars, picks, wedges, brooms and light power tools as approved by the GOVERNMENT.
2. Earth Moving Equipment: Graders and dozers capable of "benching" into natural materials on sloping surfaces steeper than 3:1 (horizontal to vertical) and into the existing embankment to remove loose and dry material and to provide adequate bonding with new embankment material.

C. Scarifying Equipment: Provide graders and dozers equipped with scarifying teeth and rippers capable of roughening horizontal surfaces and loosening of dry materials to a depth of 6 inches for the addition of water and conditioning necessary to provide a good bond with the first layer of fill.

D. Spreading Equipment: Provide graders, dozers, and other equipment to assist in spreading, aerating, and water conditioning of soil foundations and fill materials, as required.

E. Screening Equipment: As required to make materials meeting gradation and classification requirements.

F. Watering Equipment: Provide water trucks, water wagons, sprinklers, sprayers, misters, or other approved equipment with a distributor bar or approved device to ensure uniform water application.

G. Discing Equipment: Provide discs to break up, mix, aerate, and assist in water-conditioning fill in both in stockpiles and on the embankment fill, in place; and to assist in preparing soil-foundation surfaces, equipped with:

1. Depth adjustments and cut-angle adjustments that can be operated without dismounting vehicle.
2. At least two gangs hooked together in tandem in a total cutting width not less than 10 feet; and with serrated blades at least 24-inch in diameter, spaced no more than 16 inches

apart.

3. Self-cleaning scrapers to prevent the accumulation of material on and between discs.

H. Compacted Embankment: Provide:

1. Self-propelled sheepsfoot or tamping foot compactors equipped with:
 - a. Staggered, uniformly spaced tamping feet with an individual end area not greater than 32 square inches, and tamping-feet length maintained not less than 6-inches (measured from drum to tamping head).
 - b. Maximum drum width: 84 inches
 - c. Two-drum or four-drum units separated by cab must have its static weight equally distributed to all compaction drums.
 - d. Static weight (on the drum) not less than 3,000 lb/ft of drum width.
 - e. Acceptable compactors include Caterpillar Model 815F, Dynapac CT3000 or other compactors approved by the GOVERNMENT.

I. Filter Sand and Drain Gravel:

1. Horizontal Blankets Provide:
 - a. Self-propelled, smooth-drum vibratory rollers:
 - i. 72-inch maximum drum width
 - ii. Static-operating weight (on the drum) of not less than 1,400 pounds per foot of drum width
 - iii. Centrifugal force not less than 6,000 pounds
 - iv. Means to adjust and maintain the frequency of vibration, as determined during construction.
2. Chimneys or Diaphragms Provide:
 - a. Vibratory plate compactor:
 - i. Static-operating weight of not less than 160 pounds
 - ii. Centrifugal force not less than 2,900 pounds
3. The self-propelled, smooth-drum vibratory rollers will generally be operated in vibratory mode. Limit passes in vibratory mode to avoid the breakdown of sand particles.

J. Special Compaction Equipment: Provide hand-operated impact tampers, vibrating plate tampers, or small tamping equipment for compacting fill against concrete surfaces, in restricted spaces and at locations where large compacting equipment cannot operate effectively.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify Conditions

1. Prior to placing material in any section of the embankment, verify the foundation for that

portion has been cleared, stripped, and water-conditioned. No material will be placed on the foundation without prior written approval of GOVERNMENT.

2. Scarify soil foundation areas so the surface will bond and compact with the first layer of fill. Scarify by ripping and discing material 3 to 6 inches in depth, as determined by the GOVERNMENT, so no smooth surfaces remain.
3. Condition the foundation surface and previous lift surfaces to uniform water content prior to placing fill. Pay particular attention to edges of fill against sloping ground and against abutment surfaces.
4. Do not allow water to accumulate on fill surfaces.
5. Do not place frozen materials, or place fill on frozen ground.
6. Do not place fill below water.

3.02 EXCAVATION

A. Classification: No classification as to class, nature, or condition will be made of any soil or rock from required excavations or borrow areas. However, the GOVERNMENT will determine the difference between General and Foundation Excavation. If the CONTRACTOR believes the excavation class has changed CONTRACTOR shall notify GOVERNMENT.

B. Excavate: Excavate according to the lines and grades shown on the Drawings, or written field clarifications.

C. Required excavation lines shown on the Drawings are the final or actual lines unless otherwise directed by the GOVERNMENT. Final quantities may be higher or lower than those estimated in the Bid Schedule.

D. Strip the embankment foundation area, borrow areas and all areas to receive fill to a minimum depth of 6 inches or as shown on the Drawings and as required to remove all organic soils, vegetative matter, roots, and other perishable, loose or objectionable material that might interfere with compaction of embankment lift or the bonding of embankment to foundation. Objectionable material will be as determined by the GOVERNMENT. Topsoil at the project site is limited. Perform stripping operations in a manner to conserve all topsoil that contains organics.

E. Transport stripped materials to stockpile areas outside of waterways and wetlands, subject to approval. Transport with a scraper or other approved wheeled vehicle. Locate piles so as not to affect the operation of the reservoir, spillway, or outlet works, interfere with the work, or detract from the appearance of the completed work.

F. Backfill unauthorized excavations, excessive soil excavations, and excavations for CONTRACTOR convenience with approved materials as determined by the GOVERNMENT, at no additional cost to GOVERNMENT.

G. Take precautions to preserve, in a sound condition, the material below and beyond the lines of all excavations.

H. Perform operations so the excavations will yield as much suitable material for construction purposes as practicable.

I. Reuse, as much as is practicable, all suitable materials from required excavation in the

permanent construction. Separate unsuitable materials and remove them from the work area as soon as practicable.

J. Excavated materials shall be stockpiled and water conditioned in temporary stockpiles prior to being placed in the embankment.

K. Water-condition materials that are too dry for immediate compaction. Use water, discs, blades and other equipment to water condition materials directly in borrow areas and in temporary stockpiles prior to their being transported to fill areas. Water conditioning is incidental to the work.

L. Spread out and allow materials to dry that are too wet for immediate compaction until the water content is reduced sufficiently for placement in the embankment. Aerating and drying is incidental to the work.

M. Level and trim waste piles to reasonably regular lines, and slope to drain. Finished slopes shall be no more than 3H:1V.

N. Remove soft, wet, dry, objectionable, or otherwise unsuitable materials from the dam foundation, as determined by the GOVERNMENT.

3.03 PREPARATION

A. Soil Foundation/Fill-Surface Preparation

1. Foundation preparation and water conditioning is critical. Maintain all exposed foundation surfaces in a moist condition (not saturated) at all times. Use sprayers, sprinklers, misters, or other effectively demonstrated method(s) to maintain the water content of the foundation soils near optimum for compaction and for bonding to the first lift of fill. Do not allow water to accumulate in ponds or puddles. Scarify or otherwise roughen all exposed temporary cut slopes to facilitate infiltration of water.
2. Check for soft areas by proof-rolling all foundation surfaces with a minimum of four passes by rubber-tired equipment weighing a minimum of 10,000 pounds per tire.
3. Excavate overly dry, overly wet, soft/yielding, or otherwise unsuitable subgrade as determined by the GOVERNMENT, and replace with appropriate, compacted, embankment fill for the location.
4. Scarify fill slopes and foundation surfaces flatter than 2.5H:1V (horizontal to vertical) so the surface will bond and compact with the first layer of fill. Scarify by ripping and discing material 3 to 6 inches in depth, as determined by the GOVERNMENT, so no smooth surfaces remain. Moisture condition as necessary to meet requirements of adjacent fill.
5. Bench each lift into natural materials and into the existing embankment on sloping surfaces 2.5H:1V or steeper to remove loose and dry material and to provide adequate bonding with the new embankment material. Benches shall be 3 feet horizontal as determined by the GOVERNMENT.
6. Water all slopes to at least 5 feet above and ahead of benching operations.
7. Condition the foundation surface and previous lift surfaces to a uniform water content prior to placing fill. Pay particular attention to edges of fill against sloping ground and

against abutment surfaces.

8. Moisten and disc or aerate the compacted surface of any foundation or fill layer that is respectively too dry or too wet for bonding prior to placement of a subsequent layer.
9. Grade areas to receive fill to the lines and grades shown prior to fill placement.
10. Dewater foundations to prevent the accumulation against or the intermingling of water with the fill.
11. Maintain in a moistened condition all earth surfaces against which earth fill, filter drain or concrete is to be placed.

B. Rock Foundation Preparation

1. Where a rock is encountered at the embankment foundation remove all debris, loose rock, sand, silt, and unsuitable materials by hand tools followed by compressed air, air/water jets or combination thereof as approved or directed by the GOVERNMENT.
2. Clean rock foundation surfaces within the core footprint with water and/or air as appropriate, and remove water sufficiently to provide a condition where joints, fractures, and springs can be identified, mapped, and treated, if necessary, by filters, dental mortar, and/or dental concrete, as determined by the GOVERNMENT. Water jetting should not be applied excessively to clay shale foundation areas
3. Clean soil and bedrock foundation surfaces within downstream embankment footprint to a “blade-clean” condition. Use small dozers or graders without crabbing to minimize disturbance to geologic features.
4. Remove unsuitable material from depressions, crevices and open rock joints where the width of opening is greater than 1/4 inch.
5. Use only rubber-tired equipment or crane with bucket. Track-mounted equipment should not be operated on prepared rock surfaces.
6. Use picks, bars, and hand tools to obtain a prepared rock surface free from loose, drummy, or shattered material.
7. Rock surfaces in which loose, drummy, or shattered material exists may be covered with shotcrete or dental concrete treatment as approved and directed by the GOVERNMENT.
8. Trim irregularities to a uniform slope not steeper than 0.5H:1V. Overhangs and vertical surfaces are not permitted.
9. GOVERNMENT will map core trench bottom (rock) and soil foundation surfaces in sufficient detail to identify open joints (in rock) and an open-work of sand or gravel (in soil) – that will require treatment by a filter drain blanket.
10. Place one 9-inch thick lift of “wet” embankment material (wetted to between 2% and 5% above optimum for compaction) on bedrock-surfaces beneath the embankment footprint, and compact with rubber-tired equipment before placing first lift of fill.

3.04 PLACEMENT

A. General: Construct dam embankment to the lines and grades shown on the Drawings, subject

to:

1. The use of suitable materials as specified for each zone.
2. Place each lift of each zone in a continuous, approximately horizontal layer over the full length and width of the embankment.
3. The performance of excavation and placement operations such that the embankment materials are mixed and blended to provide the most homogeneous section within each zone and best degree of compaction and stability practical. Provide primary water conditioning, mixing and blending in stockpiles. Distribute the fill in any given zone and lift such that it is free from lenses, pockets, streaks or layers differing materially in texture or gradation from the surrounding fill. Spread and compact embankment fill on an axis parallel to the dam centerline to the maximum extent practicable.
4. The proper preparation and water conditioning of foundation surfaces, lift surfaces, abutment surfaces and excavation surfaces. Maintain lift surfaces in a moist and roughened condition. Restrict the use of rubber-tire and smooth drum equipment on Compacted Embankment surfaces to prevent formation of smooth lift surfaces. Disc smooth surfaces to ensure proper bonding between lifts.
5. The maintaining of slopes of longitudinal surfaces (parallel to the dam axis) in the dam embankment between the previously completed dam and that to be placed to slopes no steeper than shown on the drawings. Bench each lift into the slope of the existing embankment to remove loose, dry material, and to facilitate bonding. Water all slopes to at least 5 feet above and ahead of benching operations.
6. Do not bench between filter sand and the compacted new embankment or between filter sand and the existing embankment or foundation.
7. Restrict equipment operation adjacent to structures to avoid damage to the subsurface features. Compact fill within two feet vertically and four feet laterally of structures with approved rubber-tire equipment and hand-operated tampers.

B. Compacted Embankment

1. Distribute water uniformly throughout each lift of material prior to and during compaction.
2. Water-condition Compacted Embankment material in stockpiles and borrow areas to a water content between optimum and 2% above optimum for compaction before transporting to the embankment. Add supplemental water to the material on the embankment prior to compaction by uniform sprinkling followed by uniform mixing by discs and graders.
3. Spread materials in uniform layers not to exceed 8 inches in loose thickness, and compact by rolling with approved sheepsfoot or tamping foot rollers. Spread and compact in directions parallel to the dam axis.
4. Place at a water content between 2% below and 3% above optimum for compaction (ASTM D698). Compact to not less than 97% of the laboratory maximum dry density (ASTM D698).
5. Finish the front slope of the Compacted Embankment to the final lines and grades shown

on the Drawings with a motor patrol.

C. Outlet Conduit Fill

1. Restrict equipment operation adjacent to structures and over outlet conduit to avoid damage to the structures. Do not operate equipment over the outlet conduit until the concrete has cured for at least three days and the concrete is covered with a minimum of two feet of fill.
2. Spread materials in uniform layers not to exceed 8 inches in loose thickness.
3. Place at a water content between 2% below and 3% above optimum for compaction (ASTM D698). Compact to not less than 97% of the laboratory maximum dry density (ASTM D698).
4. Spread and compact in directions parallel to the outlet conduit.
5. Do not operate sheepsfoot or tamping foot compacting equipment directly over pipes until at least two feet of fill is in place over the pipe encasement. Compact fill within four feet laterally of structures with approved rubber-tire equipment and hand-operated tampers.

D. Filter Sand and Drain Gravel

1. Control and conduct all transporting, stockpiling, excavation, production, and placement operations to minimize contamination, segregation, and particle breakdown.
2. Traffic shall not be permitted to cross filter zones at random. Equipment crossovers shall be maintained, and the location of crossovers shall be established before beginning chimney or diaphragm placement. Each crossover shall be cleaned of contaminating material and approved by the GOVERNMENT before the placement of additional filter or drain material.
3. Prevent soils from adjacent embankment from being tracked onto the filter sand.
4. The upper surface of the drain material, constructed concurrently with adjacent zones of the embankment, shall be maintained a minimum of 1 foot above the upper surface of the adjacent fill to provide drainage away from the drain material.
5. Place drain material in continuous, horizontal lifts not to exceed 12 inches in loose thickness on the filter blanket and chimney drain, free from segregation and contamination.
6. Thoroughly wet filter-drain material during placement and compaction. Clean, potable water, from a source approved by the GOVERNMENT, shall be used to wet filter zones to avoid contamination from clay fines.
7. Compact each lift of the filter sand blanket with 2 passes of the approved vibratory roller operated in vibratory mode. A combination of static mode vs. vibratory mode, or compaction by spreading equipment may be utilized to control particle breakdown, subject to testing and review by the GOVERNMENT.
8. Compact filter sand chimneys and diaphragms with 4 passes of approved vibratory plate compactor.
9. Carry out coverages of compaction equipment so the compactive effort is distributed

uniformly and in a systematic manner over the entire lift.

10. Use cut and fill method to construct vertical chimneys and diaphragms.
 - a. Place sand or gravel by cutting into previously placed earth embankment to create a trench that will be backfilled with filter material.
 - b. After the next interval of earth embankment is placed, cut into the underlying filter zone and remove all contamination from the top surface of the sand or gravel before placing the next lift of filter material.
11. Prior to construction of filter blanket or chimney the CONTRACTOR shall construct a test fill section. The test fill shall be observed by the GOVERNMENT to verify satisfactory results will be obtained with the equipment and methods planned for use in constructing the embankment filter and drain zones.

E. Road Gravel

1. Place gravel road surfacing with a motor patrol on prepared areas in uniform lifts, and moisten prior to compaction. Compact to not less than 95% of the laboratory maximum dry density (ASTM D698).
2. Finish the compacted material to the final specified lines and grades with a motor patrol.

3.05 LIME TREATMENT

A. All equipment used to convey or transport lime to or on the work site shall be covered or enclosed to prevent lime dust problems.

B. The CONTRACTOR shall prepare, mix, and cure the lime-treated soil in the area shown on the Drawings or as directed by the GOVERNMENT. Before lime treatment operations start, the processing area used for mixing and curing shall be stripped of topsoil. After the area is no longer required for mixing and curing, it shall be regraded as needed, disked to a depth of 4 inches, and covered uniformly with the removed topsoil.

C. Application

1. Lime shall be added to the soil in the mixing area at a rate of 3% lime by dry weight. (Assuming dry soil weight of 90 lbs/cubic foot, the lime application rate is 2.25 lbs/cubic foot of soil). The dry weight of the soil stockpiled in the mixing area shall be verified to determine the correct lime application weight.
2. Lime shall not be applied when the temperature is below 40 degrees Fahrenheit or is expected to drop below 40 degrees Fahrenheit within 24 hours. Lime shall not be applied during high wind conditions that hinder effective application.
3. Method 1 – Lime shall be uniformly applied in dry form on the soil surface at a rate that will attain the specified proportioning and lightly sprinkled with water to minimize dusting and blowing.
4. Method 2 – Lime shall be mixed with water to form a slurry before application to the soil surface. Agitation shall be accomplished through integral paddles, recirculating pumps, or a combination of devices. The lime and water shall be maintained as a uniform mixture until application to the soil. The slurry shall be uniformly applied to the soil surface at a

rate that will attain the proportioning specified in Paragraph 3.05 C. 1. Apply the slurry under pressure through spray bars.

D. Mixing

1. The soil, lime, and water shall be mixed by disking or using rotary mixers until a uniform mixture is obtained. During initial mixing with dry lime, a minimum of two cycles of water application followed by mixing with disks or rotary mixers shall be performed.
2. The thickness of the soil for treatment shall be no greater than the depth that can be effectively mixed by the equipment. The cycle of watering and mixing shall continue until the soil, lime, and water are thoroughly processed into a uniform mixture without lumps of soil or lime. When mixing is complete, the water content of the mix shall not be less than 2% above the optimum moisture content for the soil-lime mixture, when tested in accordance with ASTM D698.
3. After mixing, the lime-treated layer shall be sealed to minimize evaporation loss, lime carbonation, and excess wetting from precipitation. Sealing shall be performed by lightly compacting the surface of the treated layer with a pneumatic tire or smooth wheel roller or by other methods approved by the GOVERNMENT.

E. Curing

1. The lime and soil mixture shall be cured at least 7 calendar days. The water content of the mixture shall be maintained above optimum water content during the curing period by sprinkling with water, remixing, and resealing.
2. After the required cure time has occurred and before use as fill, the treated material shall be thoroughly remixed. Final mixing shall be accomplished so all nonslaked lime particles retained on the No. 4 sieve are removed.

F. Placement and Compaction

1. Place and compact the lime treated soil mixture in the same manner as Compacted Embankment, in accordance with Section 3.04.

3.06 QUALITY CONTROL

- C. The CONTRACTOR is responsible for adequate quality control and to develop and submit a quality control plan describing how the WORK will be performed in compliance with these specifications. The CONTRACTOR must conduct tests and inspections to document that their process control is achieving the specification requirements.
- D. The CONTRACTOR will employ an Independent Laboratory to provide onsite quality control services. The CONTRACTOR will prepare a plan describing the CONTRACTOR'S quality control testing program, including identification of qualifications of the company and personnel performing the field sampling and testing, qualifications of the laboratory that will perform the laboratory testing, and field procedures that will be followed to ensure testing is being performed within the specified testing frequencies.

E. Testing

1. Testing will generally use the standard test methods listed in Table 02300-4 Standards

for Testing for determining conformance with specification requirements.

2. The CONTRACTOR will perform tests as required to demonstrate that type of material used, material placement, and material compaction conform to contract requirements.
3. Notify the GOVERNMENT 24 hours before compaction work begins and before significant changes in compaction operations (major change in equipment or procedure used).
4. Notify the GOVERNMENT immediately of equipment change due to breakdown, or re-deployment.

Table 02300-4 Standards for Testing

Procedure	ASTM Standard(s)
Soil Classification	ASTM D2487 ASTM D2488
Gradation Analysis	ASTM D6913, D7928
Atterberg Limits	ASTM D4318
Dispersive Characteristics by the Crumb Test	ASTM D6572
Laboratory Maximum Density and Optimum Moisture Content	ASTM D698
Maximum Relative Density of Soil Using a Vibratory Table	ASTM D4253
Water (Moisture) Content	ASTM D2216
In-Place Moisture and Density: Nuclear	ASTM D6938
In-Place Density: Low-Activity Nuclear Method	ASTM D8167
Water Content of Soil by Calcium Carbide Gas Pressure Tester	ASTM D4944
Rapid Construction Control	ASTM D5080

F. Testing Frequency

1. Provide tests in the quality control program to demonstrate process control of all products during production and placement to achieve the specification requirements, including, but not limited to, embankment, gravel surfacing, filter sand, and drain gravel.
2. Greater frequency of testing is normally performed at the beginning of new work, new work crew, or new equipment.
3. The minimum testing frequencies for construction of the dam embankment and appurtenant elements and features is defined in Table 02300-5.

Table 02300-5 Minimum Testing Frequencies for Quality Control Testing

Feature	Section	Test	Minimum Frequency
Embankment	2.02	Moisture Density Gradation/Atterberg/Classification	Each soil type 1 test / 2,000 CY
		In-Place Density	1 test / 500 CY or 1 test / lift if the lift is less than 500 CY
Filter Sand	2.04	Gradation	1 test / 500 CY
Drain Gravel	2.05	Gradation	1 test / 100 CY
Road Gravel	2.03	Moisture Density Gradation	Each soil type 1 test / 500 CY
		In-Place Density	1 test / 250 SY

3.07 QUALITY ASSURANCE

A. Testing

1. The GOVERNMENT will perform independent quality assurance tests at their discretion to verify the type of material used, material placement, and material compaction conforms to contract requirements and the CONTRACTOR's quality control testing program complies with specification requirements.

B. Test Pits

Excavate test pits in embankment fill material, at the discretion of the GOVERNMENT, to depths and at locations as required to permit examination of fill lifts, possible lamination between lifts caused by compaction equipment, and contact against structures. Payment is included in the unit prices for the respective embankment fill. For bidding purposes, assume four (4) test pits, 3 feet wide by five feet deep, and two hours each for evaluation/documentation/testing.

C. Support/Access

1. Provide timely access to areas for density testing and/or for examination of in-place materials. Excavate and level an area in compacted material to provide a surface for testing.
 - a. Fills compacted by sheepfoot rollers are normally tested one or two lifts below surface.
 - b. Test pits for examination purposes are normally excavated four or five lifts below the surface.
2. Backfill test pits to original compaction requirements. Vertical sides of test pits should be excavated back to slopes that accommodate normal compaction methods.

3. Provide warning lights, flags, or other safety devices as needed by testing personnel.
4. Provide adequate lighting for performing tests, if required.

END OF SECTION 02300

PART 1 - GENERAL

1.01 DESCRIPTION:

This work consists of furnishing and placing a geotextile fabric as a permeable separator between dissimilar materials, such as between subgrade and gravel base, permanent erosion control measure, sediment control device or subsurface drainage fabric, and to aid in stabilizing poor subgrade soils.

1.02 SUBMITTALS:

A. Submit the following in accordance with Section 01300 – Submittals.

B. Submittal 02370-1, Geotextile Filter Fabric Product Data

1. Manufacturer's certification that the geotextile fabric meets specified chemical, physical, and manufacturing requirements.
2. Include the manufacturer's test results for the properties listed in Table 02370-1—Geotextile Filter Fabric Physical Properties.
3. Manufacturer information:
 - a. Name and address.
 - b. Evidence of manufacturer experience.
 - c. Total square yardage of installation, installation date, project manager, designer, fabricator (if any), and installer.
 - d. Descriptive documentation of the manufacturer's quality control program.

C. Submittal 02370-2, Geotextile Stabilization Fabric Product Data

1. Manufacturer's certification that the reinforcement fabric meets specified chemical, physical, and manufacturing requirements.
2. Include the manufacturer's test results for the properties listed in Table 02370-2—Stabilization Fabric Physical Properties.
3. Manufacturer information:
 - a. Name and address.
 - b. Evidence of manufacturer experience.
 - c. Total square yardage of installation, installation date, project manager, designer, fabricator (if any), and installer.

D. Geotextile samples.

PART 2 - PRODUCTS

2.01 FILTER FABRIC

A. Nonwoven geotextile is to be Mirafi 180N manufactured by TenCate, Pendergrass, GA, or approved equal.

B. Edges: Selvaged or otherwise finished to prevent outer material from pulling away.

C. Conform to roll values listed in Table 02370-1.

1. Values listed are minimum average roll values (MARV's), unless otherwise noted.
2. Test results for weaker principal direction shall meet or exceed the minimum values listed in the table.

Table 02370-1 Geotextile Filter Fabric Properties		
Property	Test Method	Minimum Required Value
Grab Tensile Strength	ASTM D4632	205 lbs
Grab Tensile Elongation	ASTM D4632	50%
Trapezoid Tear Strength	ASTM D4533	80 lbs
CBR Puncture Strength	ASTM D6241	500 lbs
Max. Apparent Opening Size (U.S. Sieve Size)	ASTM D4751	#80
Permittivity	ASTM D4491	1.4 sec ⁻¹
Flow Rate	ASTM D4491	95 gal/min/ft ²
UV Resistance (min. % retained tensile strength at 500 hours)	ASTM D4355	>70%

2.02 STABILIZATION FABRIC

A. Woven geotextile is to be Mirafi HP370 manufactured by TenCate, Pendergrass, GA, or approved equal.

B. Edges: Selvaged or otherwise finished to prevent outer material from pulling away.

C. Conform to roll values listed in Table 31 32 32B – Reinforcement Fabric Properties.

1. Values listed are minimum average roll values (MARV's), unless otherwise noted.
2. Test results for weaker principal direction shall meet or exceed minimum values listed in the table.

Table 02370-2 Geotextile Stabilization Fabric Properties		
Property	Test Method	Minimum Required Value
Tensile Strength (Ultimate)	ASTM D4595	3,600 lbs/ft
Tensile Strength (at 5% strain)	ASTM D4595	1,500 lbs/ft
Grab Tensile Strength	ASTM D4632	400 lbs
Grab Tensile Elongation	ASTM D4632	10%
Trapezoid Tear Strength	ASTM D4533	135 lbs
CBR Puncture Strength	ASTM D6241	1450 lbs
Max. Apparent Opening Size (U.S. Sieve Size)	ASTM D4751	#30
Permittivity	ASTM D4491	0.9 sec ⁻¹
Flow Rate	ASTM D4491	60 gal/min/ft ²
UV Resistance (min. % retained tensile strength at 500 hours)	ASTM D4355	>80%

2.03 PHYSICAL AND CHEMICAL REQUIREMENTS:

Fibers used in the manufacture of geotextiles and the threads used in joining geotextiles by sewing shall consist of long-chain synthetic polymers composed of at least 95% by-weight polyolefins, polyesters, or polyamides. They must be formed into a network so the filaments on yarns retain dimensional stability relative to each other, including selvages. All non-woven geotextile shall be needle punched. Heat-bonded or resin-bonded geotextiles shall not be used. Geotextile materials shall meet the physical requirements of Table 02370-1.

2.04 SHIPMENT AND STORAGE:

During shipment and storage, protect the fabric from direct sunlight, ultra-violet rays, temperatures exceeding 140° F (60° C), mud, dust, and debris. Keep the fabric in the manufacturer's wrapping until just before use. Include in each shipping document a certification that the geotextile meets the manufacturer's certificate and guarantee previously submitted.

PART 3 - EXECUTION

3.01 GENERAL:

Replace or repair all geotextile that is torn, punctured, or muddy. Remove the damaged material and place a patch of the same type of geotextile overlapping 3 feet (0.9m) beyond the damaged area.

3.02 FILTER FABRIC APPLICATIONS:

Store, handle, and deploy filter fabric in accordance with manufacturer recommendations and as specified herein. Place filter fabric between the prepared subgrade and riprap, where shown on the Drawings, and other locations as shown on the Drawings. Subgrades will be subject to approval by the GOVERNMENT prior to placement of filter fabric. Place filter fabric in the stilling basins by unrolling the fabric from the top of the slopes to the toe areas. Overlap successive geotextile sheets in such a manner the upstream sheet is placed over the downstream sheet and/or upslope over downslope. Overlap adjacent seams a minimum of 2 feet when over-compacted subgrade (e.g., upstream slope of dam and compacted subgrade soils on stilling basin sides). Make seam overlaps 3 feet in areas of poor subgrade compaction (e.g., under stilling basin bottoms). Provide a minimum offset of 5 feet between adjacent roll ends. Anchor fabric or otherwise hold firmly in place to prevent movement during construction. Pins or other methods that puncture the fabric are not allowed. Place filter fabric and secure it against the prepared subgrade surfaces such that the geotextile is in continuous (smooth) contact between the geotextile and subgrade with no void areas or wrinkles.

3.03 STABILIZATION APPLICATIONS:

Storage, handle, and deploy filter fabric in accordance with manufacturer recommendations and as specified herein. Utilize stabilization fabric as directed by the GOVERNMENT to assist in subgrade stabilization during the placement of embankment fill. Utilize fabric between soft clay subgrades and clay or granular fill materials. Fabric placed without the GOVERNMENT's approval is at CONTRACTOR expense. Minimum seam overlaps shall be 3 feet. Repair geotextiles damaged during installation in accordance with manufacturer's recommendations at CONTRACTOR expense

END OF SECTION 02370

PART 1 - GENERAL

1.01 DESCRIPTION:

This work consists of furnishing and placing the concrete block mat revetment system in accordance with this specification and conforming with the lines, grades, design, and dimensions shown on the Drawings.

1.02 SUBMITTALS:

A. Submit the following in accordance with Section 01300 – Submittals.

B. Submittal 02380-1, Concrete Block Mat Product Data

1. Manufacturer's certification that the geotextile fabric meets specified chemical, physical, and manufacturing requirements.
2. Include the manufacturer's test results for the properties listed in the tables below.
3. Manufacturer information:
 - a. Name and address.
 - b. Evidence of manufacturer experience.
 - c. Total square yardage of installation, installation date, project manager, designer, fabricator (if any), and installer.
 - d. Descriptive documentation of the manufacturer's quality control program.

PART 2 - PRODUCTS

2.01 CONCRETE BLOCK MAT

A. Concrete Block Mat is to be Flexamat 10NW manufactured by Flexamat, Cincinnati, OH, or approved equal.

B. Concrete Blocks: Furnish blocks manufactured with concrete conforming to cement requirements of ASTM C150 and to the aggregate requirements of ASTM C33. Furnish blocks that have a minimum weight of 3 pounds per block and placed no further than 2 inches apart.

Table 02380-1	
Concrete Compressive Strength Requirements	
Age	Required Compressive Strength
28 – Day	5,000 psi

C. Polypropylene Bi-Axial Geogrid: The interlocking geogrid shall be an open knitted fabric composed of high tenacity, multifilament polypropylene yarns knitted and coated in tension with an acrylic based coating which is designed to resist degradation in environments with exposure to water.

Table 02380-2 Polypropylene Bi-Axial Geogrid Requirements			
Property	Unit	Test	Requirement
Mass/Unit Area	oz/sy	ASTM D5261	6.5 oz/sy
Aperture Size	English Units	Measured	1.4 x 1.4 inch
Ultimate Wide Width Tensile Strength (MD x CMD)	lb/ft	ASTM D6637	2,055 lb/ft
Elongation at Ultimate Tensile Strength (MD x CMD)	%	ASTM D6637	6%
Wide Width Tensile Strength at 2% (MD x CMD)	lb/ft	ASTM D6637	822 lb/ft
Wide Width Tensile Strength at 5% (MD x CMD)	lb/ft	ASTM D6637	1,640 lb/ft
Tensile Modulus at 2% (MD x CMD)	lb/ft	ASTM D6637	41,100 lb/ft
Tensile Modulus at 5% (MD x CMD)	lb/ft	ASTM D6637	32,800 lb/ft

D. Geotextile Underlayment: Mats shall have a 10-ounce non-woven geotextile cast onto the back of the blocks, adhering to the back of each concrete block. Puncturing holes through the geotextile to hog ring or tie through other means to the Concrete Block Mat is unacceptable. The geotextile shall not be installed separately or loosely under the Concrete Block Mat. Each concrete block shall be cast and adhered to the geotextile.

Table 02380-3 Non-Woven Geotextile Requirements			
Property	Unit	Test	Requirement
Weight – Typical	oz/sy	ASTM D5261	10 oz/sy
Tensile Strength	lbs	ASTM D4632	250 lbs
Elongation at Break	%	ASTM D4632	50%
Mullen Burst	psi	ASTM D3786	500 psi
Puncture Strength	lbs	ASTM D4833	155 lbs
CBR Puncture	lbs	ASTM D6241	700 lbs
Trapezoidal Tear	lbs	ASTM D4533	100 lbs
Apparent Opening Size	US Sieve	ASTM D4751	No. 100
Permittivity	Sec-1	ASTM D4491	1.2 Sec-1
Water Flow Rate	gal/min/sf	ASTM D4491	80 gal/min/sf
UV Resistance at 500 Hours	70%	ASTM D4355	70%

2.02 SHIPMENT AND STORAGE:

All mats are to be inspected upon delivery. Assure that all units are sound and free of defects that would interfere with the proper placing of the unit or impair the strength or permanence of the construction. Chipping or missing concrete resulting in a weight loss exceeding 15% of the average weight of a concrete unit is grounds for rejection. Replace, repair, or patch the damaged areas per the manufacturer's recommendations. Cover the mats or otherwise protect them during long periods of storage to protect against degradation of the backing material as recommended by the manufacturer.

PART 3 - EXECUTION

3.01 GENERAL:

Prior to installing the concrete block mats, prepare the subgrade as detailed in the Drawings. All subgrade surfaces are to be smooth and free of rocks, roots, debris, or other protrusions that would prevent intimate contact between the mats and the subgrade. Ensure the prepared subgrade provides a smooth, firm, and unyielding foundation for the mats. Subgrades will be subject to approval by the GOVERNMENT prior to placement of the concrete block mats.

Install concrete block mats to the line and grade shown on the Drawings and per the manufacturer's guidelines. The manufacturer or authorized representative will provide technical assistance during the slope preparation and installation of the concrete block mats as needed.

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. Contractor shall furnish all labor, tools, and equipment and perform all work necessary for, or incidental to, the supply and installation of toe drains as shown in the drawings and specified herein. This work includes excavation, as well as placement of filter sand, drain rock, and HDPE pipe to drain water from the embankment toe.

1.02 REFERENCE STANDARDS

A. American Association of State Highway and Transportation Officials (AASHTO)

- | | |
|----------------|--|
| 1. AASHTO M330 | Standards Specification for Polypropylene Pipe, 300- to 1200-mm (12 to 60 inches) Diameter |
| 2. AASHTO SSHB | Standard Specifications for Highway Bridges, Seventeenth Edition |

B. ASTM International (ASTM)

- | | |
|---------------|--|
| 1. ASTM D3212 | Standard Specification for Joints for Drain and Sewer Plastic Pipes using Flexible Elastomeric Seals |
| 2. ASTM F2881 | Polypropylene (PP) Dual Wall Pipe and Fittings for Non-Pressure Storm Sewer Applications |

1.03 SUBMITTALS

A. 02610-1, Product Data:

1. Submit the pipe manufacturer's product data demonstrating that raw materials and pipe meet specification requirements.

B. 02610-2, Installation & Testing Plan

1. Installation plan:
 - a. Manufacturer's installation instructions.
 - b. Method to protect the pipe and hold the pipe in place during backfill.
 - c. Method for working backfill material under haunches.
 - d. Method for cleaning pipe after installation.
 - e. If the Contractor proposes to use construction equipment that exerts a load greater than H-20 loading on the top of the pipe, submit details of the proposed methods,

crossing location, equipment type, loading conditions, and other pertinent details.

2. Testing and inspection plan:
 - a. Details, procedures, and sequences of testing and inspection of drain and outfall.

C. 02610-3, Survey Reports:

1. Location (northing, easting, station, offset, and elevation) of pipe, cleanout, and outlet.

D. 02610-4, Inspection Video:

1. Provide a USB flash drive or electronic file transfer with video files documenting the drain inspection.

1.04 DELIVERY, STORAGE, AND HANDLING

A. Pipe:

1. Store and handle in accordance with manufacturer's recommendations.
2. Do not use rope, cable, or chain slings for handling pipe. Canvas slings not less than 6 inches wide may be used for handling pipe.
3. Protect the pipe from damage and prevent foreign matter from getting inside it when lowering it into the trench, laying it, and positioning it.

PART 2 - PRODUCTS

2.01 HIGH-DENSITY POLYETHYLENE PIPES

A. Perforated and Non-perforated High-Density Polyethylene Pipe, ADS model N-12 Dual Wall Pipe, manufactured by Advanced Drainage Systems, Inc., Hilliard, OH, www.adspipe.com; or approved equal, having the following characteristics:

1. Pipe: ASTM D3212 or AASHTO M252.
2. Watertight joining system: Bell and spigot gaskets meeting the requirements of ASTM D3212.
3. Dual-walled with a corrugated exterior wall and smooth interior wall.

B. Pipe Perforations: Perforated pipe shall be provided with class II perforations as defined in AASHTO M252.

2.02 DRAIN CLEANOUT

A. Steel Pipe: ASTM A53, standard weight, galvanized, type E or S, grade B.

B. Locking Cap and Cover:

1. Locking well caps manufactured by Royer Quality Casting, Inc.,

royerqualitycastings.com, or approved equal, with the following characteristics:

- a. Aluminum alloy.
- b. Two-piece locking well cap; sized to fit cleanout pipe.
 - i. Lower half attached to the steel pipe with set screws.
 - ii. Assembly locks with a single padlock.

PART 3 - EXECUTION

3.01 PIPE INSTALLATION

- A. Use excavation methods for pipe installation in accordance with Section 02300 – Excavation.
- B. Install pipe at locations indicated on drawings using manufacturer's recommended installation procedures.
- C. Lay pipe to the elevations, lines, and grades shown on the drawings.
- D. Cap or plug ends of pipe during periods of no work.
- E. Protect pipe ends from damage. Remove or replace broken, cracked, or unsuitable pipe and replace at no additional cost to the Government.
- F. Before and during assembly of a joint, keep all parts free of mud, oil, or grease. Keep the pipe interior free from deposits of mud, sand, gravel, or other foreign matter and in good working condition until the contract is complete and accepted.
- G. Joining Pipe:
 1. The couplings shall have a close fit with the pipe and shall maintain alignment of the pipe and prevent separation of the joints.
 2. Join pipe in accordance with ASTM D 2620 and manufacturer's instructions. Examine joints for defects before installation.
- H. Maintain in correct position and alignment during installation and subsequent construction operations.
- I. Angle Points and Curvilinear Drain Alignments:
 1. Construct vertical bends using manufacturer's standard pipe bends. Maximum bend shall be 22.5 degrees. A minimum length of 6 inches of straight pipe shall be installed between bends.
 2. Construct horizontal bends using manufacturer's standard pipe bends. Maximum bend shall be 22.5 degrees. A minimum length of 5 feet of straight pipe shall be installed between bends.
 3. Making curvilinear alignments in the field by pulling pipe into radius bends will not be

allowed.

J. Cover the outfall end of the drainpipe with an Agridrain (or approved equal) rodent guard.

3.02 BACKFILL

A. Maintain pipe grade and alignment during placement of material adjacent to and over pipe. Provide complete circumferential support for pipe to prevent uneven pressures and unacceptable ring deflections.

B. Use compaction techniques adjacent to pipe that will prevent deformation or damage to the pipe.

C. Fill trenches above drain materials.

D. Do not backfill until installed pipe has been inspected and approved by the Government.

E. Do not permit equipment travel over pipe until backfill material has been placed to a minimum depth of 3 feet over the top of the pipe, or to a greater depth when recommended by the pipe manufacturer.

1. After minimum earth covers are in place, the maximum equipment loading allowed over the pipe shall be H-20 loading in accordance with the AASHTO SSHB.
2. Before using construction equipment that exerts a load larger than H-20 loading on the top of the pipe, submit details of the proposed methods, crossing location, equipment type, and loading conditions for Government approval.

3.03 CLEANING TOE DRAIN PIPE

A. Maintain inside of pipes clean during construction.

B. Before performing final inspection, clean drain lines and outfalls to remove deposits of mud, sand, gravel or other foreign matter.

C. If inspection indicates the presence of foreign matter in the drain pipes, the system must be flushed and re-inspected at no additional cost to Government.

3.04 FIELD QUALITY ASSURANCE

A. Contractor shall inspect the interior of the toe drain system using closed circuit television (CCTV) equipment.

B. Coordinate inspections with the Government. Notify Government, in writing, of date and time pipe will be ready for inspection at least 10 days before inspection is required.

C. Provide safe access and working conditions for inspections.

END OF SECTION 2610

PART 1 - GENERAL

1.01 WORK INCLUDED:

The work covered by this section of the specifications consists of providing and installing concrete encased precast reinforced concrete pipe for the outlet conduit, concrete inlet structure, precast reinforced concrete manhole sections for the outlet works tower, and other appurtenances for the Outlet Works as indicated on the Drawings.

1.02 SUBMITTALS:

A. Piping Shop Drawings & Laying Diagrams: Provide Shop Drawings and pipe laying drawings for all pipe materials provided. Submit manufacturer's technical data in sufficient detail so the pipe can be verified to meet the specified requirements.

B. Anchor Details: Submit details for anchoring pipes that are subject to flotation when backfilled with concrete bedding.

C. Waterproofing products: Provide technical data and material data safety information for waterproof coatings and waterstops.

D. Manhole Shop Drawings: Submit shop drawings for all manhole materials provided. Submit manufacturer's technical data so the manhole materials can be verified to meet the specified requirements. Include manhole placement information regarding sequence, materials, and equipment.

E. Flap Gate Shop Drawings and Product Data: Submit shop drawings and manufacturer technical data for the outlet flap gate.

F. Steel and PVC Pipe Shop Drawings and Data: Submit shop drawings and manufacturer technical data for all steel and PVC pipe and fittings.

G. Vent Pipe Materials and Encasement Plan: Submit details for forming, supporting, and placing the vent pipe encasement.

1.03 MATERIAL STORAGE:

Proper storage of materials is the responsibility of the CONTRACTOR, as is the protection of said materials from adverse conditions that would disqualify them from use under this specification.

1.04 CLASSIFICATION OF TRENCH EXCAVATED MATERIAL:

No classification of trench-excavated materials will be made by the GOVERNMENT. Excavation and trenching includes the removal and subsequent handling of all earth, shale, loose or cemented gravel, loose rock, solid rock, and/or other material or materials excavated or otherwise removed in performance of the Contract work, regardless of the type, character, composition, or condition thereof. It also includes the removal and disposal of all vegetation, debris, broken asphalt pavement, broken concrete, brick, stone, and other materials encountered within the trench excavation.

1.05 TRENCH SLOPES AND SHORING:

The CONTRACTOR is solely responsible for designing and constructing stable, temporary excavations and should shore, slope or bench the sides of the excavations as required to maintain stability of both the excavation sides and bottom. Comply with applicable local, state, and federal safety regulations including the current OSHA Excavation and Trench Safety Standards. Construction site safety is the sole responsibility of the CONTRACTOR, who is also solely responsible for the means, methods, and sequencing of construction operations.

PART 2 - PRODUCTS

2.01 4-INCH STAINLESS STEEL VENT PIPE:

A. Vent pipe shall be Type 304 or 316 stainless steel Schedule 40 pipe with welded joints conforming to ASTM A999 – Standard Specification for General Requirements for Alloy and Stainless Steel Pipe.

2.02 OUTLET CONDUIT – REINFORCED CONCRETE PIPE

A. Class V, Wall B: meeting requirements of ASTM C76 – Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe.

B. Joints: Bell-and-spigot ends with rubber gasket designed to provide a watertight seal in conformance with ASTM C443 – Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets and ASTM C1619 – Standard Specification for Elastomeric Seals for Joining Concrete Structures.

1. Pipe joints, where indicated on the drawings, shall be secured with adjustable tie bolts. Tie bolts shall be manufactured from carbon structural steel conforming to ASTM A36 and hot-dip galvanized according to ASTM A153.

2.03 OUTLET WORKS TOWER – PRE-CAST REINFORCED CONCRETE MANHOLES:

A. Furnish manholes with dimensions shown on Drawings and meeting ASTM C478 – Precast Reinforced Concrete Manhole Sections.

B. Spillway Tower Joints: single off-set joint with rubber gasket designed to provide a watertight seal in conformance with ASTM C443.

1. Wrap outside of each manhole joint with external concrete joint sealing bands that conform to ASTM C877, Type II. Sealing bands shall consist of rubberized mastic with an outer polyethylene layer, polypropylene reinforcing, and steel tightening straps. Sealing bands shall be 12-inch wide MacWrap External Sealing Band manufactured by Mar Mac Construction Products Co., Inc., McBee, SC, or approved equal.
2. Seal the inside circumference of each manhole joint by first applying a polyurethane-based, elastomeric sealant. Polyurethane-based, elastomeric sealant is to be Sikaflex-1A or approved equal.
3. After the interior joint sealant has cured, apply a flexible polyolefin waterproofing tape over the full length of the interior of each joint. Waterproofing tape is to be Sikadur Combiflex SG System or approved equal. The system is to consist of Sikadur 31 Hi Mod

adhesive with tape that has a minimum width of 8 inches.

C. Coat Exterior surface of concrete manhole with the following Waterproofing System: HEY'DI K-11, cement based, crystalline waterproofing system, Tamms Industries, Kirkland IL; or approved equal. Do not coat the manhole surface on either side of joints where an external sealing band will be placed.

2.04 WATERSTOP

A. Install rubber waterstops meeting the requirements of ASTM C923 and ASTM C1478 around the RCP that is to be cast into the tower manhole wall and the Vent Pipe that is to be cast to the outside of the tower manhole as shown on the Drawings. The RCP and Vent Pipe waterstops shall be WS Series manufactured by Press-Seal Gasket Corporation, Fort Wayne, IN or approved equal.

2.05 INLET STRUCTURE:

Cast-in-place or pre-cast structural concrete meeting the requirements of Section 03300 – Cast-In-Place Concrete.

2.06 CONCRETE CONDUIT ENCASEMENTS

Cast-in-place concrete bedding and concrete cap meeting the requirements of Section 03300 – Cast-In-Place Concrete.

2.07 CONDUIT OUTLET FLAP GATE

A. The hinged steel flap gate located at the downstream of the outlet conduit is to be manufactured by Agri Drain Corporation, <https://www.agridrain.com/>, or approved equal.

B. Non-stainless steel components are to be galvanized or powder coated.

C. All fasteners and hardware are to be stainless steel.

PART 3 - EXECUTION

3.01 BARRICADES, GUARDS AND SAFETY PROVISIONS:

To protect persons from injury and to avoid property damage, place adequate barricades, construction signs, warning lights, and guards, as required, and maintain during the progress of the construction work and until it is safe to resume use of the trench area. Observe rules and regulations of the local authorities respecting safety provisions.

3.02 OUTLET TRENCH EXCAVATION:

A. Meet current OSHA Safety and Health Standards for all excavation, trenching, shoring, and related work.

B. Excavate at the specified locations for pipeline installations and appurtenant structures.

C. Use excavator bucket without teeth to provide smooth trench bottom and walls.

D. During excavation, stockpile backfill materials away from the trench banks to assure trench

wall stability. Assure surface drainage of adjoining areas is unobstructed.

- E. Remove and dispose of excess or unsuitable excavated materials in areas designated for waste.
- F. Prevent surface water from flowing into excavations. Promptly remove all water accumulating in trench excavations. Do not permit water to accumulate in any open trench. Remove and re-lay all pipe out of alignment or grade caused by trench flooding.
- G. Grade the trench bottoms to the specified lines and grades. Assure bedding material provides uniform bearing and support for each pipe section along its entire length. Backfill over-excavation with approved material.

3.03 SOFT OR UNSTABLE TRENCH SUBGRADE:

- A. When soft or unstable material is encountered at the trench subgrade which will not uniformly support the pipe, excavate the material to the depth directed by the GOVERNMENT and backfill to trench subgrade elevation with approved material.
- B. Make excavations for manholes, structures and other appurtenances of the size and depth to permit compacting of backfill on all sides to the specified density. The requirements for removing water and other applicable portions of these specifications apply to excavation for appurtenances.
- C. Remove all groundwater encountered in trench excavations. Do not place pipe, bedding or backfill materials below the groundwater elevation established by dewatering operations.
- D. The stability of construction excavations and associated worker safety, including slope geometry and shoring/bracing considerations, are the responsibility of the CONTRACTOR. Meet current OSHA regulations.
- E. Backfill all trenches as specified immediately after grade, alignment and piping jointing has been inspected and approved by the GOVERNMENT. Conduct any pipe testing as specified in the respective sections.

3.04 PIPE INSTALLATION:

- A. Handle pipe and place in accordance with the manufacturer's recommendations and in a manner as to prevent damage to the piping. Protect pipe from displacement during bedding and backfill operations. Special considerations are necessary to prevent floating of the pipe when placing concrete backfill.
- B. Finished placement tolerances: Place such that the pipe will have no adverse grade and the plan and profile alignments are within the following tolerances of the lines shown on the Drawings at any point in the pipe - three inches for pipes larger than 24-inch diameter and two inches for 24-inch and smaller.

3.05 FILL ADJACENT TO OUTLET PIPE:

- A. Perform fill placement and compaction in conformance with Section 02300 – Earth Dam.
- B. Spread materials in uniform layers between 6 and 8 inches in loose thickness. Compact to specified density for Compacted Embankment by rolling with approved rubber-tired equipment. Spread and compact in directions parallel to the pipe.
- C. Do not operate sheepsfoot or tamping foot compacting equipment directly over pipes until a

minimum of two feet of fill is in place over the pipe.

3.06 DRAINAGE CONSIDERATION:

A. Backfill trenches in such a manner that water will not accumulate in unfilled or partially filled trenches. Remove all material deposited in roadway ditches or other watercourses crossed by the trench immediately after backfilling is completed. Restore the section, grades and contours of ditches or watercourses to their original conditions.

B. Adequately protect backfill from surface erosion where trenches are constructed in or across roadway ditches or other watercourses.

END OF SECTION 02724

PART 1 - GENERAL

1.01 WORK INCLUDED:

This section covers the furnishing and installation of galvanized corrugated steel pipe culverts, connecting bands, and appurtenances that must be placed as shown in the Contract Documents.

1.02 RESPONSIBILITY FOR MATERIAL:

The CONTRACTOR is responsible for all such material furnished and shall replace, at his own expense, all such material found defective in manufacture or damaged in handling after delivery by the manufacturer. This shall include the furnishing of all materials and labor required for the replacement of installed material discovered defective prior to the final acceptance of the work or during the one-year correction period.

The CONTRACTOR is responsible for the safe storage of material furnished by him or to him and accepted by him and intended for the work. The interior of all pipes and other accessories shall be kept free from dirt and foreign matter at all times.

1.03 SUBMITTALS

A. 02800-1, Product Data:

1. Submit the pipe manufacturer's product data demonstrating that raw materials and pipe meet specification requirements.

PART 2 - PRODUCTS – NONE

2.01 MATERIALS

A. CORRUGATED METAL PIPE (CMP)

This pipe shall conform to the requirements of the following specifications: AASHTO M-218 or ASTM A 929. The pipe shall be manufactured in accordance with the applicable requirements of AASHTO M-36 or ASTM 760. The pipe sizes, gages, and corrugations shall be as shown on the project plans.

B. CMP FLARED ENDS AND CONNECTING BANDS

The galvanized material used in the fabrication of the end sections shall conform to the applicable material requirements of AASHTO M 218 and ASTM A 929 or other pre-approved AASHTO or ASTM material specifications. Connecting bands to be used on the corrugated metal pipe shall be fabricated from galvanized steel.

PART 3 - EXECUTION

3.01 EXCAVATION

The completed channel bottom shall be firm for its full length and width. The channel bottom shall be scarified and recompacted in accordance with Section 02300 Earth Dam.

Excavated material shall be stockpiled for use as backfill, incorporated into other portions of work, or wasted as directed by the ENGINEER. Material to be used in the backfill of the pipe shall be stockpiled a minimum of two feet from the edge of the trench.

3.02 PIPE BED PREPARATION

The pipe bed shall be shaped to fit the pipe bottom and uniformly compacted. It shall be clear of stones and cobbles, lumps, organics, or failing subgrade. If bedrock, hardpan, or fragmented material exists, excavate the trench below the pipe six inches and backfill with granular material.

3.03 INSTALLATION

A. RIGID PIPE

1. Begin pipe installation at the downstream end. Ensure the full length of each pipe segment bottom is in contact with the shaped bedding and place bell or groove ends facing upstream.
2. Ensure that the ends are clean and properly lubricated. Join sections with the ends tight and the inner surfaces flush and even.

B. FLEXIBLE PIPE

1. Flexible pipes shall be firmly jointed by coupling bands. Begin pipe installation at the downstream end of the line. The lower half of the pipe shall be in contact with the shaped trench bottom throughout its entire length.
2. The outside circumferential laps of flexible pipe shall be placed facing upstream. Place the pipe with longitudinal laps or seams at the sides.

3.04 BACKFILL

Backfill material placed under, adjacent to, and over the pipe shall consist of fill conforming to the requirements of Section 2300. Backfill material shall not contain frozen lumps, organics, highly plastic clay, or large stones. Backfill shall be placed in compactable layers of approximately 8 inches loose measurement, and each layer compacted to 95% of optimum dry unit weight in accordance with ASTM D698 and within 2% of the moisture content. Backfill adjacent to or over pipe culverts shall be brought up simultaneously on both sides of the pipe to avoid lateral displacement. Backfill under the haunches of the pipe shall be tamped or rammed into place between the pipe and the channel bottom.

The contractor will exercise during backfilling, operations to prevent damage to the pipe, either through unequal pressure exerted on the sides or by backfill being dropped directly onto the pipe. Pipe

shall be inspected before any backfill is placed. Any pipe found to be out of alignment, unduly settled or damaged shall be taken up and re-laid or replaced.

END OF SECTION 02800

PART 1 - GENERAL

1.01 WORK INCLUDED:

- A. Stripping and stockpiling of topsoil.
- B. Spreading and grading of topsoil in preparation for seeding.
- C. Seeding of restoration areas.

1.02 SUBMITTALS

- A. Seeding Plan:
 - 1. Qualifying experience of person responsible for supervision of seeding. Include names, addresses, and telephone numbers of references.
 - 2. Describe equipment and methods to be used for seedbed preparation and seeding.
 - 3. Plan describing approach to establish vegetation and CONTRACTOR's Quality Control plan.

PART 2 - PRODUCTS

2.01 TOPSOIL:

- A. Strip "topsoil" from within the limits of disturbance to a depth of six (6) inches or as shown on the Drawings. Remove all construction debris.

2.02 SEED

- A. All seed and seed mixes will be provided by the GOVERNMENT.

PART 3 - EXECUTION

3.01 STRIPPING:

- A. Strip and stockpile all areas to be excavated for borrow and waste areas and all areas that are to receive fill. Any additional topsoil required to be placed on disturbed areas shall be excavated from the reservoir basin in areas approved by the GOVERNMENT.

3.02 PLACING TOPSOIL:

- A. Do not spread topsoil when the ground or topsoil is frozen, excessively wet or otherwise in a condition detrimental to the Work. Lightly scarify designated surfaces prior to the spreading operation. Place 6-inches of topsoil over all areas disturbed by the work. After placement is completed, finish the topsoil surface to a uniform surface. Do not compact topsoil and

minimize equipment travel over placed topsoil.

3.03 AREAS TO BE SEEDED

A. Seed all vegetated areas impacted during construction. Seeding is to be accomplished using a seed drill. Where slopes are too steep for a seed drill, those areas may be seeded using a broadcast seeder. The broadcast seed application rate is twice the drill seed rate.

3.04 SEEDBED PREPARATION

A. The CONTRACTOR shall provide 72hour notice prior to seedbed preparation to allow for proper GOVERNMENT review and inspection.

B. Prepare a clean and firm seedbed.

1. Disk a maximum of two times.
2. Create suitable soil conditions for seed adherence.
3. Break down dirt clods.
4. Remove stiff clods, lumps, roots, litter, stones, and other foreign material greater than 6 inches in size from the surface.

C. Scarify or harrow and rake topsoil to minimum depth of three inches in a direction which follows the natural contour of the land on slopes of three to one (3:1) or less. Soils on slopes greater than three to one (3:1) shall be prepared for planting in a manner approved by the GOVERNMENT. Any irregularities in the ground surface resulting from soil preparation operations shall be corrected and sloped to drain.

D. Contour area so furrows cross water flow caused by rainfall or snowmelt.

E. Fill and smooth topsoil surface to remove rills, gullies and depressions.

F. Protect prepared seedbed from erosion and washouts. Repair damaged surfaces before seeding.

3.05 SEEDING

A. If seeding is required between May 15 and October 1, plant a temporary cover crop of oats or barley. Re-seed the area with the specified seed mixture after October 1, but before the ground freezes.

B. Do not seed when ambient temperature is below 38 degrees Fahrenheit without approval of the GOVERNMENT.

C. Do not seed when ground is snow covered.

D. Do not seed when wind velocities prevent uniform application of materials or would mobilize and drift placed materials.

3.06 DRILL SEED

A. The seed shall be uniformly distributed with a grass drill seeder to a depth of ½ inch. The area shall be drilled from two directions, with at least a 45° difference in angle of approach. If the drill seeder does not include a packer wheel system, all seeded areas shall be cultipacked

until a minimum of 85% of the seed is covered, or otherwise accepted by the GOVERNMENT.

3.07 BROADCAST SEEDING

A. Mechanical broadcasting:

1. Equipment:

- a. Centrifugal type.
- b. Pull type with cultipacker.

2. Designed and regulated to apply seed uniformly at proper rate per acre.

B. Hand Broadcasting (in areas inaccessible to a mechanical type broadcaster):

1. By hand broadcaster.

2. By hand.

C. Cover seed with soil to depth of 1/4-inch to 1/2-inch immediately after broadcasting.

1. Use hand rake or float.
2. Do not use log chain or similar device.
3. Press seed into soil using a cultipacker.

3.08 CLEANUP OF AREA:

A. Upon completion of the Work, clear the entire project site of all debris. Finish ground surfaces to smooth, uniform slopes and present in a neat, workmanlike appearance. Repair any utilities, structures, landscaping, grass areas, etc., that are damaged due to negligence of the CONTRACTOR at the CONTRACTOR'S expense, and as directed by the GOVERNMENT.

END OF SECTION

DIVISION 3

CONCRETE

PART 1 - GENERAL

1.01 SUMMARY

A. Structural Concrete Elements: All work necessary to construct structural concrete elements, including construction of formwork, installation of reinforcement steel, and producing, delivering, testing, placing, curing, and protecting concrete.

1.02 REFERENCE STANDARDS

A. American Concrete Institute (ACI)

- | | | |
|-----|------------|---|
| 1. | ACI 117 | Specifications for Tolerances for Concrete Construction and Materials |
| 2. | ACI 212.3R | Chemical Admixtures for Concrete |
| 3. | ACI 214 | Guide to Evaluation of Strength Test Results of Concrete |
| 4. | ACI 301 | Standard Specification for Structural Concrete |
| 5. | ACI 304R | Guide for Measuring, Mixing, Transporting and Placing Concrete |
| 6. | ACI 305R | Hot Weather Concreting |
| 7. | ACI 306.1 | Cold Weather Concreting |
| 8. | ACI 308R | Guide to Curing Concrete |
| 9. | ACI 309 | Guide for Consolidation of Concrete |
| 10. | ACI 315 | Details and Detailing of Concrete Reinforcement |
| 11. | ACI 318 | Building Code Requirements for Structural Concrete and Commentary |
| 12. | ACI 347 | Guide to Formwork for Concrete |
| 13. | ACI 504 | Guide to Sealing Joints in Concrete Structures |
| 14. | ACI 546R | Concrete Repair Guide |

B. ASTM International (ASTM)

- | | | |
|----|------------|---|
| 1. | ASTM A 36 | Standard Specification for Carbon Structural Steel |
| 2. | ASTM A 615 | Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement |
| 3. | ASTM A 706 | Standard Specification for Deformed and Plain Low-Alloy Steel Bars for Concrete Reinforcement |
| 4. | ASTM A 775 | Standard Specification for Epoxy-Coated Steel Reinforcing Bars |

5.	ASTM C 31	Making and Curing Concrete Test Specimens in the Field
6.	ASTM C 33	Concrete Aggregates
7.	ASTM C 39	Compressive Strength of Cylindrical Concrete Specimens
8.	ASTM C 41	Obtaining and Testing Drilled cores and Sawed Beams of Concrete
9.	ASTM C 94	Ready-Mixed Concrete
10.	ASTM C 114	Chemical Analysis of Hydraulic Cement
11.	ASTM C 136	Sieve analysis of Fine and Coarse Aggregates.
12.	ASTM C 138	Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete.
13.	ASTM C 143	Slump of Hydraulic-Cement Concrete
14.	ASTM C 150	Portland Cement
15.	ASTM C 171	Sheet Materials for Curing Concrete.
16.	ASTM C 172	Standard Practice for Sampling Freshly Mixed Concrete
17.	ASTM C 231	Air Content of Freshly Mixed Concrete by the Pressure Method.
18.	ASTM C 260	Air-entraining Admixtures for Concrete
19.	ASTM C 295	Petrographic Examination of Aggregates for Concrete
20.	ASTM C 309	Liquid Membrane-Forming Compounds for Curing Concrete.
21.	ASTM C 494	Chemical Admixtures for Concrete
22.	ASTM C 618	Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
23.	ASTM C 1017	Chemical Admixtures for Use in Producing Flowing Concrete.
24.	ASTM C 1064	Temperature of Freshly Mixed Hydraulic Cement Concrete.
25.	ASTM C 1074	Estimating Concrete Strength by the Maturity Method
26.	ASTM C 1077	Standard Practice for Laboratories Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Laboratory Evaluation
27.	ASTM C 1260	Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)

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- b. Test reports (less than 6 months old) demonstrating compliance with the specified material properties.
 5. Manufacturer data sheets for admixtures to be incorporated into the mix.
 6. Re-submit mix design for change in material source or type.
- B. Manufacturer data sheets for materials to be incorporated into the work, including, but not limited to:
 1. Curing materials
 2. Bonding agents
 3. Expansion joint material
- C. Reinforcement Shop and Assembly Drawings
 1. Shop Drawings: Details shop fabrication requirements (e.g., length and bending requirements) for reinforcement at splices and joints and identifies reinforcement quantities.
 2. Assembly Drawings: Details how the reinforcement is to be installed
- D. Reinforcement Certified Materials Test Report (CMTR) demonstrating compliance with the material properties for reinforcement steel.
- E. Formwork Design and Assembly Drawings
- F. Concrete Placement Drawings:
 1. Drawings for each individual concrete placement. An individual concrete placement is defined as a portion of concrete work placed in one continuous operation between specified lines or joints.
 2. Show locations, dimensions, blockouts, openings, recesses, waterstops, and finishes.
 3. Show details of items embedded in or associated with placement except reinforcing steel.
 4. Include a separate drawing showing placement sequence.
 5. List reference drawings from which details shown on placement drawing were obtained on each drawing.
 6. Reference related steel reinforcement drawings associated with placement on each drawing.
- G. Concrete Placement Schedule:
 1. Written plan describing location, sequence, and the scheduled date for concrete placements.
 2. Complete, detailed concrete placement schedule showing the CONTRACTOR's plan for placement of individual features, units, and other elements of concrete work.

H. Concrete Placement Plan(s)

1. Written plan(s) describing how each concrete placement will be executed. Include items such as, but not necessarily limited to:
 - a. Anticipated haul times to the jobsite including, if necessary, provisions to account for long hauls.
 - b. Provisions to account for adverse environmental conditions (e.g., heat, cold, wind)
 - c. Method of concrete placement (e.g., direct placement from delivery truck discharge chute, pump truck).
 - d. Provisions to ensure adequate concrete placement and consolidation in areas of rebar congestion or limited accessibility of the concrete discharge, such as “bird mouths” in wall placements.
 - e. Identify the method of curing and how the completed work will be protected against damage from environmental conditions and construction activities.

I. Cold Weather Placement Plan

1. Submit detailed procedures for the production, transportation, placement, protection, curing, and temperature monitoring of concrete during cold weather. Include procedures to be implemented upon abrupt changes in weather conditions or equipment failures.

J. Hot Weather Placement Plan

1. Submit detailed procedures for the production, transportation, placement, protection, curing, and temperature monitoring of concrete during hot weather.

K. Certifications:

1. Batch Plant: National Ready Mixed Concrete Association (NRMCA) Certification of Production Facilities.
2. Concrete Materials Testing Laboratory: Accredited testing laboratory, including laboratory and field technicians, meeting the criteria of ASTM C 1077.

L. Concrete Batch Tickets

1. Provide concrete batch tickets from each concrete delivery truck at the time of delivery.

M. Protection Period Temperature Records

1. Submit temperature records documenting the temperature of the curing environment during the protection period.

1.04 DELIVERY, STORAGE, AND HANDLING

A. With each load of concrete to the jobsite, furnish batch ticket to GOVERNMENT. Batch tickets shall comply with the requirements of ASTM C 94.

B. Concrete Reinforcement Steel:

1. Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid rusting.
2. Protect from contaminants such as grease, oil, and dirt.
3. Provide for identification after bundles are broken and tags removed.

PART 2 - PRODUCTS2.01 CEMENTITIOUS MATERIALS

A. Cement:

1. Portland Cement in accordance with ASTM C 150, Type V.
2. Low-alkali Blended Hydraulic Cement in accordance with ASTM C 595, Type IL, IP, IS, or IT.
 - a. Blended hydraulic cement mixtures must meet the physical requirements for High Sulfate (HS) special properties. Blended hydraulic cement is to be tested in accordance with ASTM C 1012 and submitted for approval. Acceptance will be based on an expansion of less than 0.10% at 18 months.

2.02 WATER

A. Potable water is acceptable without testing. Where non-potable water will be used (wholly or blended with potable water), the water shall comply with ASTM C 1602, including the optional requirements of Table 2.

2.03 AGGREGATES

A. Fine Aggregate: ASTM C 33, except allowable fines content (material passing No. 200 sieve) less than 3 percent.

B. Coarse Aggregate: ASTM C 33; nominal maximum size aggregate (NMSA) as identified in Table 03300A – Mix Design Table

2.04 ADMIXTURES

- A. Admixtures are limited to those approved.
- B. Do not use chemical admixtures, which contain more than 0.1 percent chloride, by weight.
- C. Admixtures shall be compatible with each other as certified by the manufacturers or demonstrated through trial batches.
- D. Air-Entraining Admixture shall comply with ASTM C 260.
- E. Chemical admixtures shall comply with ASTM C 494, and plasticizing admixtures shall be tested in accordance with ASTM C 1017.

2.05 STEEL REINFORCEMENT

- A. Reinforcing Bars:
 - 1. Plain Dowels: ASTM A 36.
 - 2. Deformed steel bars: Grade 60; ASTM A 615, ASTM A 7006, or ASTM A 775.
- B. Mechanical Splices and Terminations: Mechanical splices and terminations may be approved to increase assembly efficiency and increase the clear space around reinforcement steel. Use of such features must be specifically identified on the reinforcement assembly drawings and approved by the GOVERNMENT.

2.06 WATERSTOP

- A. Hydrophilic waterstop shall be a preformed joint sealant that swells in the presence of water, such as Hydro-Flex Waterstop HF302, or approved equal.
- B. Hydrophobic waterstop shall be a preformed joint sealant that does not swell in the presence of water, such as Synko-Flex Waterstop SF302, or approved equal.

2.07 CURING MATERIALS

- A. Water: ASTM C 1602, including optional requirements of Table 2 within ASTM C 1602.
- B. Curing Compound:
 - 1. Curing Compound: ASTM C 1315, type II (white pigmented).
 - 2. ASTM C 309, Type 1-D.
- C. Sheet Materials: ASTM C 171.

2.08 BONDING AGENT:

- A. Non-rewettable type for use in water retaining structures.
- B. Available Product: Subject to compliance with requirements, products which may be incorporated into the work include, but are not limited to, the following:
1. "Weld-Crete"; Larson Products
 2. "Everbond"; L&M Construction Chemicals
 3. "AKKRO 7T"; Euclid Chemical Co.
 4. Euco #452 Epoxy System; Euclid Chemical Co.
 5. "Daraweld C"; W.R. Grace
 6. "Acryl 10"; BASF

2.09 ACCESSORIES

- A. Sponge Rubber Joint Filler:
1. ASTM D 1752, Type I, except as specified.
 2. Test specimen compression load: 50 to 150 pounds per square inch.
- B. Joint Filler Adhesive: Nonbituminous adhesive recommended by filler manufacturer.

2.10 CONCRETE MIX PROPORTIONS

- A. The CONTRACTOR shall design concrete mixes.
- B. Design mixes in accordance with Table 03300A – Mix Table:

Table 03300A – Mix Table

Mix No.	Feature	f'c (lb/in ²)	Max. w/cm ¹	NMSA ² (inches)	Slump ^{3,4,5} (inches)	Air Content ⁵
1	Typical Structural Concrete	4,500 @ 28 days	0.42	3/4	4 (max)	6.5%, ± 1.5%

¹Maximum water/cementitious ratio.

²Nominal Maximum Size Aggregate.

³As-delivered slump. Slump >4 inches may be achieved using water reducing admixture added onsite.

⁴Concrete Bedding/Riprap Grout: Slump to be not less than 6" and not more than 10". Use water reducing admixture for consolidation in areas of difficult placement (slump > 6 inches).

⁵Measured at point of placement

1. Slump: In accordance with ASTM C 143.

2. Air Entrainment: Percent air by volume of concrete as discharged at placement, in accordance with ASTM C 231.
3. Use set-retarding admixtures only as approved by the GOVERNMENT.

2.11 CONCRETE TEMPERATURE

- A. Batch concrete as necessary to achieve placement temperatures between 50°F and 90°F.
- B. Use materials and/or procedures necessary to achieve required placement temperatures; including, but not limited to, the following:
 1. Replace all or a part of the mixing water with chilled water.
 2. Replace part of the mixing water with ice.
 3. Introduce liquid nitrogen into the concrete as it is batched.
- C. When air temperature has fallen, or is expected to fall, below 40 degrees Fahrenheit, prepare ingredients and mix in accordance with ACI 306.1.

PART 3 - EXECUTION

3.01 FORMWORK

- A. Design, construct, and remove formwork in accordance with ACI 347.
- B. Tolerances: Tolerances are based on the finished concrete surfaces as described in ACI 117.

3.02 STEEL REINFORCEMENT

- A. Clean reinforcement surfaces of heavy, flaky rust; loose mill scale; dirt; grease; and other foreign substances before placement.
- B. Accurately place and secure in position so that reinforcement will not be displaced during concrete placement.
- C. Field bending is not allowed, unless specifically approved by the GOVERNMENT.
- D. Do not weld (or tack weld) reinforcing bars unless shown on drawings or specifically approved by the GOVERNMENT.
- E. Place reinforcement with a clear distance of 1-inch, minimum, between surrounding reinforcement and anchor bolts, form ties, or other embedded metalwork, unless otherwise shown on drawings.
- F. Place plain dowels parallel to each other and to concrete surface.
- G. Tolerances: In accordance with ACI 117

3.03 CONCRETE

A. Hot Weather: In accordance with ACI 305R.

B. Cold Weather: In accordance with ACI 306.1.

C. BATCHING, MIXING, AND TRANSPORTING

1. Batch, mix, and transport concrete to site in accordance with ASTM C 94. The CONTRACTOR must allow GOVERNMENT access to the batch plant to observe batching operations, sequence of batching, and to confirm batch component tolerances.
2. Chemical admixtures shall be incorporated into the mix in accordance with ACI 212.3R and the manufacture's recommendations.
3. For hauls longer than 1-1/2 hours, extended set control admixtures may be used as approved during the mix design approval process.

D. CONCRETE PLACEMENT

1. In accordance with ACI 304R.
2. In accordance with the approved Concrete Placement Plan.
3. Seven days (minimum) between adjacent placements, or as approved by GOVERNMENT.
4. Clean and roughen surfaces of existing concrete to be covered with fresh concrete.
 - a. Remove laitance, loose, or defective concrete, coatings, sand, curing compound, and other foreign material.
 - b. Sandblast, steel shotblast, or high-pressure water jet surfaces, or use other methods approved by the GOVERNMENT, to create a surface profile equivalent to, or larger than, CSP 5 as defined in ICRI 310.2.
 - c. Wash surface thoroughly.
5. Provide three days advance notification to GOVERNMENT for each concrete placement.
6. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed. Place concrete in presence of the GOVERNMENT.
7. On-site adjustments to the fresh concrete mixture shall be as allowed by ASTM C 94 and as approved by the GOVERNMENT.
 - a. The concrete supplier may add additional air-entraining admixture to achieve the desired air content, provided the slump and time restrictions have not been exceeded. Additional air-entraining admixture shall be recorded on the batch ticket to the nearest ounce.
 - b. Upon arrival at the job site, the CONTRACTOR may execute a one-time water addition to the batched concrete if:
 - i. Slump and time restrictions have not been exceeded

- ii. No concrete has been discharged,
 - iii. The additional water will not result in a W/C ratio greater than the W/C ratio of the approved mix design, and
 - iv. The volume of additional water shall be recorded on the batch ticket to the nearest gallon.
- c. After the one on-site addition of air-entraining admixture or water, the concrete shall be mixed for at least 30 revolutions of the mixer drum at mixing speed.
- d. CONTRACTOR quality control testing and quality assurance tests by the GOVERNMENT will be taken after the on-site adjustment and additional mixing.
- 8. Thoroughly moisten absorptive surfaces, such as earth and concrete, to be covered with concrete to reduce moisture loss from the fresh concrete into the substrate/adjoining surfaces.
- 9. Convey concrete from mixer to the place of final deposit rapidly by methods that prevent segregation or loss of ingredients and will ensure the required quality of concrete. Use conveying equipment, conveyors, hoppers, baffles, chutes, pumps that are sized and designed to prevent cold joints from occurring and prevent segregation in discharged concrete. Clean conveying equipment before each placement. Do not use aluminum pipes and chutes for placing or pumping concrete.
- 10. Do not use concrete which has started to set and cannot be properly placed.
- 11. Deposit concrete in forms in horizontal layers with proper consolidation into previous layers and in a manner to avoid inclined construction joints. Place each layer while preceding layer is still plastic, to avoid cold joints. For walls greater than six feet in height, concrete shall be placed using a tremmie inserted into the formwork to prevent a concrete free-fall discharge of more five feet. For wall pours greater than 12 feet in height, the CONTRACTOR shall include specific placement methods and procedures in the Concrete Placement Plan submittal.
 - a. Consolidate placed concrete with mechanical vibrating equipment. Use equipment and procedures for consolidating concrete recommended by ACI 309R.
 - b. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations no farther than the visible effectiveness of the vibrator. Place vibrators to rapidly penetrate placed layer and at least six inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mix constituents to segregate.
 - c. Make construction joints only where located on Drawings unless otherwise approved by the GOVERNMENT. Plan placements to continuously place concrete from one construction joint to another.
- 12. Deposit concrete for slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.

- a. Consolidate concrete during placement operations such that concrete is thoroughly worked around reinforcement and other embedded items and into corners.
 - b. Vibrate concrete until concrete has been consolidated to maximum practical density, is free from pockets of coarse aggregate, and is thoroughly worked around reinforcement and other embedded items and closes snugly into corners and against surfaces of forms.
 - c. Maintain reinforcement in position on masonry chairs made of durable, non-corroding materials such as plastic or precast concrete during concrete placement. Metal and/or wood chairs are not allowed.
 - d. Screed slab surfaces with a straightedge and strike off to correct elevations.
 - e. Slope surfaces uniformly to drains where required
 - f. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, free of humps or hollows, before excess moisture or bleed-water appears on the surface. Do not further disturb slab surfaces before starting finishing operations.
13. Protect concrete work from physical damage or reduced strength that could be caused by environmental conditions, such as cold, heat, and wind.

3.04 EMBEDDED ITEMS

- A. At the time of concrete placement, embedded items shall be clean and free from mud, oil, and other coatings that may adversely affect bonding capacity. Aluminum embedments shall be coated with a bituminous material to prevent electrolytic action between the embedded item and reinforcing steel that results in concrete deterioration. Embedment items shall be accurately placed and securely fastened to the forms in such a manner that the placing of concrete shall not alter their alignment or location. Contact between embedded items and reinforcing steel or tendon ducts is not permitted.

3.05 JOINTS AND EDGES

A. Construction joints (CJ):

1. Construction joints are joints which are purposely placed in concrete to facilitate construction, reduce initial shrinkage stresses and cracks, allow time for installation of embedded metalwork, or allow for subsequent placing of other concrete.
2. Bond is required at construction joints regardless of whether or not reinforcement is continuous across joint.
3. Locate construction joints where shown on drawings.
4. Relocation, addition, or elimination of construction joints will be subject to approval by the GOVERNMENT.
5. Clean, roughen, and surface dry surfaces of construction joints to be covered with fresh concrete.
6. Do not use a mortar layer on construction joints.

B. Control Joints (CTJ):

1. Control joints are joints placed in concrete to provide for control of initial shrinkage stresses and cracks of monolithic units.
2. Saw-cut control joints are to have a minimum depth of one inch, but they must be kept shallow enough to avoid cutting into the reinforcing steel.
3. Saw-cut joints are to be executed as soon as final set has been achieved and the saw cutting operation will not ravel or dislodge the coarse aggregate out of the slab.
 - a. In hot weather, final set is typically reached within 4 hours after the slab has been finished.
 - b. In cold weather, final set is typically reached within 12 hours after the slab has been finished.

C. Edges:

1. Chamfer edges of permanently exposed concrete, except slabs and top edges of curbs, with a 45 degree bevel 3/4-inch by 3/4-inch; unless otherwise shown of drawings.
2. Tooled Edges: Tool exposed edges of slabs and top edges of curbs to a radius of 1/4-inch.
3. Preformed joints consisting of plastic or metal strips not allowed.

3.06 FINISHING

A. Formed Surfaces:

1. Below Grade Surfaces: Smooth-form finish as defined in ACI 301
2. Exposed Surfaces: Smooth-rubbed finish as defined in ACI 301

B. Unformed Surfaces and Flatwork:

1. Below Grade Surfaces: Float finish as defined in ACI 301.
2. Water Conveying Surfaces: Float finish as defined in ACI 301.
3. Exposed Surfaces: Broom or belt finish as defined in ACI 301.
 - a. Treading surfaces: Broom/belt surfaces perpendicular to the typical direction of traffic.
 - b. Slope exterior surfaces, or interior surfaces identified on the drawings, for drainage:
 - i. Slope narrow surfaces, such as tops of walls, approximately 3/8-inch per foot of width, unless other slopes or level surface requirements are identified on the drawings.
 - ii. Slope broad surfaces, such as walks, platforms, and decks approximately 1/4-inch per foot, unless other slopes or level surface requirements are identified on the drawings.

C. Finishing flatwork with steel trowels is not allowed.

3.07 TOLERANCES

A. Tolerances: In accordance with ACI 117.

B. Specified tolerances are consistent with modern construction practices and governed by effects that permissible variations may have upon a structure. Tolerances for individual components and activities are not cumulative; the completed construction must be functional and suitable for its intended use, including compatibility with the surrounding grades and structural elements and functionally interface with other project features, such as mechanical or electrical appurtenances.

C. Prevention of Repeated Failure to Meet Tolerances: When concrete placements result in hardened concrete that does not meet specified tolerances, submit to the GOVERNMENT an outline of preventive actions, such as modifications to forms, modified procedures, and/or modified finishing techniques to be implemented to avoid repeated failures.

3.08 CONCRETE PATCHING AND REPAIR

A. Repair concrete surfaces that are not within specified tolerances.

B. Minor surface defects: Repair minor surface defects in accordance with ACI 301.

C. Major surface defects: Surface defects penetrating deeper than (or equal to) the thickness of concrete over the reinforcement are considered major surface defects. For major surface defects, the CONTRACTOR shall propose a repair procedure in accordance with the guidelines presented in ACI 546R. The repair proposal shall be reviewed and approved by the GOVERNMENT.

D. Other defects: Where the finished product is outside the tolerances for alignment, plane, or grade, or cannot be used for its intended purpose, the CONTRACTOR shall propose a repair procedure to resolve the deficiency. The repair proposal shall be reviewed and approved by the GOVERNMENT.

E. Acceptance: The GOVERNMENT shall inspect and provide final acceptance of repairs.

3.09 CURING

A. Cure concrete in accordance with ACI 308R.

B. Water Curing:

1. Keep concrete surface wet for 7 days, minimum, from time concrete has attained sufficient set to prevent detrimental effects to surface.
2. Cure methods:
 - a. Water-saturated material.
 - b. System of perforated pipes, mechanical sprinklers, or porous hose.
 - c. Other methods that will keep surfaces wet.

C. Curing with Curing Compound:

1. Apply in accordance with manufacture's application recommendations.
2. Apply to concrete surface to provide a water-retaining film. As necessary, decrease application rate on rough surfaces to obtain a continuous film. Reapply as necessary to maintain a continuous, water-retaining film on surface for 28 days.

D. Sheet Materials Curing Method:

1. Thoroughly moisten concrete surface by lightly spraying with water as soon as concrete has hardened sufficiently to prevent damage.
2. Completely cover concrete surface with sheet material to provide an airtight, water-retaining film over entire surface.
3. Lap edges of sheets to seal adjacent sheets.
4. Place tightly against concrete surface at extreme edge of curing area.
5. Secure film to withstand wind and prevent circulation of air inside curing film.

E. Form Curing: Concrete forms with documentation demonstrating that the form surface will limit the rate of moisture loss to less than 0.08 lb/ft² in 72 hours may be left in place for the duration of the cure period. Alternatively, forms lined with a sheet material complying with ASTM C 171 may also be left in place for the duration of the curing period.

3.10 PROTECTION

A. Protect concrete from damage until final acceptance by the GOVERNMENT.

1. Do not load, remove forms or shoring, or backfill against concrete until concrete has gained sufficient strength to safely support its weight and imposed loads.
2. Protect fresh concrete against erosion from rain, hail, sleet, or snow; contamination from foreign materials; and damage from foot traffic until the concrete has hardened.
3. Protect concrete from heavy foot traffic and other construction activities by covering with plywood or other suitable material.

B. Protect Concrete when Freezing Temperatures are Imminent:

1. Maintain concrete at a temperature of 50 degrees Fahrenheit or greater for 72 hours, minimum, after placement. Vent heater and prevent concrete from drying where artificial heat is employed.
2. When water curing, protect concrete from freezing during. After discontinuance of water curing, maintain at a temperature of 50 degrees Fahrenheit or greater for next 72 hours.
3. Discontinue protection against cold weather such that the drop in temperature of the concrete will be gradual and will not exceed 5 degrees Fahrenheit per hour and 40 degrees Fahrenheit in 24 hours for thin sections, and 5 degrees Fahrenheit per hour and 20 degrees Fahrenheit in 24 hours for massive sections (smallest dimension greater than 36 inches).

3.11 FIELD QUALITY CONTROL

A. Testing Agency: The CONTRACTOR is responsible for field quality control. The CONTRACTOR will use a qualified independent testing and inspecting agency to sample materials, perform tests, and submit test reports during concrete placement according to the requirements specified herein.

B. Testing Services: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:

1. Minimum Quality Control Testing Frequency:
 - a. Obtain one composite sample from the first load each day for each concrete mix placed.
 - b. After the first load, obtain one composite sample from each day's pour for each 50 yd³ (or fraction thereof). Randomly select loads to be sampled in accordance with ASTM D 3665.
2. Slump: ASTM C 143; one test at point of placement for each composite sample. Perform additional tests when concrete consistency appears to change.
3. Air Content: ASTM C 231, pressure method; one test at point of placement for each composite sample.
4. Concrete Temperature: ASTM C 1064; one test hourly when air temperature is 40 deg F, or below, and when 80 deg F, or above, and one test for each composite sample.
5. Unit Weight: ASTM C 138; one test for each composite sample.
6. Test Specimens: ASTM C 31; cast set of a minimum of 5 cylinder specimens for laboratory curing (4-in. x 8-in. cylinder test specimens) for each composite sample.
 - a. Additional field cure cylinders may be taken at the CONTRACTOR's discretion or as directed by the GOVERNMENT.
7. Compressive-Strength Tests: ASTM C 39
 - a. One test cylinder will be tested at 7 Days and 3 at 28 Days. The remaining cylinder(s) will be held to verify test results, if needed. The 7-Day cylinder will be used as an indication of normal strength gain and will not have associated acceptance criteria.

C. Test results shall be reported in writing to the GOVERNMENT, concrete producer, and CONTRACTOR within 48 hours of testing. Concrete testing reports shall contain placement identification name and number, date of placement, concrete delivery truck number, batch ticket number, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mix design identification name/number, and strength result(s), including type of break.

D. Evaluation of strength test results and acceptance of concrete shall be in accordance with ACI 214 and ACI 301.

1. If any concrete fails to meet these requirements, immediate corrective action shall be taken to increase the compressive strength for subsequent batches of the type of concrete affected.
2. Concrete that fails to meet the ACI requirements and these Specifications is subject to removal and replacement at no additional cost to the GOVERNMENT.

E. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other specified requirements have not been met, or as directed by the GOVERNMENT.

END OF SECTION

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DIVISION 5

METALS

PART 1 - GENERAL

1.01 DESCRIPTION OF WORK:

A. Definition: Metal fabrications include items made from iron and steel shapes, plates, bars, strips, tubes, pipes, and castings that are not part of structural steel or other metal systems specified elsewhere.

B. Types of work in this section include metal fabrications for:

1. Inlet structure steel grating trashrack and channels.
2. Outlet works tower trashrack.
3. Stainless steel vent pipe.
4. Deflector plate and support ring at conduit inlet.
5. Rough hardware.
6. Loose bearing and leveling plates.
7. Loose steel lintels.
8. Miscellaneous framing and supports.

1.02 QUALITY ASSURANCE:

A. Field Measurements: Take field measurements prior to preparation of shop drawings and fabrication, where possible. Do not delay job progress; allow for trimming and fitting where taking field measurements before fabrication might delay Work.

B. Shop Assembly: Preassemble items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for reassembly and coordinated installation.

1.03 SUBMITTALS:

A. Product Data: Submit manufacturer's specifications, anchor details and installation instructions for products used in miscellaneous metal fabrications, including paint products and grout.

B. Shop Drawings: Submit shop drawings for fabrication and erection of miscellaneous metal fabrications. Include plans, elevations and details of sections and connections. Show anchorage and accessory items. Provide templates for anchor and bolt installation by others.

Where materials or fabrications are indicated to comply with certain requirements for design loadings include structural computations, material properties and other information needed for structural analysis.

PART 2 - PRODUCTS

2.01 MATERIALS:

A. Metals:

1. Metals, Surfaces, General: For fabrication of miscellaneous metal work which will be exposed to view, use only materials which are smooth and free of surface blemishes including pitting, seam marks, roller marks, rolled trade names and roughness.
2. Steel Plates, Shapes and Bars: ASTM A 36.
3. Steel Bar Grating: Welded bar type galvanized, safe load at 100 psf 300 pound concentrated load for stair treads, serrated, ASTM A 1011.
4. Steel Tubing: Cold formed, ASTM A 500; or hot rolled, ASTM A 501.
5. Structural Steel Sheet: Hot-rolled, ASTM A 1031; or cold-rolled ASTM A 1008, Class 1; of grade required for design loading.
6. Galvanized Structural Steel Sheet: ASTM A 924 of grade required for design loading.
7. Steel Pipe: ASTM A 53; Type and grade (if applicable) as selected by fabricator and as required for design loading; black finish unless galvanizing is indicated; standard weight (Schedule 40), unless otherwise indicated.

B. Fasteners: Use fasteners made of same basic metal as fastened metal except use galvanized fasteners complying with ASTM A 153 for exterior steel and aluminum units, unless otherwise indicated. Do not use metals which are corrosive or incompatible with metals joined. Provide fasteners of proper type and size required for application indicated.

C. Grout:

1. Metallic Non-Shrink Grout: Pre-mixed, factory-packaged, ferrous aggregate grout complying with CE CRD-C588, Type M.
2. Non-Shrink, Non-Metallic Grout: Pre-mixed, factory-packaged, non-staining, non-corrosive, non-gaseous grout complying with CE CRD-C621. Provide grout specifically recommended by manufacturer for interior and exterior applications of type specified in this section.

D. Paint:

1. Primer selected must be compatible with finish coats of paint. Coordinate selection of metal primer with finish paint requirements.
2. Galvanizing Repair Paint: High zinc dust content paint for regalvanizing welds in galvanized steel, complying with the Military Specifications MIL-P-21035 (Ships).

2.02 FABRICATION, GENERAL:

A. Workmanship: Use materials of size and thickness shown or, if not shown, of required size and thickness to produce strength and durability in finished product. Work to dimensions shown or accepted on shop drawings, using proven details of fabrication and support. Use type of materials shown or specified for various components of Work.

B. Form exposed work true to line and level with accurate angles and surfaces and straight sharp

edges. Ease exposed edges to a radius of approximately 1/32" unless otherwise shown. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing Work.

C. Weld corners and seams continuously, complying with AWS recommendations. At exposed connections, grind exposed welds smooth and flush to match and blend with adjoining surfaces.

D. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of type shown or, if not shown, Phillips flat-head (countersunk) screws or bolts.

E. Provide for anchorage of type shown, coordinated with supporting structure. Fabricate and space anchoring devices to provide adequate support for intended use.

F. Cut, reinforce, drill and tap miscellaneous metal work as indicated to receive finish hardware and similar items.

G. Galvanizing: Provide a zinc coating for those items shown or specified to be galvanized, as follows:

1. ASTM A 153 for galvanizing iron and steel hardware.
2. ASTM A 123 for galvanizing iron and steel products

H. Fabricate joints which will be exposed to weather in a manner to exclude water or provide weep holes where water may accumulate.

I. Shop Painting: Shop paint miscellaneous metal work, except members or portions of members to be embedded in concrete or masonry, surfaces and edges to be field welded, and galvanized surfaces, unless otherwise specified.

1. Remove scale, rust and other deleterious materials before applying shop coat. Clean off heavy rust and loose mill scale in accordance with SSPC SP-2 "Hand Tool Cleaning", or SSPC SP-3 "Power Tool Cleaning", or SSPC SP-7 "Brush-Off Blast Cleaning".
2. Immediately after surface preparation, brush or spray on primer in accordance with manufacturer's instructions, and at a rate to provide uniform dry film thickness of 2.0 mils for each coat. Use painting methods which will result in full coverage of joints, corners, edges and exposed surfaces.

2.03 MISCELLANEOUS METAL FABRICATIONS:

A. Rough Hardware: Furnish bent or otherwise custom fabricated bolts, plates, anchors, hangers, dowels and other miscellaneous steel and iron shapes as required for framing and supporting woodwork, and for anchoring or securing woodwork to concrete or other structures.

1. Manufacture or fabricate items of sizes, shapes and dimensions required. Furnish malleable iron washers for heads and nuts which bear on wood structural connections; elsewhere, furnish steel washers.

B. Loose Steel Lintels: Provide loose structural steel lintels for openings and recesses in masonry walls and partitions as shown. Weld adjoining members together to form a single unit where indicated. Provide not less than 8" bearing at each side of openings, unless otherwise shown.

C. Miscellaneous Framing and Supports: Provide miscellaneous steel framing and supports which

are not a part of structural steel framework, as required to complete Work.

1. Fabricate miscellaneous units to sizes, shapes and profiles shown or, if not shown, of required dimensions to receive adjacent other Work to be retained by framing. Except as otherwise shown, fabricate from structural steel shapes and plates and steel bars, of welded construction using mitered joints for field connection. Cut, drill and tap units to receive hardware and similar items.
2. Equip units with integrally welded anchors for casting into concrete or building into masonry. Furnish inserts if units must be installed after concrete is placed.
3. Galvanize miscellaneous frames and supports where indicated.

D. Brackets, Flanges, Fittings and Anchors: Provide wall brackets, end closures, flanges, miscellaneous fittings and anchors for interconnection of pipe and attachment of railings and handrails to other Work. Furnish inserts and other anchorage devices for connecting railings and handrails to concrete or masonry Work.

1. For railing posts set in concrete provide sleeves of galvanized steel pipe not less than 6" long and with an inside diameter not less than 1/2" greater than the outside diameter of pipe. Provide steel plate closure welded to bottom of sleeve and of width and length not less than 1" greater than outside diameter of sleeve.

PART 3 - EXECUTION

3.01 PREPARATION:

A. Coordinate and furnish anchorages, setting drawings, diagrams, templates, instructions, and directions for installation of anchorages, such as concrete inserts, sleeves, anchor bolts and miscellaneous items having integral anchors, which are to be embedded in concrete or masonry construction. Coordinate delivery of such items to project site.

1.01 INSTALLATION:

B. General:

1. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing miscellaneous metal fabrications to in-place construction; including, threaded fasteners for concrete and masonry inserts, toggle bolts, through-bolts, lag bolts, wood screws and other connectors as required.
2. Cutting, Fitting and Placement: Perform cutting, drilling and fitting required for installation of miscellaneous metal fabrications. Set Work accurately in location, alignment and elevation, plumb, level, true and free of rack, measured from established lines and levels. Provide temporary bracing or anchors in formwork for items which are to be built into concrete, masonry or similar construction.
 - a. Fit exposed connections accurately together to form tight hairline joints. Weld connections which are not to be left as exposed joints, but cannot be shop welded because of shipping size limitations. Grind exposed joints smooth and touch-up shop paint coat. Do not weld, cut or abrade the surfaces of exterior units which have been hot-dip galvanized after fabrication, and are intended for bolted or

screwed field connections.

3. Field Welding: Comply with AWS Code for procedures of manual shielded metal-arc welding, appearance and quality of welds made, and methods used in correcting welding Work.

C. Setting Loose Plates: Clean concrete and masonry bearing surfaces of any bond-reducing materials, and roughen to improve bond to surfaces. Clean bottom surface of bearing plates.

1. Set loose leveling and bearing plates on wedges, or other adjustable devices. After the bearing members have been positioned and plumbed, tighten the anchor bolts. Do not remove wedges or shims, but if protruding, cut-off flush with the edge of the bearing plate before packing with grout. Use metallic non-shrink grout in concealed locations where not exposed to moisture; use non-metallic non-shrink grout in exposed locations, unless otherwise indicated.
2. Pack grout solidly between bearing surfaces and plates to ensure no voids remain.
3. Weld non-removable units to supporting members or framework.

3.02 ADJUST AND CLEAN:

A. Touch-Up Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with same material as used for shop painting. Apply by brush or spray to provide a minimum dry film thickness of 2.0 mils.

B. For galvanized surfaces: Clean field welds, bolted connections and abraded areas and apply two coats of galvanizing repair paint.

END OF SECTION 05500

APPENDIX A

**GEOTECHNICAL
DATA REPORT**

Bar Island Dam Improvements Geotechnical Data Report

Prepared For:



Bureau of Land Management

Montana State Office

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and

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December 2025

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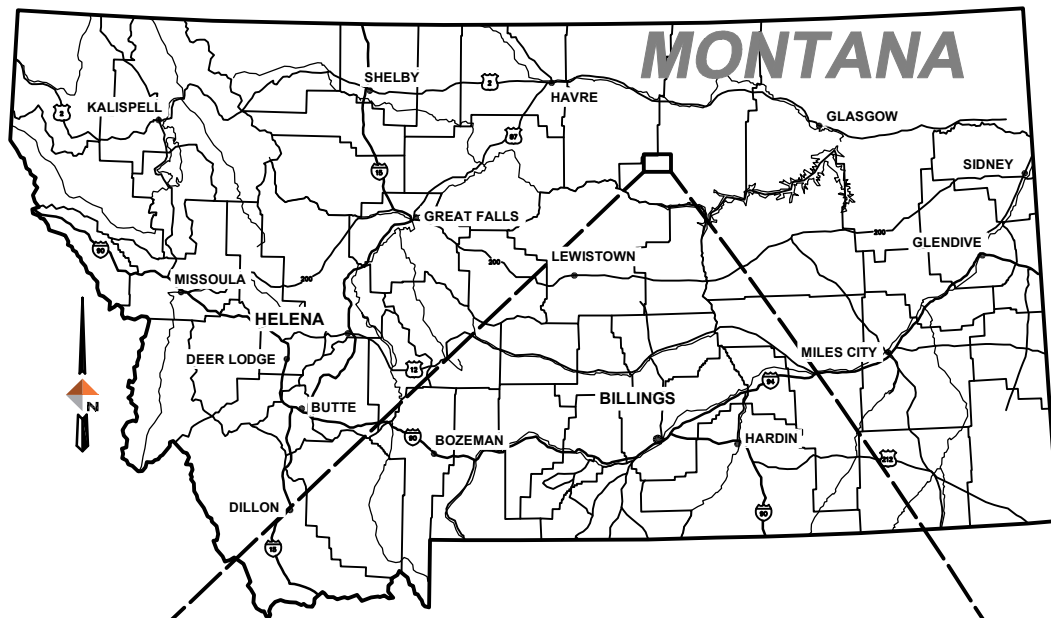
- Appendix A: Exploration Logs
- Appendix B: Photograph Log
- Appendix C: Laboratory Test Results

1.0 INTRODUCTION

DOWL completed a geotechnical investigation for Bar Island Dam in Phillips County, Montana. The scope of geotechnical services consisted of reviewing existing geotechnical and geological information, field observations, subsurface exploration, and preparing this Geotechnical Data Report. The purpose of this investigation was to evaluate the existing dam's geotechnical material properties and potential borrow areas to support the dam rehabilitation as outlined in the scope of work (SOW).

2.0 PROJECT DESCRIPTION

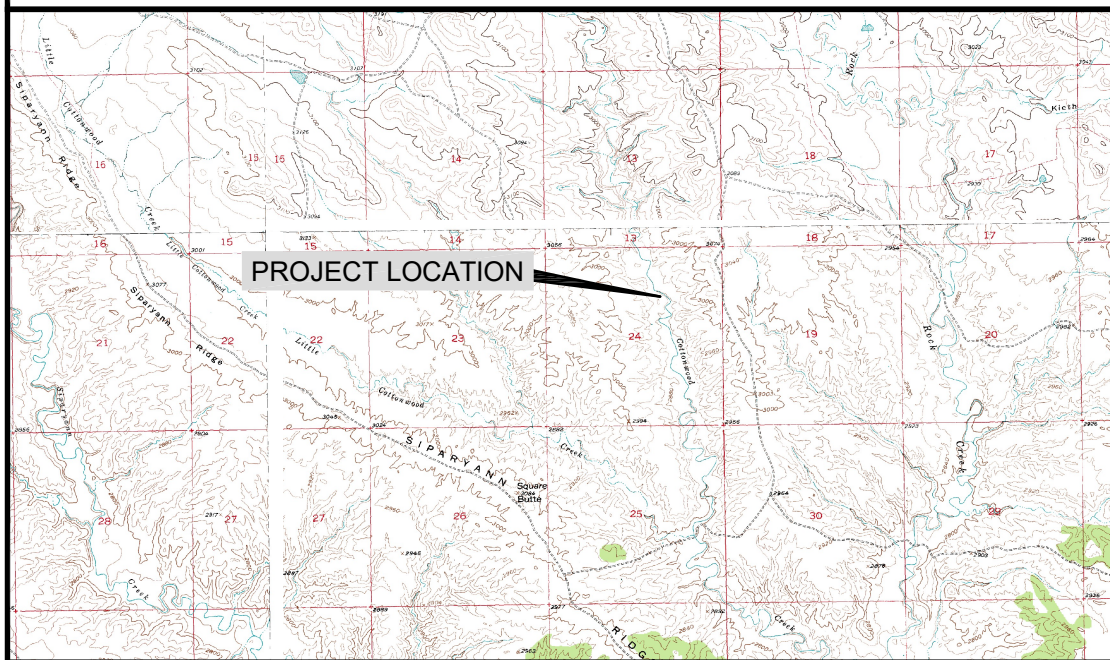
Bar Island Dam is an earthen embankment dam located approximately 57 miles southwest of Malta, Montana, in Phillips County. The dam is located on Cottonwood Creek, a tributary of Rock Creek, which is a minor tributary of the Missouri River. The dam is surrounded by rangeland vegetated with sagebrush and prairie grasses. Wetland-type plants and vegetation are present along the banks of the reservoir and outlet works. The Bar Island Dam Road crosses the dam embankment.



VICINITY MAP

NOT TO SCALE

BAR ISLAND DAM - ZORTMAN, MONTANA



LOCATION MAP



BAR ISLAND DAM RECONSTRUCTION GEOTECHNICAL INVESTIGATION VICINITY AND LOCATION MAP

PROJECT 4691.12652.01
DATE 01/14/2025

FIGURE 1

The Dam is classified as a low-hazard class dam. The compacted earth embankment dam was constructed in 1967 as a peak discharge control structure and an erosion and sediment control structure. The existing principal outlet structure consists of a 30-inch diameter welded metal pipe (WMP) drop inlet approximately 220 ft from the right abutment. The existing auxiliary spillway is comprised of an earthen trapezoidal channel at the right abutment. There is a 10-foot deep headcut within the spillway that has progressed upstream to within about 15 feet of the dam crest.

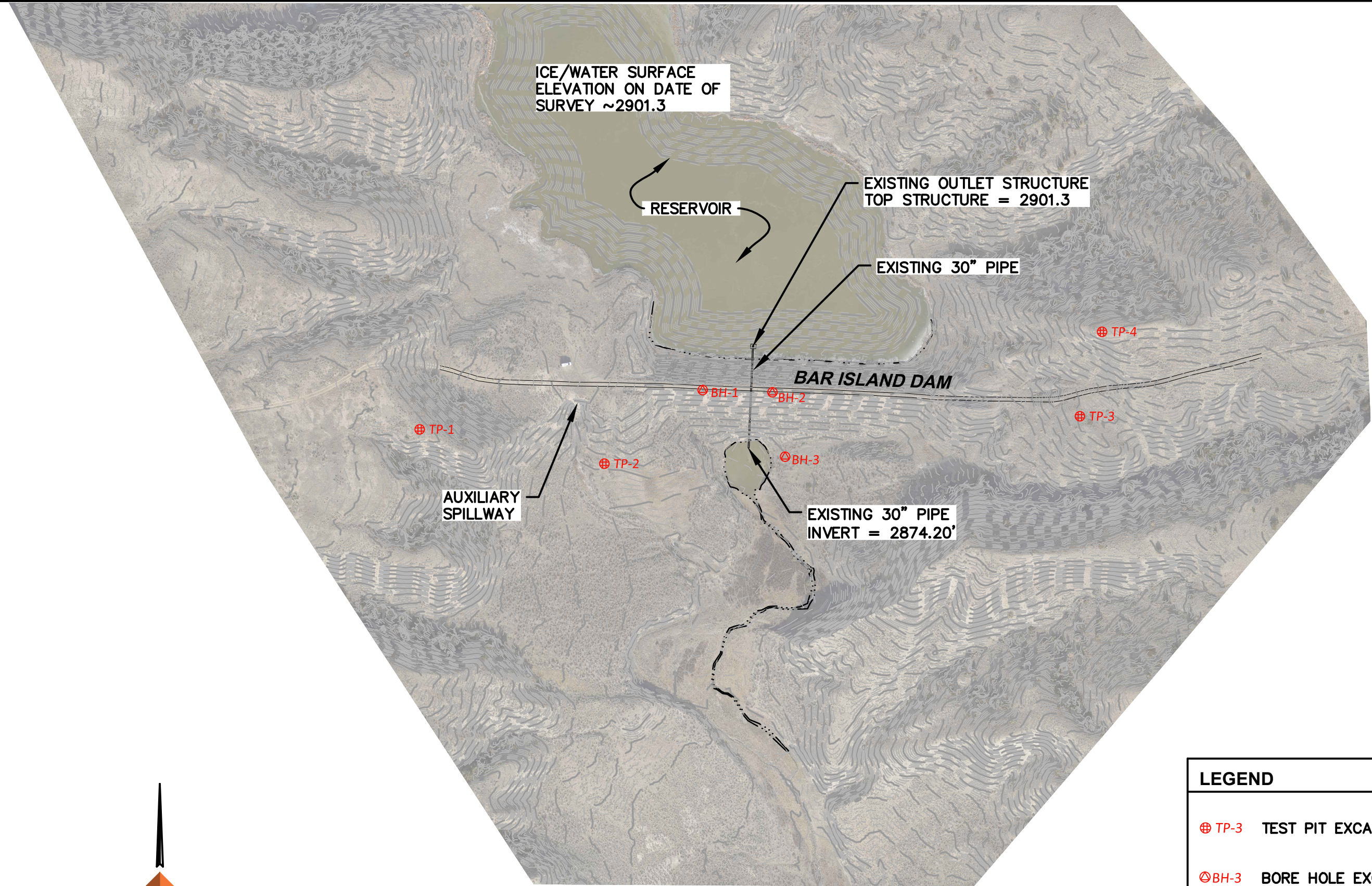
The dam also serves as a habitat for native fish, sage grouse, and many other wildlife species, and is a general recreation use area. Furthermore, the dam consistently has water available for the control of area wildland fires. Physical characteristics associated with the Bar Island Dam are summarized in Table 1.

Table 1: Bar Island Dam Existing Characteristics

Characteristic	Value
Structural Dam Height	46 ft
Minimum Dam Crest Elevation	41 ft
Dam Crest Length	620 ft
Minimum Dam Crest Width	10 ft
Upstream Slope	3.5H:1V
Downstream Slope	2H:1V – 3H:1V
Reservoir Capacity at Dam Crest	744 acre-feet
Reservoir Capacity at Maximum Controllable Water	553 acre-feet
Reservoir Storage Capacity	146 acre-feet

-Values were obtained from Original Design Drawings unless noted otherwise

J:\91\12652-01\65Gad\Civil\SA-CS-EX-FIG-FIG-12656-01.dwg PLOT DATE 2025-1-14 13:26 SAVED DATE 2025-01-14 13:25 USER: jsvelson



LEGEND	
⊕ TP-3	TEST PIT EXCAVATED NOVEMBER 2024
⊕ BH-3	BORE HOLE EXCAVATED NOVEMBER 2024



BAR ISLAND DAM MODIFICATIONS
EXISTING CONDITIONS
BORE HOLE AND TEST PIT LOCATIONS

PROJECT	4961.12652.01
DATE	01/14/2025
FIGURE 2	

The dam embankment has a crest elevation of approximately 2,920.0 feet and a height of approximately 33 feet from the bottom of the outlet works. There is no riprap on the upstream or downstream slopes of the embankment. Figure 2 shows the features of the dam embankment. A large scour hole is located at the outlet conduit's downstream end. An earth-lined auxiliary spillway is about 250 feet south of the right abutment. The auxiliary spillway has started to erode/headcut into the shale bedrock.

- The reconstruction design will consider the dam's documented deficiencies, which include failing principal spillway outlet works, with the riser undermined and the barrel perforated, evidence of piping, and internal erosion.
- The outlet works has a scour hole, moderate undercutting, and minor embankment erosion and sloughing.
- The spillway has severe head cutting and erosion in the discharge channel.
- Minor wave action erosion on the upstream slope of the dam embankment.
- Minor head cutting at the left abutment of the dam (upstream and downstream groins).
- Moderate seepage below downstream dam embankment along left and right groins.

The proposed project will involve removing and replacing the existing outlet conduit and outlet works and improving the auxiliary spillway.

3.0 INVESTIGATION

3.1 FIELD INVESTIGATION

In November 2024, DOWL performed fieldwork, which consisted of site observations, drilling three borings, and excavating four test pits.

Boland Drilling drilled the borings to depths ranging from 25.0 to 65.0 feet below the existing ground surface under the direction of a DOWL geologist. Jackola surveyed the boring locations, and the coordinates are reported in Montana State Plane Coordinates. Elevations are reported with respect to NAVD88.

2 Diggers, Inc. dug the test pits using a CAT 315 track hoe at the direction of a DOWL geologist to depths ranging from 4 to 10 feet. A summary of exploration locations and depths is presented in Table 2. As the excavation progressed for each test pit, soil samples were taken for geotechnical laboratory testing. Following the investigation, test pits were backfilled with excavated soil. Soil samples were transported to the DOWL laboratory, where engineering property tests were conducted.

Jackola surveyed and staked the boring prior to the fieldwork beginning. The survey is reported in the Montana State Plane Coordinate system and the NAVD88 vertical datum. The test pits were surveyed at the time of the fieldwork by a handheld GPS. Exploration locations are shown in Figure 2 and displayed in Table 2.

Table 2: Exploration Summary

Exploration	Depth (feet)	Surface Elevation (feet)	Northing (feet)	Easting (feet)	Location
B-1	65.0	2920.0	1,275,070.7	2,195,306.6	Right middle of the Crest
B-2	26.5	2920.1	1,275,066.2	2,195,446.1	Left middle of the Crest
B-3	25.0	2879.0	1,274,937.4	2,195,471.6	Toe in the middle of the Dam
TP-1	12.0	2959.3	1,274,991.9	2,194,740.2	Right, up the hill from the dam
TP-2	11.0	2887.1	1,274,923.3	2,195,109.9	Spillway
TP-3	12.0	2962.6	1,275,018.3	2,196,061.3	Left Up the hill from the dam
TP-4	12.0	2995.4	1,275,187.3	2,196,105.7	Left Up the hill from the dam

The exploration logs in Appendix A include soil descriptions and observed groundwater conditions. Stratification boundaries on the test pit logs represent the approximate location of changes in soil types; in situ, the transition between materials may be gradual and vary. Photographs of the test pits in the field investigation are included in Appendix B.

Boland Drilling drilled the borings under the direction and observation of a DOWL geotechnical engineer. Boland Drilling drilled the borings B-1 through B-3 with a track-mounted CME 45 drill rig and 3.5-inch Hollow-Stem Augers. DOWL conducted the field exploration activities referencing the following ASTM standards:

- ASTM D6151 Standard Practice for Using Hollow-Stem Augers for Geotechnical Exploration and Soil Sampling,
- ASTM D1586 Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils,
- ASTM D1587 Standard Practice for Thin-Walled Tube Geotechnical Sampling of Soils,
- ASTM D2113 Standard Practice for Rock Core Drilling and Sampling for Rock.

The NQ rock core system consists of a triple-tube core barrel with a diamond coring bit that is raised and lowered by drilling rods. Rotary drilling advances the coring. Water is circulated through the drill rod during coring to remove rock cuttings from the boring. The NQ core system recovers a 1.875-inch diameter core while being advanced into the bedrock using a 3.0-diameter diamond core bit. As the drillers recovered core samples, DOWL recorded the run length, recovery, and Rock Quality Designation (RQD) along with the rock type and associated geologic formation, the rock core's field strength, weathering, fracture index, and structural parameters referencing DOWL's Rock Classification Description presented in Appendix A.

The exploration logs in Appendix A include soil and groundwater conditions, as well as field SPT blow count data. Stratification boundaries on the boring logs represent the approximate location of changes in soil types; in situ, the transition between materials may be gradual and may vary. Photographs of the drive samples acquired during drilling are included in Appendix B.

The soil descriptions shown on the boring logs for field and laboratory testing are based on ASTM Standards D2487 and D2488. The soil and groundwater conditions depicted are only for the specific dates and locations reported and may not necessarily be representative of other locations or times. All borings were backfilled with a bentonite chip hole plug.

3.2 LABORATORY TESTING

Representative samples were selected for laboratory testing. DOWL performed tests for index and engineering soil properties in Billings, Montana. Laboratory tests were performed in general accordance with the most recent ASTM or other standards listed in Table 3.

Table 3: Laboratory Tests

Test	Purpose
Natural Moisture Content ASTM D 2216	Provides a measure of natural (in-situ) water content.
Unit Weight ASTM D7263	Provides unit weight (density) of soil.
Atterberg Limits ASTM D 4318	Provides an indicator of the consistency and swell potential of fine-grained soils.
Particle-Size Distribution ASTM D 421	Provides a measure of grain sizes of the soils for classification and identification of physical characteristics.
Moisture-Density Relationship ASTM D698	Provides a measure of the relationship of water content to soil density during compaction.
Consolidated Undrained Triaxial Compression Test ASTM D4767	Provides the strength and stress-strain relationships of a cylindrical specimen of either an intact, reconstituted, or remolded saturated cohesive soil
Hydraulic Conductivity ASTM D5084	Provides a measure of the hydraulic conductivity (permeability) of soil.
Unconfined Compression ASTM D2166	Provides an estimate of the undrained shear strength of the soil.
Double Hydrometer ASTM D4221	Provides an estimate of the dispersion potential of soil.
Pin Hole Dispersion ASTM D4647	Provides an estimate of the erodibility of soil.
Crumb Dispersion ASTM D6572	Provides an estimate of the dispersion potential of soil.
Na, Mg, K, Ca cations SW6010B	Used to calculate Sodium Adsorption Ratio (SAR) and Total Dissolved Salts (TDS), which correlate to a soil's dispersive characteristics.
pH, Resistivity, and Soluble Sulfates ASTM G 51-23/G 187/D 8459-23	Determines the potential for corrosive interaction of soils with concrete and metal.

3.2.1 Moisture-Density Relationship Results

Two standard Proctor (moisture-density relationship) tests were performed to determine the optimum moisture and unit weight for compaction tests. One sample was from a potential borrow area, and the other was taken from the embankment. Table 4 summarizes those tests and their results. Lab results are in Appendix C.

Table 4: Proctor Results

Test Pit	Depth (feet)	Optimum Moisture (%)	Unit Weight (pcf)	Material
TP-1	5.0 to 10.0	11.5	122	Clayey Gravel
TP-3	5.0 to 10.0	25.4	95	Fat Clay with Sand

3.2.2 Triaxial Test

Two consolidated undrained triaxial shear tests were performed on a Shelby tube soil sample collected from the Boring B-2 in the embankment material and remolded soil from TP-3 of the foundation clay. The results are summarized in Table 5. Effective shear strengths represent shear strengths where pore pressures dissipate, while total strength parameters represent conditions where pore pressure does not dissipate.

Table 5: Triaxial Test Results

Sample Location	Description	Friction Angle (degrees)		Cohesion (psf)	
		Effective	Total	Effective	Total
B-2 – 20.0 to 22.0 feet	Fat Clay (Embankment)	28.6	13.8	0	234
TP-3 – 5.0 to 10.0 feet	Fat Clay (Foundation)	24.4	15.6	57	0

3.2.3 Hydraulic Conductivity

Two hydraulic conductivity tests were performed on soil samples collected from the Boring B-2 in the dam embankment and TP-3 in the foundation soil. The results are summarized in Table 6. Hydraulic Conductivity Test results will be completed in the 65% report.

Table 6: Hydraulic Conductivity Test Results

Sample Location	Description	Hydraulic Conductivity
B-2 – 20.0 to 22.0 feet	Fat Clay (Embankment)	8.4×10^{-10}
TP-3 – 5.0 to 10.0 feet	Fat Clay (Foundation)	3.0×10^{-09}

3.2.4 Unconfined Compression

Eight unconfined compression tests were performed on a rock core collected from the Boring B-1 and Boring B-3 in the foundation below the embankment. The results are summarized in Table 7.

Table 7: Unconfined Compression Test Results

Sample Location	Description	Unconfined Compressive Strength (psi)	Unconfined Compressive Strength (psf)
B-1 – 50.8 to 52.5 feet	Shale Foundation	80	11,506
B-1 – 52.2 to 52.8 feet	Shale Foundation	121	17,368
B-1 – 56.6 to 57.0 feet	Shale Foundation	244	35,086
B-1 – 57.1 to 57.6 feet	Shale Foundation	239	34,387
B-1 – 63.6 to 64.4 feet	Shale Foundation	79	11,322
B-3 – 10.4 to 11.3 feet	Shale Foundation	67	9,602
B-3 – 16.5 to 17.2 feet	Shale Foundation	77	11,089

3.2.5 Soil Chemistry and Corrosion

Based on the results shown in Table 8, concrete in contact with the corridor soils has an exposure class of S3, according to the American Concrete Institute (ACI) 318 Table 19.3.1.1 (ACI, 2014). To achieve the required protection against sulfate-related corrosion, we recommend specifying Type V plus pozzolan or slag cement. Concrete needs a maximum water-to-cement ratio of 0.45 (by weight, normal weight concrete) and a minimum compressive strength, f'_c , of 4,500 pounds per square inch (psi). Details can be found in the above ACI reference and the Portland Cement Association Publication *Design and Control of Concrete Mixtures*.

According to Corrosion/Degradation of Soil Reinforcement for Mechanically Stabilized Earth Walls (FWHA, 2009), the soil at the site is very corrosive to steel. Based on that publication and the tests above, carbon steel has a corrosion rate of over 1.2 ounces per square foot per year, and galvanized steel has a corrosion rate of 0.4 ounces per square foot per year. Test Results are presented in Appendix C.

Table 8: Soil Chemistry Test Results

Location	Sample Depth (ft)	Soluble Sulfate (ppm)	Resistivity (ohm-cm)	pH
B-1	15.0 to 20.0	16,100	217	7.5
TP-2	5.0 to 10.0	25,600	168	6.5

3.2.6 Soil Dispersion

Due to the potential for dispersive soils in the project area, tests were performed on the existing embankment materials (B-1) and potential borrow soil (TP-2) to identify whether dispersive clay soils are present at the site. These tests included the Double Hydrometer (ASTM D4221), Pinhole Test (ASTM D4647), Crumb Test (ASTM D6572), and cation chemical tests. The test results are discussed below, with test reports and calculations presented in Appendix C.

The dispersive physical properties tests indicate that the embankment sample tested as non-dispersive and the borrow samples tested as intermediate. The results of the dispersive physical tests are summarized below.

Table 9: Mechanical Dispersion Test Results

Location	Sample Depth (ft)	Pinhole Dispersion ¹	Double Hydrometer ² (%)	Crumb Dispersion
B-1	15.0 to 20.0	ND1	7.13	Non-Dispersive
TP-2	5.0 to 10.0	D1	6.29	Intermediate

¹ D1= Dispersive clay that failed rapidly, ND1 = Nondispersive clay with very slight to no colloidal erosion under 380mm (15 inches) head.

² NRCS guidelines for the double hydrometer indicate soils with less than 30% dispersion are non-dispersive, 30-60% are intermediate, and greater than 60% are dispersive.

Chemical tests were performed on the soil samples to coincide with the dispersive physical property tests (double hydrometer, crumb, and pinhole). The cation chemical tests are used to determine the individual amounts of sodium (Na), magnesium (Mg), potassium (K), calcium (Ca), total dissolved salts (TDS), and the sodium adsorption ratio (SAR). Based on research by the NRCS and others, calcium or magnesium cations dominate in non-dispersive clay, and sodium cations are dominant in dispersive clay. Additionally, one of the most common units to express the presence of sodium is the sodium adsorption ratio ($SAR = Na / (\frac{1}{2}(Ca+Mg))^{\frac{1}{2}}$). The following table summarizes SAR results and NRCS guidelines for SAR relative to a soil's dispersive characteristics.

Table 10: Sodium Adsorption Ratio (SAR) Results

SAR	NRCS Dispersion Characterization ¹	B-1	TP-2
<40	Non-dispersive	13.8	20.1
40-60	Intermediate or Transitional		
>60	Dispersive		

¹ Characterization guidelines based on Table 13-3 in NRCS NEH 633, Ch. 13, June 2013 Draft.

Another way of relating the cation concentrations to dispersion potential is by comparing the percent sodium to the total salt content ($TDS = Na+Mg+K+Ca$). Total Dissolved Salts (TDS) and percent sodium (%Na) results for the tested samples are summarized in Table 11. The TDS and percent sodium data are then plotted on the X-axis and Y-axis, respectively, of the chart in Figure 13-19 of NRCS NEH Part 633, Ch. 13. The TDS-%Na results for the embankment soils plot in the dispersive range and the proposed borrow soils plot in the dispersive range (see Figure 4).

Table 11: TDS – Percent Sodium Results

Location	TDS (meq/L)	% Na
B-1	114.8	58
TP-2	159.1	65

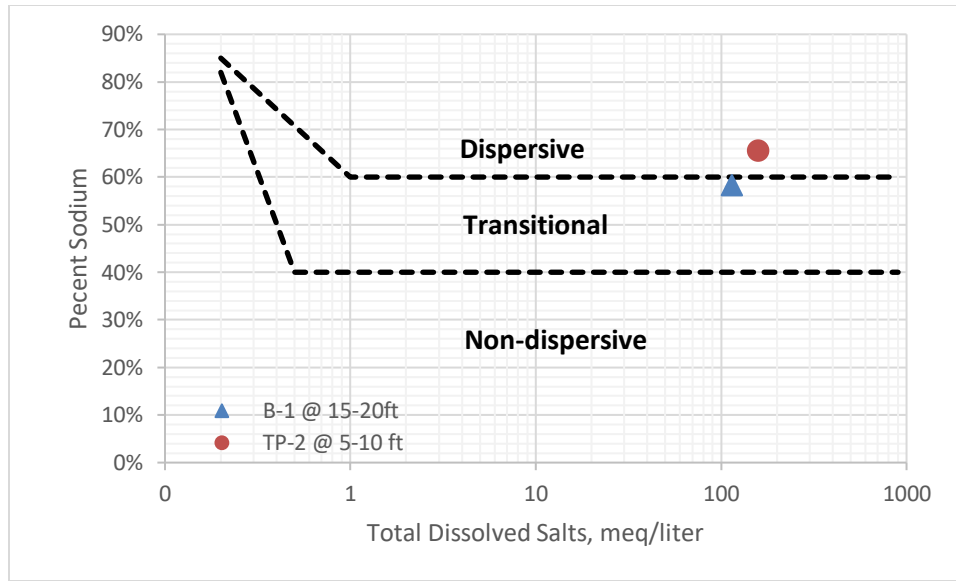


Figure 3: Dispersion Potential

Based on NRCS recommendations, SAR results indicate that the samples obtained from the existing embankment and proposed borrow locations are *nondispersive* to intermediate dispersive. In contrast, the TDS-%Na results indicate that the embankment and borrow sources are classified as transitional to dispersive. Although the TDS-%Na chart method, which is the NRCS's preferred method for conclusions regarding dispersive soil characteristics, shows the embankment and borrow samples are *transitional (intermediate)* to *dispersive*, the physical testing shows the existing embankment is *non-dispersive*, and the borrow soils are *intermediate*. The SAR results also indicate that the tested samples for the existing embankment and borrow sources were *non-dispersive*. The following table summarizes dispersive testing:

Table 12: Dispersive Test Summary

Sample Locations	Physical Test			Chemical Test	
	Pinhole	Double Hydrometer	Crumb Dispersion	SAR	%Na
B-1	ND1	7.13	Non-Dispersive	13.8	58.2
TP-2	D1	6.29	Intermediate	20.1	65.4

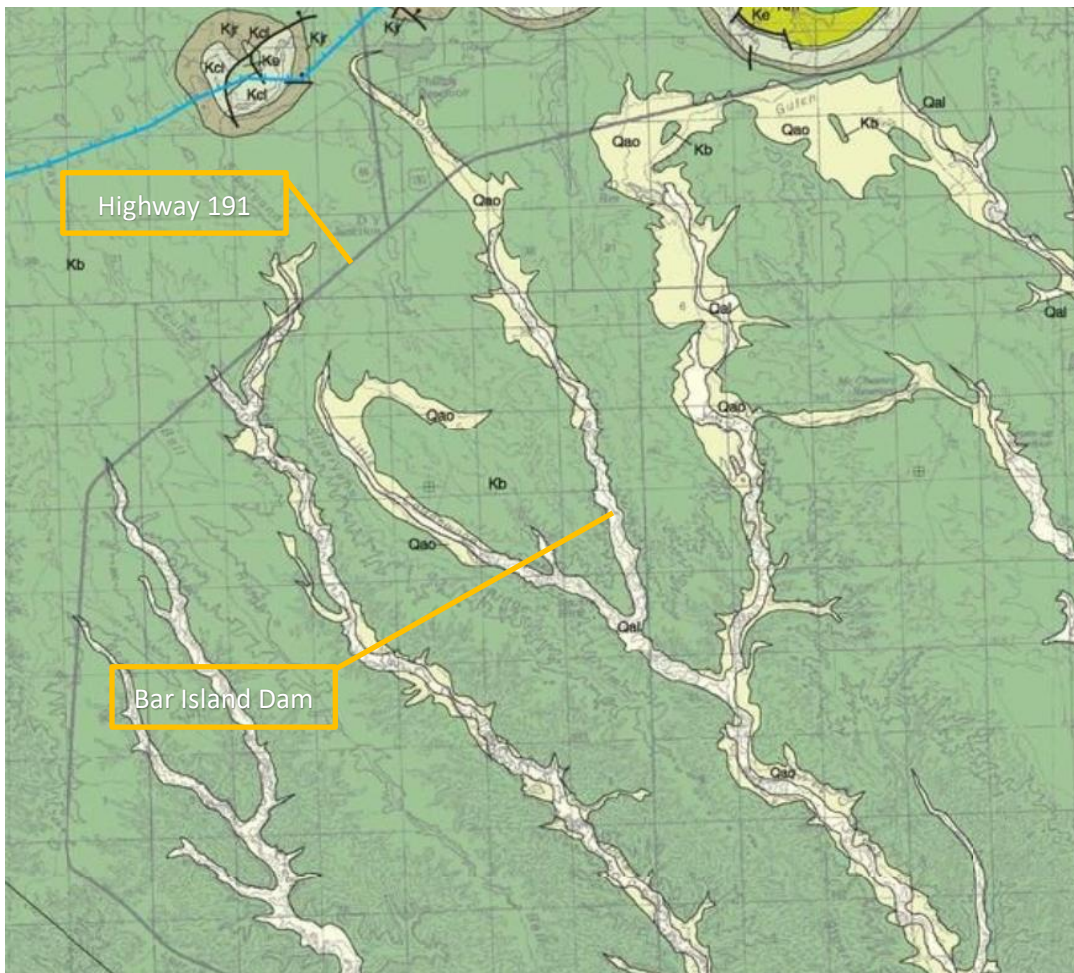
Signs of dispersion were observed during the field investigations. The large headcut in the spillway, erosion at the left abutment of the dam, and erosion at the downstream end of the outlet works are further evidence that dispersive soil may be present. In addition, evidence of salt was observed in the test pits. With these signs, testing results/discrepancies, and as a precaution, the existing embankment soils and borrow soils should be treated as potentially dispersive, so mitigation and treatment of the soils are recommended as part of the dam reconstruction. Additional considerations and recommendations regarding the dispersive soils are included in Section 5.0.

4.0 SUBSURFACE CONDITIONS

4.1 SITE GEOLOGY

The site is located on residual clayey soils overlaying the Late Cretaceous Bearpaw Shale, a dark clay shale with calcareous concretions. (Hatcher & Stanton, 1903) Marine fossils are routinely found within the

Bearpaw Shale. (Wilde & Porter, 1993) It typically overlies the Judith Riverbeds. The Bearpaw is generally a high-plasticity clay and has occasional bentonite beds. A map of surficial geology in the area of Bar Island Dam is presented in Figure 5.



Qal – Alluvium Deposits, Qac – Colluvium Deposits, Kb – Bearpaw Formation, Kjr – Judith River Formation, Kcl – Clagat Shale, Ke – Eagle Formation

Figure 4: Surficial Geologic Map

4.2 OBSERVED SOIL CONDITIONS

The generalized soil profile observed at Bar Island Dam consists of a dam embankment overlaying residual clay soil that overlies the Late Cretaceous Bearpaw Shale.

4.2.1 Dam Embankment

The embankment fill generally consists of brown fat clay with sand and fat clay with sand and gravel. The clay appears to be derived from the native shale and contains shale fragments. The moisture content ranged from 13 to 22 percent, with an average of 18 percent. The optimum moisture content (ASTM D698) was 25 percent. The clay is moderately to highly plastic, with plasticity indices ranging from 46 to 50. The

soil samples from the embankment had 55 to 82 percent passing a No. 200 sieve, and based on hydrometer tests, the clay content ranged from 37 to 56 percent.

4.2.2 Residual Clay Soil

The residual clay soil consisted of fat clay weathered from the Bearpaw Shale. Most of the clay was relatively soft, moist, and had salt stains. The clay contained shale fragments and became harder with depth. The moisture content ranged from 17 to 25, with an average of 21 percent. The optimum moisture content (ASTM D698) was 25 percent. The clay is highly plastic, with plasticity indices ranging from 35 to 46. The soil samples from the borrow area (TP-2 to TP-4) had 74 to 82 percent passing a No. 200 sieve, and based on hydrometer tests, the clay content ranged from 51 to 60 percent.

4.2.3 Clayey Gravel

The clayey gravel was relatively dense, moist, and had subrounded clasts. The moisture content ranged from 6 to 15, with an average of 11 percent. The optimum moisture content (ASTM D698) was 12 percent. The clay in the gravel is plastic, with plasticity indices ranging from 28 to 31. The soil samples from the borrow area (TP-1) had 19 percent passing a No. 200 sieve.

4.2.4 Shale Bedrock

Weathered shale of the Bearpaw Formation was observed in all borings and test pits except TP-2 and B-2. The shale was dark brown to black, slightly weathered with severe to moderate fracturing. Excavation of the shale resulted in 4- to 10-inch shale pieces. The unconfined shear strength of the core samples taken from the shale ranged from 4.8 ksf to 17.5 ksf.

4.3 GROUNDWATER/PHREATIC SURFACE

We did not encounter groundwater in any of the borings or test pits during excavation; however, groundwater fluctuations can occur due to seasonal moisture conditions, changes in land use, and many other factors. Groundwater conditions may vary from those encountered during the field investigation, depending on the influence of these factors. The elevation of the reservoir during drilling was 2901.0 feet, the same height as the outlet works, and the elevation of the pond at the toe of the dam was approximately 2876.5 feet.

5.0 DESIGN CONSIDERATIONS

This project requires multiple design considerations. Below, we summarize specific items designers and the BLM should consider during design and construction.

- The geotechnical deficiency observed during our field investigations was the headcut in the spillway. There was also seepage along the left downstream groin; however, evidence of internal or surface erosion was not observed.
- The existing embankment appears to be composed of reworked fat clay and is suitable for reuse. However, the embankment material below the phreatic surface may have a high moisture content and may be difficult to compact without drying or mixing with drier soil.

- Consider benching into the conduit replacement excavation three to four feet (horizontally) every two vertical feet to provide a stepped contact rather than a straight contact between the existing embankment and the new fill. This will increase the length of the contact, reduce the potential for differential settlement, and increase the contact between existing dry soil and the moisture-conditioned fill.
- Based on the test pit explorations, the best borrow material is located west of the right abutment, along the drainage cut near the road, and east of the left abutment, along the drainage cut near the road.
- Materials encountered in borrow areas exhibit similar physical characteristics to the existing embankment and are suitable as new embankment fill. However, using site soils will require some level of on-site processing, such as screening and moisture conditioning.
- Over-excavation and reconditioning of subgrade soils will be necessary to facilitate construction and long-term performance. Subgrade improvement is recommended beneath the new outlet works and stilling basin to provide uniform support to structures. We recommend a minimum of two feet of subgrade improvement beneath all structures. To reduce potential seepage paths, improvement should consist of excavating site soils, moisture conditioning them, and replacing them with density control.
- The contractor should anticipate the need to dewater the stilling basin to allow for subgrade improvement. Dewatering should be completed so that groundwater is a minimum of two feet below the bottom of the excavation.
- Site soils are highly corrosive to metals and concrete. Buried metals will require corrosion protection, and the proper cement type and concrete mix design must be used.
- Due to the potential for dispersive clay soil in the existing and reconstructed embankment, the reconstruction design should consider several means of mitigating this type of soil. We recommend using a filter diaphragm around the conduit, a blanket drain below the conduit, and lime treatment of the soil around the conduit.
- Lime treatment may be considered in a limited zone around the outlet conduit to reduce the dispersive characteristics of the site clay soils; however, high sulfate contents may pose challenges as discussed below. If used, the lime-amended zone should extend from the bottom of the conduit excavation to the conduit from the upstream slope to the sand chimney across the full width of the excavation. Lime-amended soil should also be placed a minimum of three feet above the conduit encasement from the inlet structure to the sand chimney across the full width of the excavation. The curing period of the lime and soil mixture is important to allow the lime/soil reaction and associated swelling to occur in the stockpile before it is placed in the embankment.
- We recommend a minimum target of 3% lime by weight for all amended soils. Additional testing is recommended prior to and during construction, before and after lime is added to the soil, to provide construction control of the lime treatment process and to verify the required lime percentage. Testing should include moisture-density relationships (proctor) and physical dispersion testing, including pinhole dispersion, crumb dispersion, and double hydrometer.

- Regarding lime amendments, high sulfates can result in very large expansion potential and accompanying swell pressures. It is typical to allow soil expansion to occur prior to placement and compaction. To allow adequate mellowing (curing) time, water in excess of the optimum moisture content is required. Laboratory testing shows that required water may greatly exceed optimum moisture contents. We recommend a minimum moisture content of 5% above optimum moisture; however, this should be confirmed during construction. Typical values are 3 to 5% above optimum moisture, but should be confirmed with laboratory testing. If there is inadequate water, expansion may occur after placement in the embankment. Mellowing time should also be examined during the recommended laboratory testing. We recommend a minimum mellowing time of 7 days unless laboratory testing dictates otherwise. Double application techniques may also be effective in treating high-sulfate soils.

6.0 REFERENCES

- ACI. (2014). *Building Code Requirements for Structural Concrete*. American Concrete Institute.
- FWHA. (2009). *Corrosion/Degradation of Soil Reinforcement for Mechanically Stabilized Earth Walls*. Publication No. FHWA-NHI-09-087. Federal Highways Administration.
- Hatcher, J., & Stanton, T. W. (1903). The stratigraphic position of the Judith River beds and their correlation with the Belly River beds. *Science, new series*, v. 18, no. 150, 211-212.
- Wilde, E. M., & Porter, K. W. (1993). *Geologic Map of the Winnett 30' x 60' Quadrangle, Central Montana*. Retrieved from the Montana Bureau of Mines and Geology.

Appendix A: Exploration Logs



Boring Log Descriptive Terminology

Key to Soil Symbols and Terms

SOIL CLASSIFICATION CHART

MAJOR DIVISIONS			SYMBOLS		TYPICAL		
			GRAPH	LETTER	DESCRIPTIONS		
COARSE GRAINED SOILS MORE THAN 50% OF MATERIAL IS LARGER THAN NO. 200 SIEVE SIZE	GRAVEL AND GRAVELLY SOILS MORE THAN 50% OF COARSE FRACTION RETAINED ON NO. 4 SIEVE	CLEAN GRAVELS (LITTLE OR NO FINES)		GW	Well-graded gravels, gravel sand mixtures, little or no fines.		
				GP	Poorly graded gravels, gravel-sand mixtures, little or no fines.		
		GRAVELS WITH FINES (APPRECIABLE AMOUNT OF FINES)		GM	Silty gravels, gravel-sand-silt mixtures.		
				GC	Clayey gravels, gravel-sand-clay mixtures.		
	SAND AND SANDY SOILS MORE THAN 50% OF COARSE FRACTION PASSING ON NO. 4 SIEVE	CLEAN SANDS (LITTLE OR NO FINES)		SW	Well-graded sands, gravelly sands, little or no fines.		
				SP	Poorly graded sands, gravelly sands, little or no fines.		
		SANDS WITH FINES (APPRECIABLE AMOUNT OF FINES)		SM	Silty sands, sand-silt mixtures.		
				SC	Clayey sands, sand-clay mixtures.		
FINE GRAINED SOILS MORE THAN 50% OF MATERIAL IS SMALLER THAN NO. 200 SIEVE SIZE	SILTS AND CLAYS	LIQUID LIMIT LESS THAN 50		ML	Inorganic silts and very fine sands, rock flour, silty or clayey fine sands or clayey silts with slight plasticity.		
				CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays.		
				OL	Organic silts and organic silty clays of low plasticity.		
	SILTS AND CLAYS	LIQUID LIMIT GREATER THAN 50		MH	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts.		
				CH	Inorganic clays of high plasticity, fat clays.		
				OH	Organic clays of medium to high plasticity, organic silts.		
			HIGHLY ORGANIC SOILS			PT	Peat and other highly organic soils.

NOTE: DUAL SYMBOLS ARE USED TO INDICATE BORDERLINE SOIL CLASSIFICATIONS

Notes

See Soil Boring Information Special Provision.

SPT (Standard Penetration Test-ASTM D1586):

The number of blows of a 140 lb (63.6 kg) hammer falling 2.5 ft (750 mm) used to drive a 2 in (50 mm) O.D. Split Spoon sampler for a total of 1.5 ft (0.45 m) of penetration.

Written as follows:

first 0.5 ft (0.15 m) - second 0.5 ft (0.15 m) - third 0.5 ft (0.15 m)
(ex: 1-3-9)

Note: if the number of blows exceeds 50 before 0.5 ft (0.15 m) of penetration is achieved, the actual penetration rounded to the nearest 0.1 ft (0.03 m) follows the number of blows in parentheses (ex: 12-24-50 (0.09 m), 34-50 (0.4 ft), or 100 (0.3 ft)). WR denotes a zero blow count with the weight of the rods only.

WH denotes a zero blow count with the weight of the rods plus the weight of the hammer.

MC=Moisture Content, LL=Liquid limit, PL=Plastic Limit
-200%=percent soil passing 200 sieve, DD=Dry Density

Soil Classifications are Based on the Unified Soil Classification System, ASTM D2487 and D2488.
Also included are the AASHTO group classifications (M145).
Descriptions are based on visual observation, except where they have been modified to reflect results of laboratory tests as deemed appropriate.

Example soil description: Sandy FAT CLAY (CH), soft, wet, brown. (A-7)

Order of Descriptors

- Group Name
- Consistency or Relative Density
- Moisture Condition
- Color
- Particle size descriptor(s) (coarse grained soils only)
- Angularity of coarse grained soils
- Other relevant notes

Criteria For Descriptors

Consistency of Fine Grained Soils

Consistency	N-Value (uncorrected)
Very Soft	< 2
Soft	2 - 4
Medium Stiff	5 - 8
Stiff	9 - 15
Very Stiff	16 - 30
Hard	> 30

Apparent Density of Coarse Grained Soils

Relative Density	N-Value (uncorrected)
Very Loose	< 4
Loose	4 - 10
Medium Dense	11 - 30
Dense	31 - 50
Very Dense	> 50

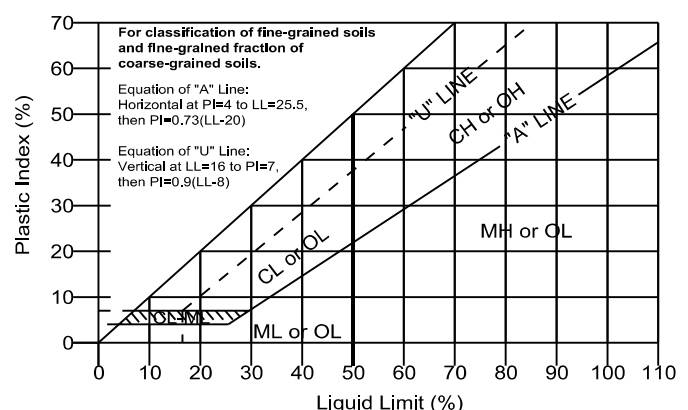
Moisture Condition

- Dry - Absence of moisture, dusty, dry to the touch.
- Moist - Damp, but no visible water.
- Wet - Visible free water.

Definition of Particle Size Ranges

Soil Component	Size Range
Boulder	> 12 in (300 mm)
Cobble	3 in (75 mm) - 12 in (300 mm)
Gravel	No. 4 Sieve (4.75 mm) to 3 in (75 mm)
Sand	No. 200 (0.075 mm) to No. 4 Sieves (4.75 mm)
Silt	< No. 200 Sieve (0.075 mm)*
Clay	< No. 200 Sieve (0.075 mm)*

*Atterberg limits and chart below to differentiate between silt and clay.



Angularity of Coarse-Grained Particles

- Angular - Particles have sharp edges and relative plane sides with unpolished surfaces.
- Subangular - Particles are similar to angular description, but have rounded edges.
- Subrounded - Particles have nearly plane sides, but have no edges.
- Rounded - Particles have smoothly curved sides and well-rounded corners and edges.

Boring Log Descriptive Terminology

Key to Rock Symbols and Terms

Rock Type	Symbol	Rock Type	Symbol	Rock Type	Symbol
Argillite		Dolomite		Quartzite	
Basalt		Gneiss		Rhyolite	
Bedrock (other)		Granitic		Sandstone	
Breccia		Limestone		Schist	
Claystone		Siltstone		Shale	
		Conglomerate			

Order of Descriptors

- Rock Type
- Color
- Grain size (if applicable)
- Stratification/Foliation (as applicable)
- Field Hardness
- Other relevant notes

Criteria For Descriptors

Grain Size

Description	Characteristic
Coarse Grained	-Individual grains can be easily distinguished by eye
Fine Grained	-Individual grains can be distinguished with difficulty

Stratum Thickness

Thickly Bedded	3-10 ft (1-3 m)
Medium Bedded	1-3 ft (300 mm - 1 m)
Thinly Bedded	2-12 in (50-300 mm)
Very Thinly Bedded	< 2 in (50 mm)

Rock Field Hardness

Very Soft	-Can be carved with knife. Can be excavated readily with point of rock hammer. Can be scratched readily by fingernail.
Soft	-Can be grooved or gouged readily by knife or point of rock hammer. Can be excavated in fragments from chips to several inches in size by moderate blows of the point of a rock hammer.
Medium	-Can be grooved or gouged 0.05 in (2 mm) deep by firm pressure of knife or rock hammer point. Can be excavated in small chips to pieces about 1 in (25 mm) maximum size by hard blows of the point of a rock hammer.
Moderately hard	-Can be scratched with knife or pick. Gouges or grooves to 0.25 in (6 mm) can be excavated by hard blow of rock hammer. Hand specimen can be detached by moderate blows.
Hard	-Can be scratched with knife or pick only with difficulty. Hard hammer blows required to detach hand specimen.
Very Hard	-Cannot be scratched with knife or sharp rock hammer point. Breaking of hand specimens requires several hard blows of a rock hammer.

Notes:

UCS = Unconfined Compressive Strength obtained from laboratory testing at the given depth.

See Soil Boring Information Special Provision.

Miscellaneous Soil/Rock Symbols and Terms

	Concrete
	Asphalt
	Water
	Boulders and Cobbles
	Coal
	Fill
	Millings
	Topsoil

Explanation of Text Fields in Boring Logs:

Material Description: Lithologic Description of soil or rock encountered.

Remarks: Comments on drilling, including method, bit type, and problems encountered.
Unless stated on logs as being surveyed by district survey, all locations are considered approximate.

General Notes

- Descriptions on these boring logs apply only at the specific boring, and at the time the borings were made. These logs are not warranted to be representative of subsurface conditions at other locations or times.
- Water level observations apply only at the specific boring, and at the time the borings were made. Due to the variability of groundwater measurements given the type of drilling used, and the stratification of the soil in the boring, these logs are not warranted to be representative of groundwater conditions at other locations or times.
- Other terms may be used as descriptors, as defined by the profession.

-Soil and Rock descriptions are based on visual observation, except where they have been modified to reflect results of laboratory tests as deemed appropriate.

Operation Types:

	Auger
	Casing Advancer
	Core Barrel
	Drive Casing

Sample Types:

	Split Spoon
	Shelby
	Bulk Sample
	Grab Sample
	Cone Penetrometer
	Vane Shear
	Special Samplers
	Testpit

Example Rock Log

SANDSTONE, gray, fine grained, thickly bedded, hard field hardness.



LOG OF BORING

Boring B-1

Sheet 1 of 3

Project: Bar Island Dam Reconstruction				Rig: TRACK-MOUNTED CME-45				Boring Location N: 1275070.71 ft Coordinates E: 2195306.6 ft							
Project Number: 4691.12652.01				Boring Diameter: 8"		System: MT S.P. (E) Datum: NAD83			Ground Elevation: 2920.0 ft						
Date Started: 12/3/24		Date Finished: 12/4/24		Drilling Fluid: Water		Location Source: Surveyed			Elevation Source: Surveyed						
Driller: B. HARDY Logger: D.DUNCAN				Notes: Right middle of Crest				Abandonment: Bentonite							
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests
5 2915.0			56		8 - 12 - 13		Fat CLAY with sand and gravel (CH), very stiff, moist, brown. (Dam Embankment)	4.0 2916.0	13						Hollow stem augers with bullet bit.
			78		6 - 9 - 10		Fat CLAY (CH), very stiff to hard, moist, dark brown. Trace sand and gravel (Dam Embankment)	17							
10 2910.0			67		6 - 13 - 18			17	70	20	68				
			61		10 - 13 - 15			14							
15 2905.0			56		9 - 9 - 16			17							
20 2900.0			89		7 - 14 - 16			19							
25 2895.0			72		10 - 11 - 15			19							
Water Level Observations															Notes:
☒ End of Drilling: Not Encountered ☒ After Drilling:															

(2) INVESTIGATION LOG - BELLEAYR - 2020.GLB - 15/1/25 15:30 - \\DOWL.COM\PROJECTS\9112652-01\50GEO\EXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ



LOG OF BORING

Boring B-1

Sheet 2 of 3

Project: Bar Island Dam Reconstruction				Rig: TRACK-MOUNTED CME-45				Boring Location N: 1275070.71 ft Coordinates E: 2195306.6 ft								
Project Number: 4691.12652.01				Boring Diameter: 8"		System: MT S.P. (E) Datum: NAD83			Ground Elevation: 2920.0 ft							
Date Started: 12/3/24		Date Finished: 12/4/24		Drilling Fluid: Water		Location Source: Surveyed			Elevation Source: Surveyed							
Driller: B. HARDY Logger: D.DUNCAN				Notes: Right middle of Crest				Abandonment: Bentonite								
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests	
30 2890.0			50		6 - 14 - 16		Fat CLAY (CH), very stiff to hard, moist, dark brown. Trace sand and gravel (Dam Embankment)(continued)	20								
35 2885.0			78		10 - 14 - 18			19	63	19	55					
40 2880.0			83		4 - 10 - 14			22								
45 2875.0			100		5 - 9 - 13		Fat CLAY (CH), very stiff, moist, dark brown to black. (Foundation)	43.0 2877.0	25	66	20	82				
50 2870.0			100		48 - 50/0.3ft		SHALE, brown to black, very thinly bedded, soft to medium field hardness, (Bear Paw). With Planar Bedding. Severly fractured and moderately weathered.	48.0 2872.0	21				80	117	Switched to NQ coring with surface set bit and water.	
55 2865.0			60	34									121	118		
Water Level Observations																Notes:
☒ End of Drilling: Not Encountered ☐ After Drilling:																

(2) INVESTIGATION LOG - BELLEAYR - 15/1/25 15:30 - \DOWL.COM\PROJECTS\9112652-01\50GEOEXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ

LOG OF BORING

Boring B-1

Sheet 3 of 3

Project: Bar Island Dam Reconstruction						Rig: TRACK-MOUNTED CME-45			Boring Location N: 1275070.71 ft Coordinates E: 2195306.6 ft								
Project Number: 4691.12652.01						Boring Diameter: 8"		System: MT S.P. (E) Datum: NAD83				Ground Elevation: 2920.0 ft					
Date Started: 12/3/24		Date Finished: 12/4/24		Drilling Fluid: Water		Location Source: Surveyed				Elevation Source: Surveyed							
Driller: B. HARDY Logger: D.DUNCAN				Notes: Right middle of Crest					Abandonment: Bentonite								
Depth (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft)	Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests	
60 2860.0			100	20			SHALE, brown to black, very thinly bedded, soft to medium field hardness, (Bear Paw). With Plannar Bedding. Severly fractured and moderately weathered.(continued)							244 239	119 120		
65 2855.0			100	82			Boring Depth: 65.0 ft, Elevation: 2855.0 ft	65.0 2855.0						79	120		
Water Level Observations									Notes:								
End of Drilling: Not Encountered						After Drilling:											



LOG OF BORING

Boring B-2

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: TRACK-MOUNTED CME-45		Boring Location N: 1275066.2 ft Coordinates E: 2195446.1 ft									
Project Number: 4691.12652.01				Boring Diameter: 8"		System: MT S.P. (E) Datum: NAD83		Ground Elevation: 2920.1 ft							
Date Started: 12/4/24		Date Finished: 12/5/24		Drilling Fluid: None		Location Source: Surveyed		Elevation Source: Surveyed							
Driller: B. HARDY Logger: D.DUNCAN			Notes: Left middle of Crest				Abandonment: Bentonite								
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests
5 2915.1			89		12 - 18 - 16		Fat CLAY with sand and gravel (CH), very stiff to hard, moist, brown. (Dam Embankment)		13	58	18	69			Hollow stem augers with bullet bit.
			72		7 - 14 - 17										
10 2910.1			83		9 - 14 - 20				14	58	18	80			
15 2905.1			100		9 - 14 - 14										
20 2900.1			89		10 - 15 - 17										
25 2895.1			89		9 - 15 - 21			26.5 2893.6	20	63	18	79			Triax EA: 28.6 ° Triax EC: 0 psf Traix TA: 13.8 ° Traix TC: 234 psf HC: 8.4e-10 cm/s
Boring Depth: 26.5 ft, Elevation: 2893.6 ft															
Water Level Observations															
Notes:															
☒ End of Drilling: Not Encountered ☐ After Drilling:															

(2) INVESTIGATION LOG - BELLEAYR - 2020.GLB - 15/1/25 15:30 - \\DOWL.COM\PROJECTS\9112652-01\50GEO\EXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ

LOG OF BORING

Boring B-3

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: TRACK-MOUNTED CME-45				Boring Location N: 1274937.4 ft Coordinates E: 2195471.6 ft																																																																																																																																																																					
Project Number: 4691.12652.01				Boring Diameter: 8"		System: MT S.P. (E) Datum: NAD83				Ground Elevation: 2879.0 ft																																																																																																																																																																			
Date Started: 12/5/24		Date Finished: 12/5/24		Drilling Fluid: Water		Location Source: Surveyed				Elevation Source: Surveyed																																																																																																																																																																			
Driller: B. HARDY Logger: D.DUNCAN				Notes: Toe in the middle of the Dam				Abandonment: Bentonite																																																																																																																																																																					
<table><thead><tr><th>Depth (ft) Elev. (ft)</th><th>Operation</th><th>Sample Type</th><th>Recovery (%)</th><th>RQD (%)</th><th>Blow Count</th><th>Lithology</th><th>Material Description</th><th>Depth (ft) Elev. (ft)</th><th>MC (%)</th><th>LL</th><th>PL</th><th>-200 (%)</th><th>Qu (psi)</th><th>DD (pcf)</th><th>Remarks and Other Tests</th></tr></thead><tbody><tr><td rowspan="4">5 2874.0</td><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4">56</td><td rowspan="4"></td><td rowspan="4">3 - 5 - 4</td><td rowspan="4"></td><td>Fat CLAY (CH), stiff, moist, brown. (Foundation)</td><td></td><td rowspan="4">25</td><td rowspan="4">59</td><td rowspan="4">21</td><td rowspan="4">56</td><td rowspan="4"></td><td rowspan="4"></td><td rowspan="4">Hollow stem augers with bullet bit.</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">5.5 2873.5</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">100</td><td rowspan="2"></td><td rowspan="2">11 - 12 - 11</td><td rowspan="2"></td><td>Clayey GRAVEL (GC), medium dense, wet, brown to multi-colored, subrounded. (Foundation)</td><td>5.5 2873.5</td><td rowspan="2">15</td><td rowspan="2">43</td><td rowspan="2">15</td><td rowspan="2">14</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">7.5 2871.5</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">67</td><td rowspan="2"></td><td rowspan="2">11 - 22 - 44</td><td rowspan="2"></td><td>SHALE, brown to black, very thinly bedded, soft to medium field hardness, (Bear Paw). With Ripple and Plannar Bedding. Moderately fractured and moderately weathered.</td><td>7.5 2871.5</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">10 2869.0</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">100</td><td rowspan="2">70</td><td rowspan="2"></td><td rowspan="2"></td><td></td><td></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">67</td><td rowspan="2">116</td><td rowspan="2">Switched to NQ coring with surface set bit and water.</td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">15 2864.0</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">95</td><td rowspan="2">67</td><td rowspan="2"></td><td rowspan="2"></td><td></td><td></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">77</td><td rowspan="2">116</td><td rowspan="2"></td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">20 2859.0</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">97</td><td rowspan="2">68</td><td rowspan="2"></td><td rowspan="2"></td><td></td><td></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td></tr><tr><td></td><td></td></tr><tr><td rowspan="2">25 2854.0</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td></td><td>25.0 2854.0</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">184</td><td rowspan="2">123</td><td rowspan="2"></td></tr><tr><td></td><td></td></tr><tr><td colspan="16">Boring Depth: 25.0 ft, Elevation: 2854.0 ft</td></tr></tbody></table>												Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests	5 2874.0			56		3 - 5 - 4		Fat CLAY (CH), stiff, moist, brown. (Foundation)		25	59	21	56			Hollow stem augers with bullet bit.							5.5 2873.5			100		11 - 12 - 11		Clayey GRAVEL (GC), medium dense, wet, brown to multi-colored, subrounded. (Foundation)	5.5 2873.5	15	43	15	14						7.5 2871.5			67		11 - 22 - 44		SHALE, brown to black, very thinly bedded, soft to medium field hardness, (Bear Paw). With Ripple and Plannar Bedding. Moderately fractured and moderately weathered.	7.5 2871.5										10 2869.0			100	70									67	116	Switched to NQ coring with surface set bit and water.			15 2864.0			95	67									77	116				20 2859.0			97	68														25 2854.0								25.0 2854.0					184	123				Boring Depth: 25.0 ft, Elevation: 2854.0 ft															
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Water Level Observations																																																																																																																																																																													
Notes:																																																																																																																																																																													
End of Drilling: 5.5 ft (2873.5 ft). Augers In																																																																																																																																																																													
After Drilling:																																																																																																																																																																													



LOG OF BORING

Boring TP-1

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: CAT 306				Boring Location N: 1274923.3 ft Coordinates E: 2195109.9 ft							
Project Number: 4691.12652.01				Boring Diameter: 24" Wide		System: MT S.P. (E) Datum: NAD83			Ground Elevation: 2887.1 ft						
Date Started: 12/5/24		Date Finished: 12/5/24		Drilling Fluid: None		Location Source: Surveyed			Elevation Source: GPS and Plans						
Driller: J. OTTUN Logger: D.DUNCAN				Notes: Right Abutment				Abandonment: Cuttings							
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests
2 2885.1							Clayey GRAVEL (GC), moist, brown to multi-colored, subrounded.								MDUW: 122 pcf OM: 11.5 %
4 2883.1															
6 2881.1															
8 2879.1															
10 2877.1															
Boring Depth: 11.0 ft, Elevation: 2876.1 ft								11.0 2876.1							
Water Level Observations															Notes:
☒ End of Drilling: Not Encountered ☐ After Drilling:															

(2) INVESTIGATION LOG - BELLEAYR - 2020.GLB - 15/1/25 15:30 - \\DOWL.COM\PROJECTS\9112652-01\50GEO\EXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ

LOG OF BORING

Boring TP-2

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: CAT 306				Boring Location N: 1274991.9 ft Coordinates E: 2194740.2 ft								
Project Number: 4691.12652.01				Boring Diameter: 24" Wide		System: MT S.P. (E) Datum: NAD83				Ground Elevation: 2959.3 ft						
Date Started: 12/5/24		Date Finished: 12/5/24		Drilling Fluid: None		Location Source: Surveyed				Elevation Source: GPS and Plans						
Driller: J. OTTUN Logger: D.DUNCAN				Notes: Next to Auxiliary Spillway				Abandonment: Cuttings								
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests	
2 2957.3							Fat CLAY (CH), moist, dark brown.		20	55	20	82				
4 2955.3																
6 2953.3																
8 2951.3																
10 2949.3																
12 2947.3	Boring Depth: 12.0 ft, Elevation: 2947.3 ft							12.0 2947.3								
Water Level Observations																Notes:
End of Drilling: Not Encountered															After Drilling:	



LOG OF BORING

Boring TP-3

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: CAT 306				Boring Location N: 1275018.328 ft Coordinates E: 2196061.361 ft								
Project Number: 4691.12652.01				Boring Diameter: 24" Wide		System: MT S.P. (E) Datum: NAD83			Ground Elevation: 2962.6 ft							
Date Started: 12/5/24		Date Finished: 12/5/24		Drilling Fluid: None		Location Source: Surveyed			Elevation Source: GPS and Plans							
Driller: J. OTTUN Logger: D.DUNCAN				Notes: Left Abutment				Abandonment: Cuttings								
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests	
2 2960.6							Fat CLAY (CH), moist, dark brown.									
4 2958.6																
6 2956.6																
8 2954.6																
10 2952.6																
12 2950.6							SHALE, brown to black, very thinly bedded, medium field hardness, (Bear Paw).	10.0 2952.6								
Boring Depth: 12.0 ft, Elevation: 2950.6 ft								12.0 2950.6							MDUW: 95.3 pcf OM: 25.4 %	
Water Level Observations																Notes:
☒ End of Drilling: Not Encountered ☐ After Drilling:																

(2) INVESTIGATION LOG - BELLEAYR 2020.GLB - 15/1/25 15:30 - I:\DOWL\COM\PROJECTS\9112652-01\50GEO\EXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ



LOG OF BORING

Boring TP-4

Sheet 1 of 1

Project: Bar Island Dam Reconstruction				Rig: CAT 306				Boring Location N: 1275187.306 ft Coordinates E: 2196105.669 ft							
Project Number: 4691.12652.01				Boring Diameter: 24" Wide		System: MT S.P. (E) Datum: NAD83			Ground Elevation: 2995.4 ft						
Date Started: 12/5/24		Date Finished: 12/5/24		Drilling Fluid: None		Location Source: Surveyed			Elevation Source: GPS and Plans						
Driller: J. OTTUN Logger: D.DUNCAN				Notes: Left Abutment				Abandonment: Cuttings							
Depth (ft) Elev. (ft)	Operation	Sample Type	Recovery (%)	RQD (%)	Blow Count	Lithology	Material Description	Depth (ft) Elev. (ft)	MC (%)	LL	PL	-200 (%)	Qu (psi)	DD (pcf)	Remarks and Other Tests
2 2993.4							Fat CLAY (CH), moist, dark brown.								
4 2991.4															
6 2989.4															
8 2987.4															
10 2985.4							SHALE, brown to black, very thinly bedded, medium field hardness, (Bear Paw).	10.0 2985.4							
12 2983.4								12.0 2983.4							
Boring Depth: 12.0 ft, Elevation: 2983.4 ft															
Water Level Observations															
Notes:															
☒ End of Drilling: Not Encountered ☐ After Drilling:															

(2) INVESTIGATION LOG - BELLEAYR 2020.GLB - 15/1/25 15:30 - \\DOWL.COM\PROJECTS\9112652-01\50GEO\EXPLORATION\BORINGS\BAR ISLAND DAM BORINGS.GPJ

Appendix B: Photograph Log



General Site Location – Boring 1 – Looking West



Boring 1 – Standard Penetration Test – 2.5 to 4.0 feet



Boring 1 – Standard Penetration Test – 5.0 to 6.5 feet



Boring 1 – Standard Penetration Test – 7.5 to 9.0 feet



Boring 1 – Standard Penetration Test – 10.0 to 11.5 feet



Boring 1 – Standard Penetration Test – 15.0 to 16.5 feet



Boring 1 – Standard Penetration Test – 20.0 to 21.5 feet



Boring 1 – Standard Penetration Test – 25.0 to 26.5 feet



Boring 1 – Standard Penetration Test – 30.0 to 31.5 feet



Boring 1 – Standard Penetration Test – 35.0 to 36.5 feet



Boring 1 – Standard Penetration Test – 40.0 to 41.5 feet



Boring 1 – Standard Penetration Test – 45.0 to 46.5 feet



Boring 1 – Standard Penetration Test – 50.0 to 50.8 feet



Boring 1 – Core Sample – 50.0 to 51.0 feet



Boring 1 – Core Sample – 51.0 to 52.0 feet



Boring 1 – Core Sample – 52.0 to 53.0 feet



Boring 1 – Core Sample – 53.0 to 55.0 feet (Lost Core)



Boring 1 – Core Sample – 55.0 to 56.0 feet



Boring 1 – Core Sample – 56.0 to 57.0 feet



Boring 1 – Core Sample – 57.0 to 58.0 feet



Boring 1 – Core Sample – 58.0 to 59.0 feet



Boring 1 – Core Sample – 59.0 to 60.0 feet



Boring 1 – Core Sample – 60.0 to 61.0 feet



Boring 1 – Core Sample – 61.0 to 62.0 feet



Boring 1 – Core Sample – 62.0 to 63.0 feet



Boring 1 – Core Sample – 63.0 to 64.0 feet



Boring 1 – Core Sample – 64.0 to 65.0 feet



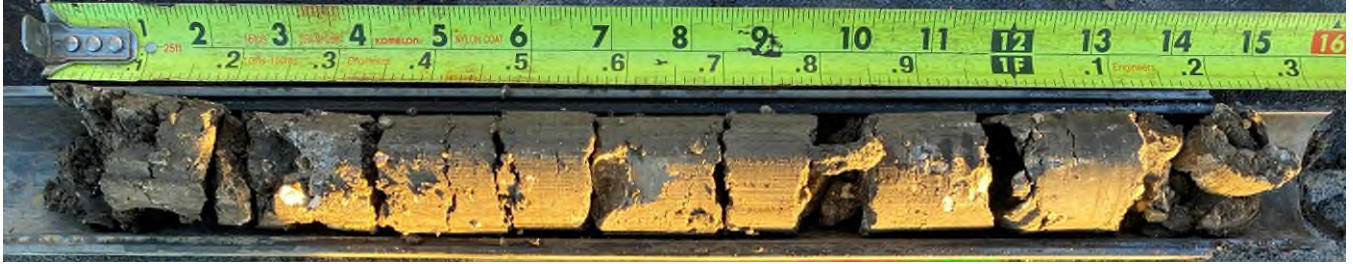
General Site Location – Boring 2 – Looking Southwest



Boring 2 – Standard Penetration Test – 2.5 to 4.0 feet



Boring 2 – Standard Penetration Test – 5.0 to 6.5 feet



Boring 2 – Standard Penetration Test – 10.0 to 11.5 feet



Boring 2 – Standard Penetration Test – 15.0 to 16.5 feet



Boring 2 – Standard Penetration Test – 22.0 to 23.5 feet



Boring 2 – Standard Penetration Test – 25.0 to 26.5 feet



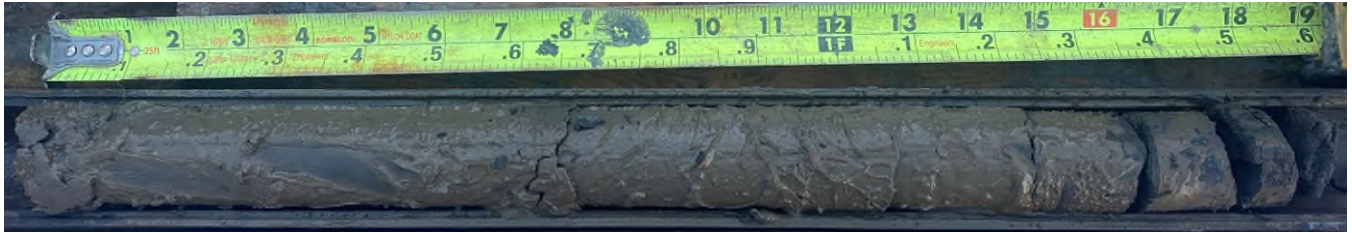
General Site Location – Boring 3 – Looking West



Boring 3 – Standard Penetration Test – 2.5 to 4.0 feet



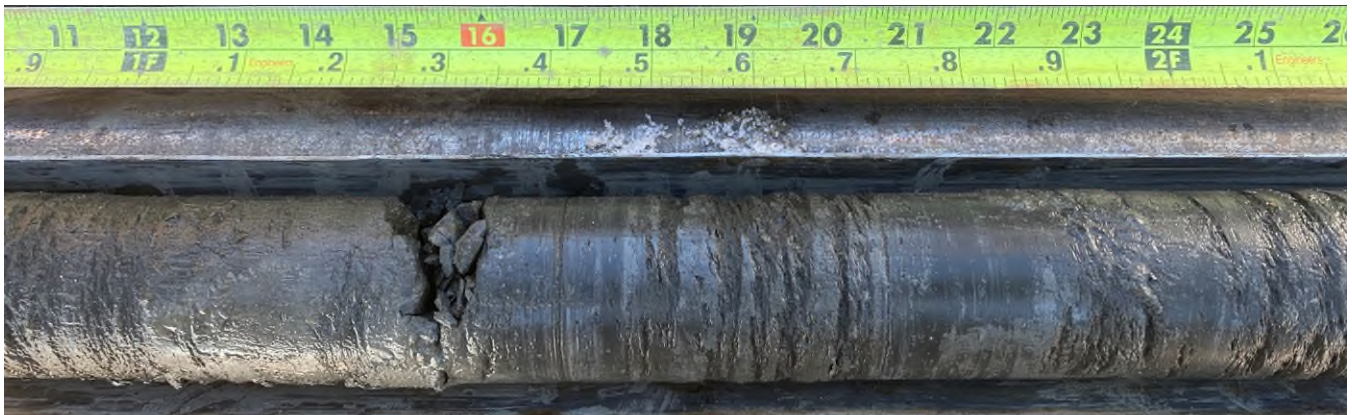
Boring 3 – Standard Penetration Test – 6.0 to 7.5 feet



Boring 3 – Standard Penetration Test – 7.5 to 9.0 feet



Boring 3 – Core Sample – 10.0 to 11.0 feet



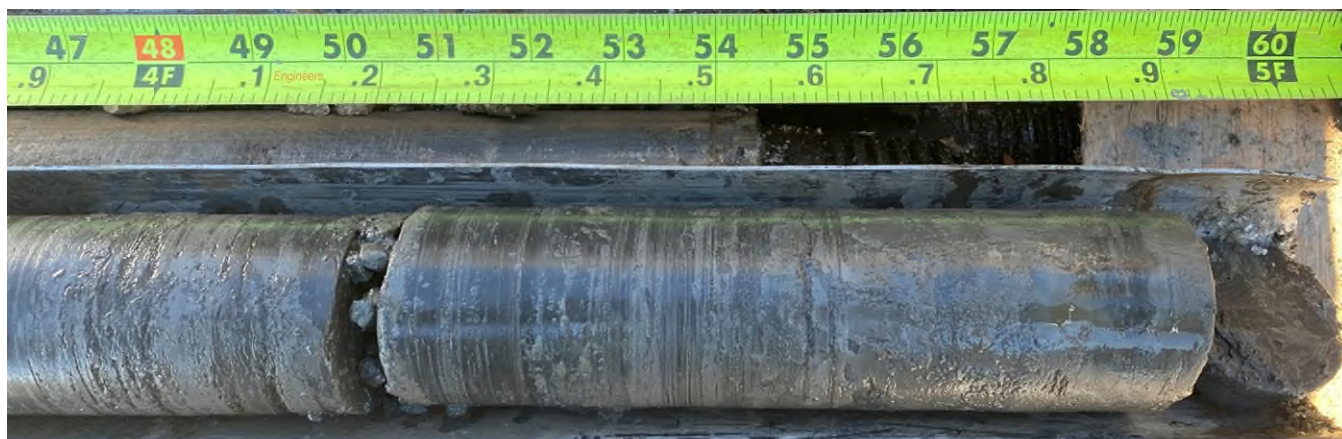
Boring 3 – Core Sample – 11.0 to 12.0 feet



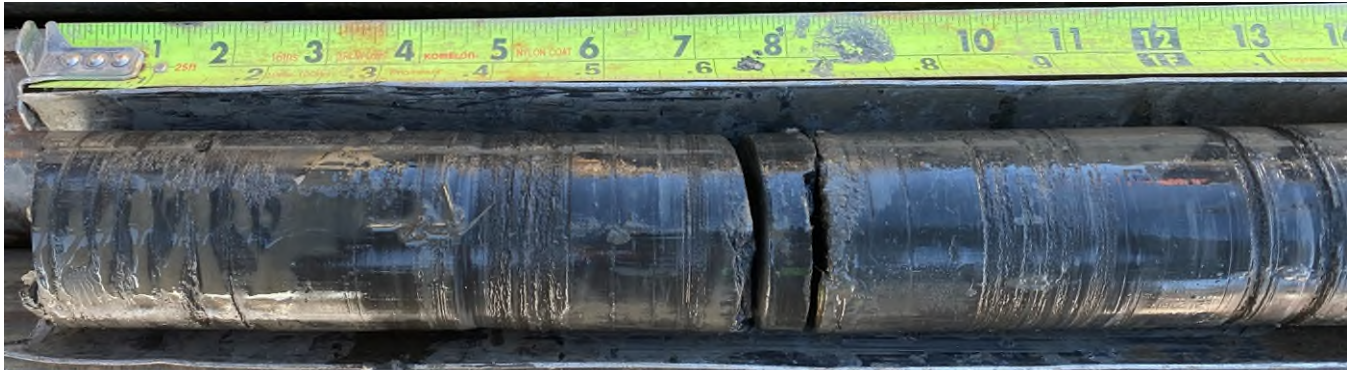
Boring 3 – Core Sample – 12.0 to 13.0feet



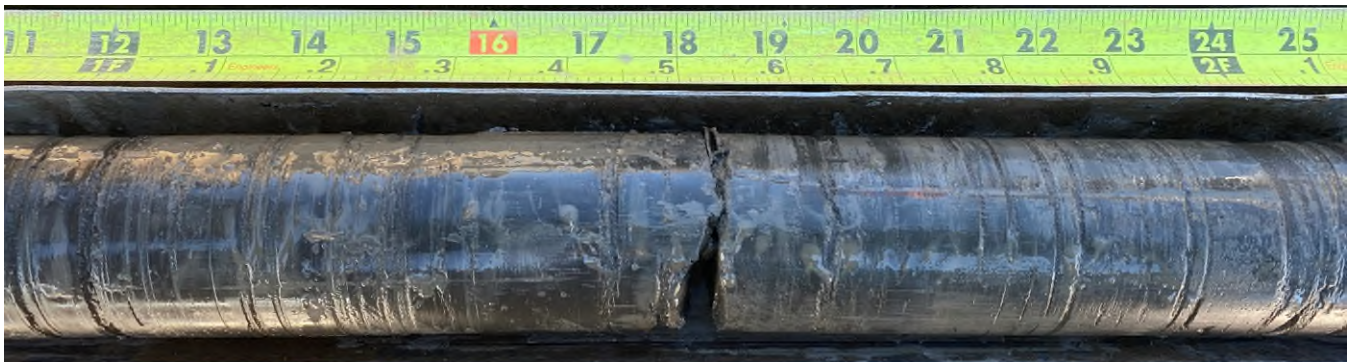
Boring 3 – Core Sample – 13.0 to 14.0 feet



Boring 3 – Core Sample – 14.0 to 15.0 feet



Boring 3 – Core Sample – 15.0 to 16.0 feet



Boring 3 – Core Sample – 16.0 to 17.0 feet



Boring 3 – Core Sample – 17.0 to 18.0 feet



Boring 3 – Core Sample – 18.0 to 19.0 feet



Boring 3 – Core Sample – 19.0 to 20.0 feet



Boring 3 – Core Sample – 20.0 to 21.0 feet



Boring 3 – Core Sample – 21.0 to 22.0 feet



Boring 3 – Core Sample – 22.0 to 23.0 feet



Boring 3 – Core Sample – 23.0 to 24.0 feet



Boring 3 – Core Sample – 24.0 to 25.0 feet



General Site Location – Test Pit 1 – Looking West



Test Pit 1



Test Pit 1



General Site Location – Test Pit 2 – Looking Northwest



Test Pit 2



Test Pit 2



General Site Location – Test Pit 3 – Looking South



Test Pit 3



Test Pit 3



General Site Location – Test Pit 4 – Looking East



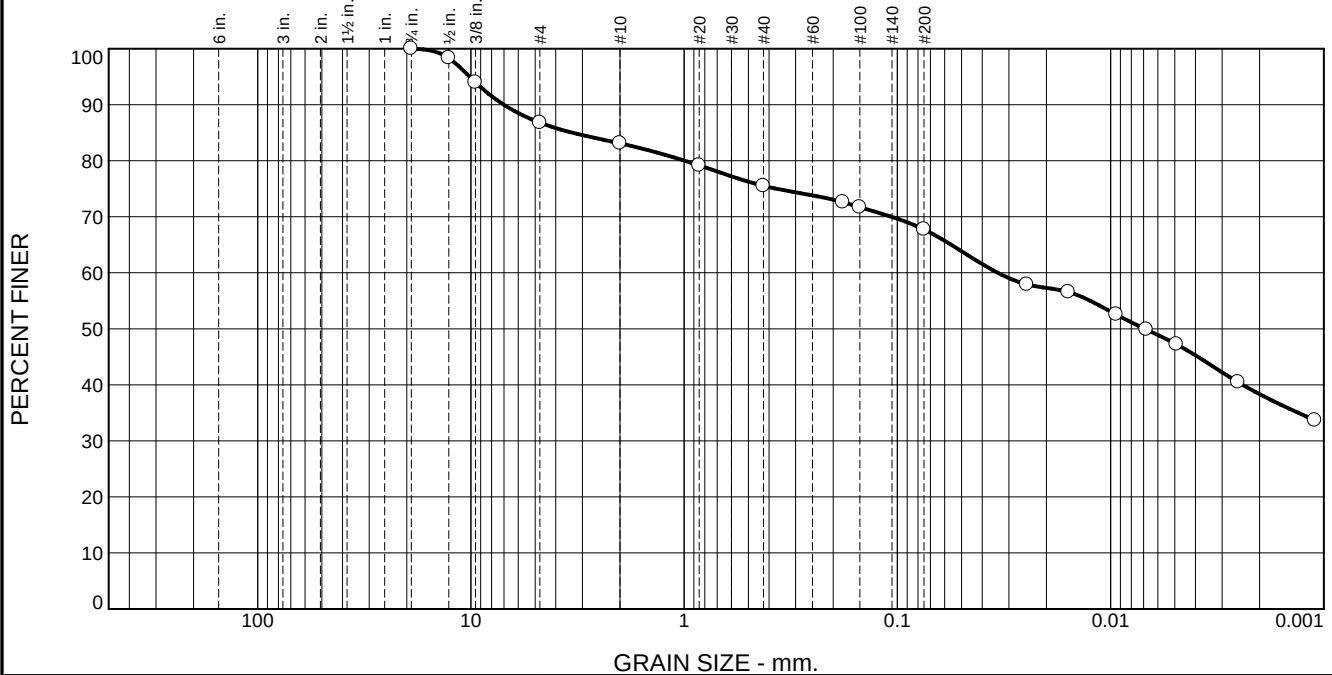
Test Pit 4



Test Pit 4

Appendix C: Laboratory Test Results

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	13.2	3.7	7.6	7.8	20.3	47.4

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.75	100.0		
.5	98.3		
.375	94.0		
#4	86.8		
#10	83.1		
#20	79.2		
#40	75.5		
#80	72.6		
#100	71.7		
#200	67.7		
0.0247 mm.	57.9		
0.0158 mm.	56.6		
0.0094 mm.	52.5		
0.0068 mm.	49.9		
0.0049 mm.	47.2		
0.0025 mm.	40.5		
0.0011 mm.	33.7		

* (no specification provided)

Material Description
Sandy Fat CLAY

Atterberg Limits (ASTM D 4318)
PL= 20 LL= 70 PI= 50

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(33)

Coefficients
D₉₀= 7.0149 D₈₅= 3.3431 D₆₀= 0.0338
D₅₀= 0.0069 D₃₀= D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 **Date Tested:** 12/20/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: B-1
Sample Number: 38285 **Depth:** 7.5-9.0 ft

Date Sampled: 12/03/2024

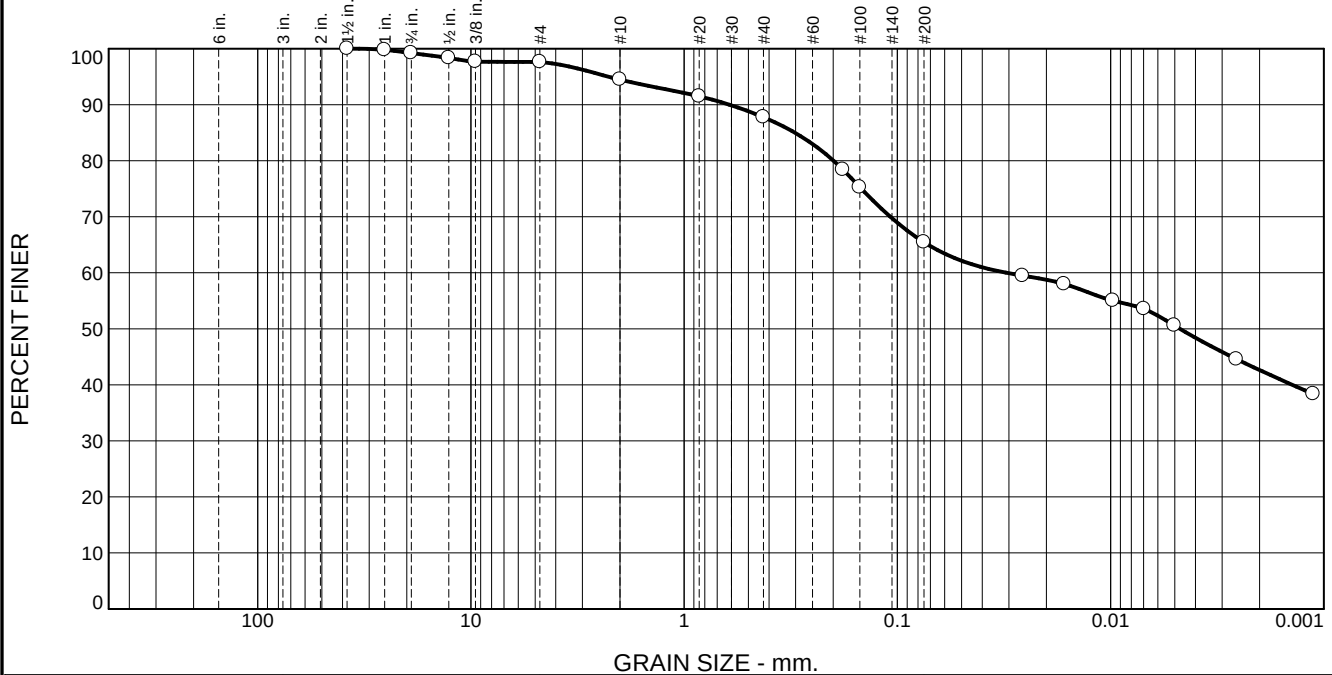


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.8	1.6	3.1	6.7	22.4	14.9	50.5

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
1.5	100.0		
1	99.8		
.75	99.2		
.5	98.3		
.375	97.7		
#4	97.6		
#10	94.5		
#20	91.5		
#40	87.8		
#80	78.4		
#100	75.3		
#200	65.4		
0.0258 mm.	59.5		
0.0165 mm.	58.0		
0.0098 mm.	55.0		
0.0070 mm.	53.5		
0.0050 mm.	50.6		
0.0026 mm.	44.6		
0.0011 mm.	38.4		

* (no specification provided)

Material Description

Sandy Fat CLAY

Atterberg Limits (ASTM D 4318)

PL= 18 LL= 62 PI= 44

Classification

USCS (D 2487)= CH AASHTO (M 145)= A-7-6(26)

Coefficients

D₉₀= 0.6135 D₈₅= 0.3024 D₆₀= 0.0308
D₅₀= 0.0047 D₃₀= C_u=
D₁₀= C_c=

Remarks

Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/17/2024

Tested By: SS

Checked By: CC

Title: Lab Supervisor

Location: B-1

Sample Number: 38288

Depth: 15.0-20.0 ft

Date Sampled: 12/04/2024



Client: Jackola Engineering

Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

DISPERSIVE CHARACTERISTICS OF CLAYEY SOILS

Double Hydrometer, Crumb & Pinhole

ASTM D6572/D4221/D4647

Project: 4691.12652.01

Lab No: 38288

Project Name: Bar Island Dam Mod.

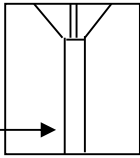
Date Sampled/By: 12/4/24 DD

Sample Location: B-1 15.0-20.0

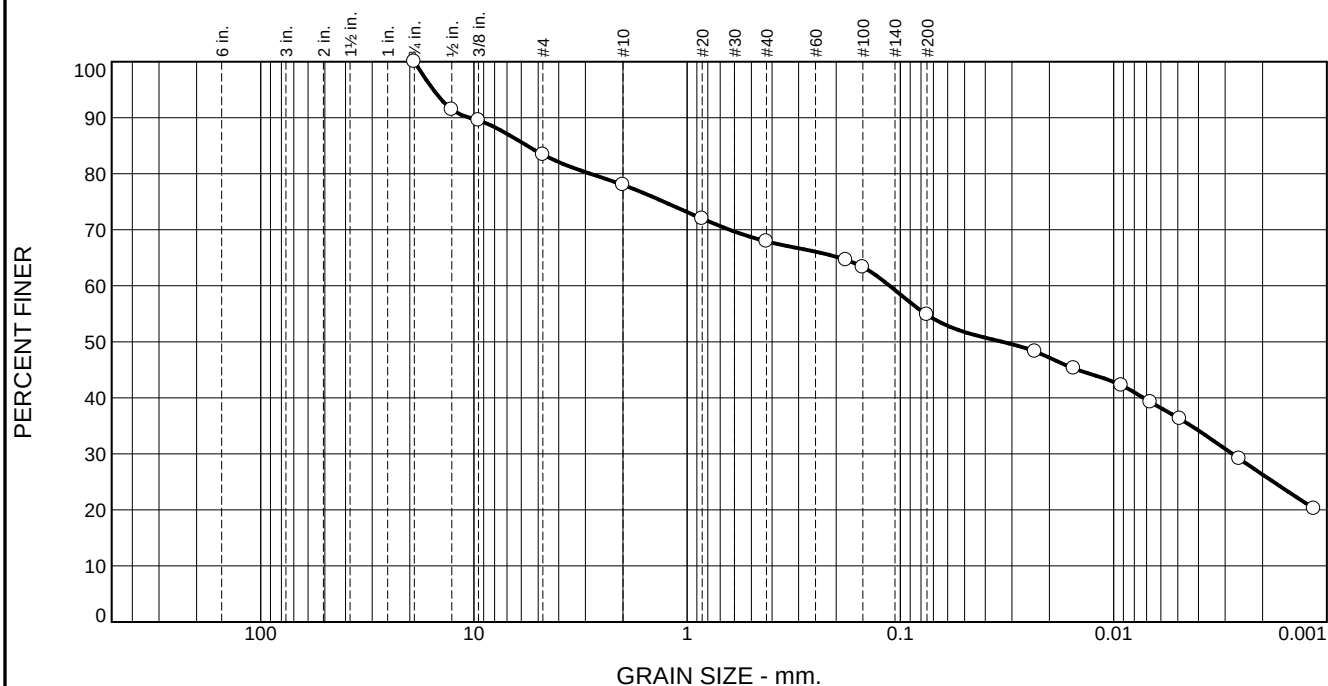
Date Test/By: 1/7/24 SS

Other:

Date Check/By: 1/7/24 CC

PINHOLE DISPERSION		GRADATION		ATTERBERG LIMITS	
Final Hole Diameter 2.0 mm  Specimen after Test		1.5	100.0	PL	18
		1	99.8	LL	62
		.75	99.2	PI	44
		.5	98.3		
		.375	97.7		
		#4	97.6		
		#10	94.5		
		#20	91.5		
		#40	87.8		
		#80	78.4		
		#100	75.3		
		#200	65.4		
		0.0258 mm.	59.5		
		0.0165 mm.	58.0		
		0.0098 mm.	55.0		
		0.0070 mm.	53.5		
		0.0050 mm.	50.6		
		0.0026 mm.	44.6		
		0.0011 mm.	38.4		
SOIL SPECIMAN DATA Nat. Moisture 20.5 Curing Time 24 Hours % Compaction 95% Test Method A				CLASSIFICATION USCS CH AASHTO A-7-6(26)	
DISPERSIVE CLASSIFICATION ND1				DESCRIPTION Sandy Fat CLAY	
CRUMB DISPERSION Type of Test ☒ Natural ☐ Remolded		PHOTOGRAPH OF CRUMB TEST AFTER 6 HOURS 			
Nat. Moisture 20.5 Curing Time 24 Hours					
DISPERSIVE CLASSIFICATION ☒ Non Dispersive ☐ Dispersive ☐ Intermediate ☐ Highly Dispersive					

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	16.6	5.4	10.1	13.1	18.3	36.5

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.75	100.0		
.5	91.5		
.375	89.5		
#4	83.4		
#10	78.0		
#20	72.0		
#40	67.9		
#80	64.6		
#100	63.3		
#200	54.8		
0.0234 mm.	48.3		
0.0154 mm.	45.3		
0.0092 mm.	42.2		
0.0067 mm.	39.3		
0.0049 mm.	36.3		
0.0026 mm.	29.1		
0.0011 mm.	20.2		

* (no specification provided)

Material Description

Sandy Fat CLAY with gravel

Atterberg Limits (ASTM D 4318)

PL= 19 LL= 63 PI= 44

Classification

USCS (D 2487)= CH AASHTO (M 145)= A-7-6(20)

Coefficients

D₉₀= 10.4791 D₈₅= 5.6334 D₆₀= 0.1122
D₅₀= 0.0332 D₃₀= 0.0028 D₁₅=
D₁₀= C_u= C_c=

Remarks

Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/19/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: B-1

Sample Number: 38292

Depth: 35.0-36.5 ft

Date Sampled: 12/24/2024



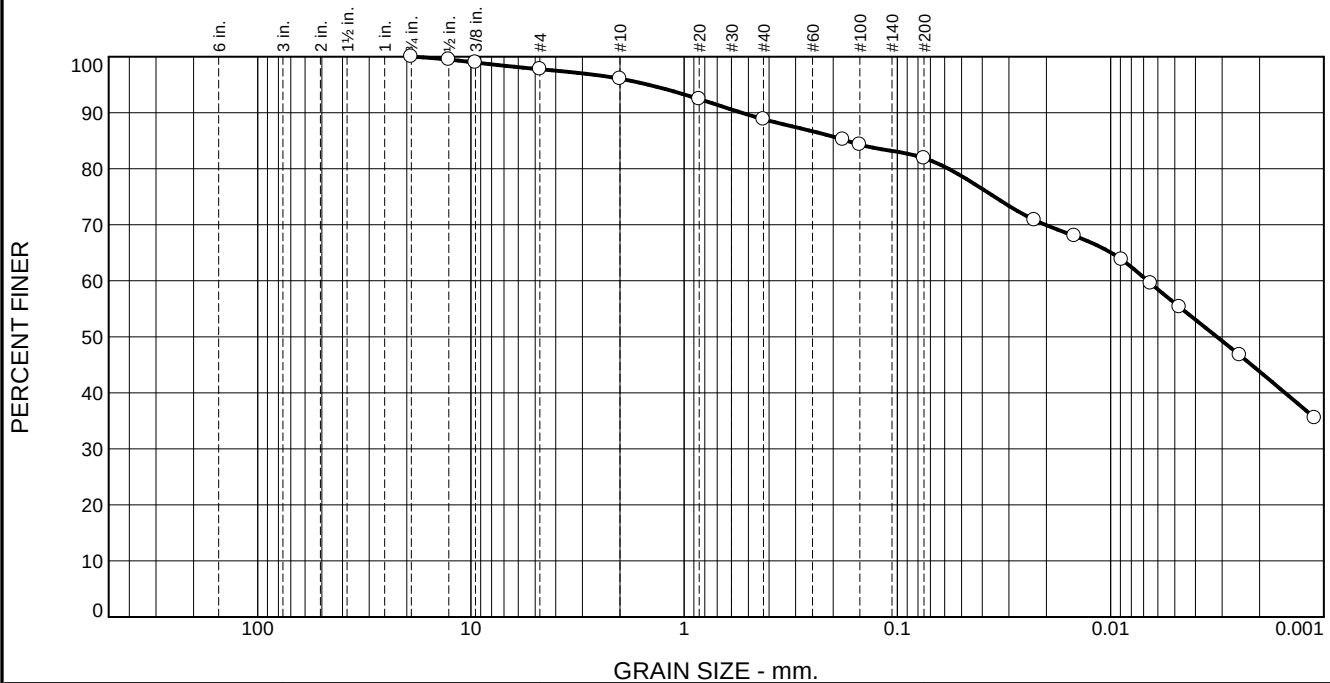
Client: Jackola Engineering

Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	2.2	1.7	7.3	6.9	25.9	56.0

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.75	100.0		
.5	99.5		
.375	98.9		
#4	97.8		
#10	96.1		
#20	92.4		
#40	88.8		
#80	85.2		
#100	84.3		
#200	81.9		
0.0227 mm.	70.9		
0.0148 mm.	68.0		
0.0089 mm.	63.8		
0.0065 mm.	59.6		
0.0047 mm.	55.3		
0.0025 mm.	46.7		
0.0011 mm.	35.5		

* (no specification provided)

Material Description

Fat CLAY with sand

Atterberg Limits (ASTM D 4318)

PL= 20 LL= 66 PI= 46

Classification

USCS (D 2487)= CH AASHTO (M 145)= A-7-6(40)

Coefficients

D₉₀= 0.5384 D₈₅= 0.1731 D₆₀= 0.0067
D₅₀= 0.0032 D₃₀= C_u= D₁₅= C_c=

Remarks

Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/19/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: B-1

Sample Number: 38294

Depth: 45.0-46.5 ft

Date Sampled: 12/04/2024



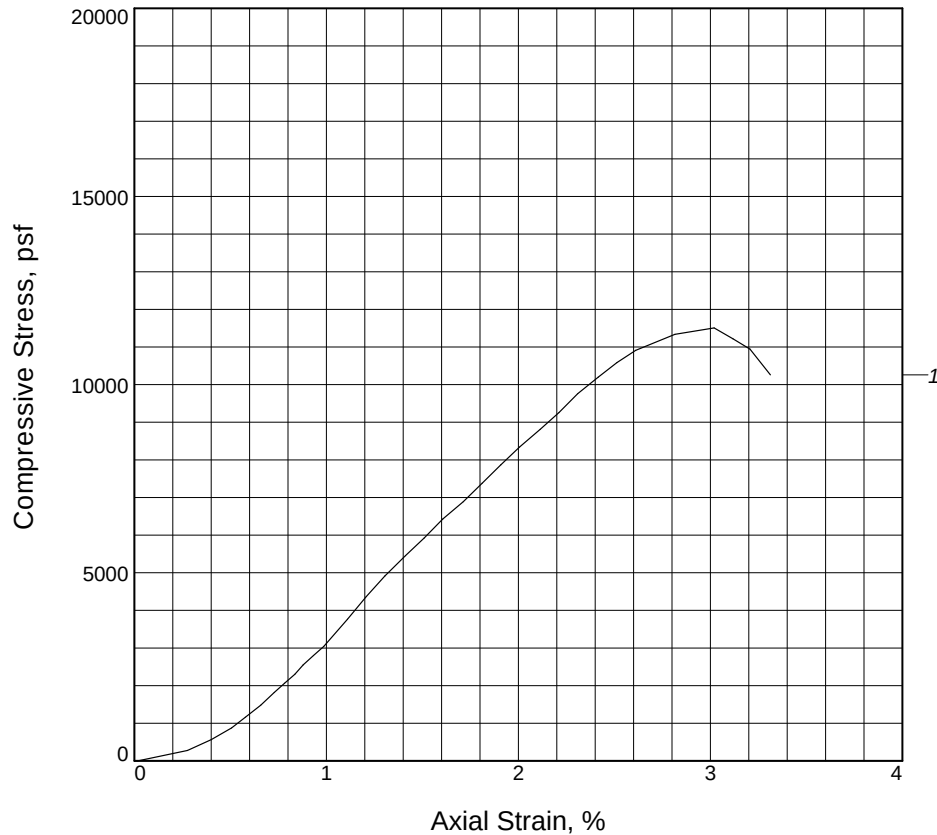
Client: Jackola Engineering

Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	11506			
Undrained shear strength, psf	5753			
Failure strain, %	3.0			
Strain rate, %/min.	0.04			
Water content, %	15.8			
Wet density, pcf	135.9			
Dry density, pcf	117.4			
Saturation, %	98.0			
Void ratio	0.4363			
Specimen diameter, in.	1.99			
Specimen height, in.	3.98			
Height/diameter ratio	2.00			

Description: Shale

LL =	PL =	PI =	Assumed GS= 2.7	Type: Intact
------	------	------	-----------------	--------------

Project No.: 4691.12652.01

Date Sampled: 12/04/24

Remarks:

Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-1

Sample Number: 38296

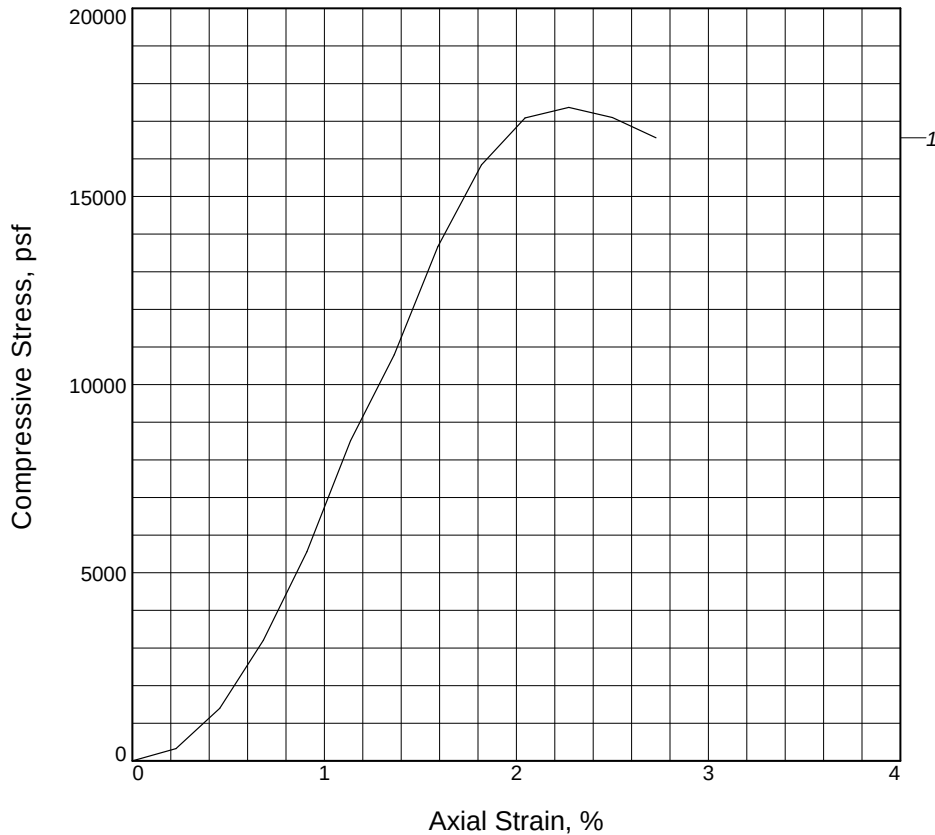
Depth: 50.8-52.5 ft

Figure _____



Tested By: SS Checked By: CC

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	17368			
Undrained shear strength, psf	8684			
Failure strain, %	2.3			
Strain rate, %/min.	0.03			
Water content, %	15.9			
Wet density, pcf	136.7			
Dry density, pcf	117.9			
Saturation, %	96.0			
Void ratio	0.4559			
Specimen diameter, in.	1.98			
Specimen height, in.	4.40			
Height/diameter ratio	2.22			

Description: Shale

LL =	PL =	PI =	Assumed GS= 2.75	Type: Intact
------	------	------	------------------	--------------

Project No.: 4691.12652.01

Date Sampled: 12/4/24

Remarks:
Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-1

Sample Number: 38297

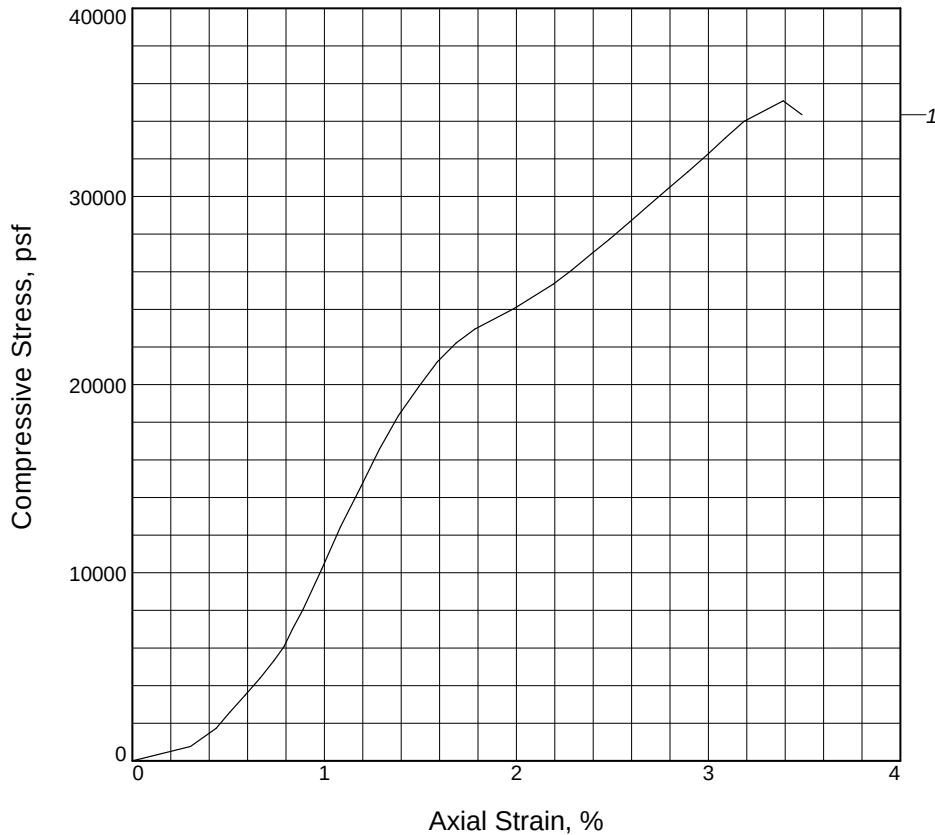
Depth: 52.2-52.8 ft

Figure _____



Tested By: SS _____ **Checked By:** CC _____

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	35086			
Undrained shear strength, psf	17543			
Failure strain, %	3.4			
Strain rate, %/min.	0.04			
Water content, %	15.4			
Wet density, pcf	137.2			
Dry density, pcf	118.9			
Saturation, %	95.6			
Void ratio	0.4443			
Specimen diameter, in.	1.98			
Specimen height, in.	3.86			
Height/diameter ratio	1.95			

Description: Shale

LL = **PL =** **PI =** **Assumed GS=** 2.75 **Type:** Intact

Project No.: 4691.12652.01

Date Sampled: 12/4/24

Remarks:
Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-1

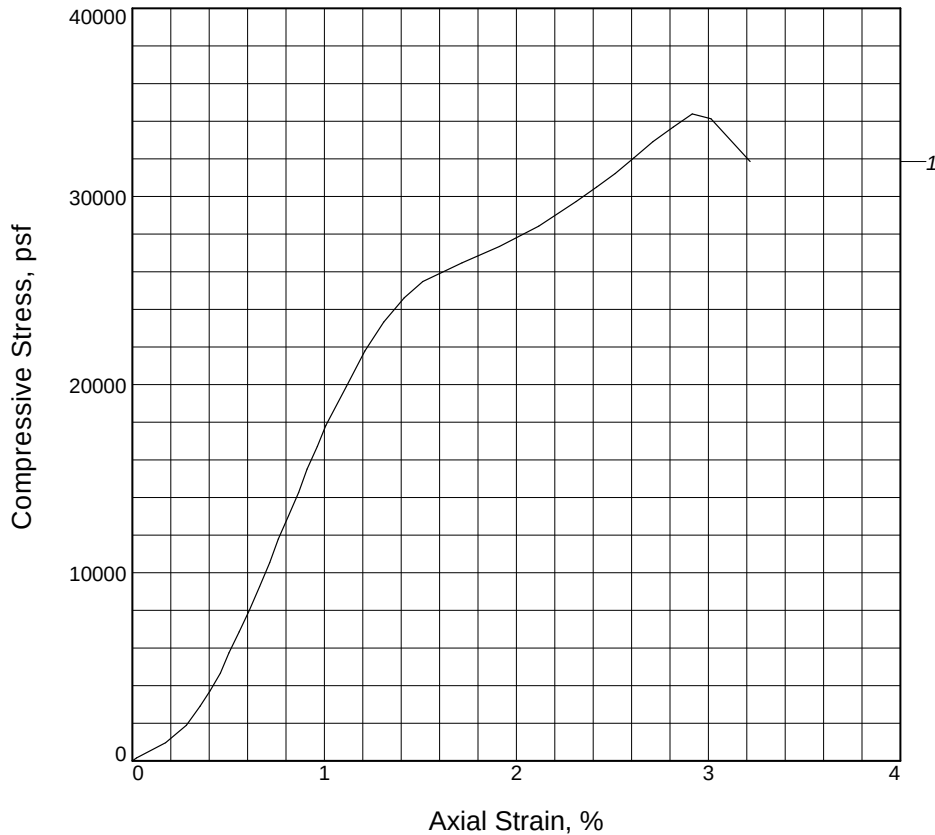
Sample Number: 38298 **Depth:** 56.5-57.0 ft

Figure _____



Tested By: SS **Checked By:** CC

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	34387			
Undrained shear strength, psf	17193			
Failure strain, %	2.9			
Strain rate, %/min.	0.04			
Water content, %	15.5			
Wet density, pcf	138.1			
Dry density, pcf	119.6			
Saturation, %	97.7			
Void ratio	0.4357			
Specimen diameter, in.	1.98			
Specimen height, in.	4.05			
Height/diameter ratio	2.05			

Description: Shale

LL =	PL =	PI =	Assumed GS= 2.75	Type: Intact
------	------	------	------------------	--------------

Project No.: 4691.12652.01

Date Sampled: 12/4/24

Remarks:
Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-1

Sample Number: 38299

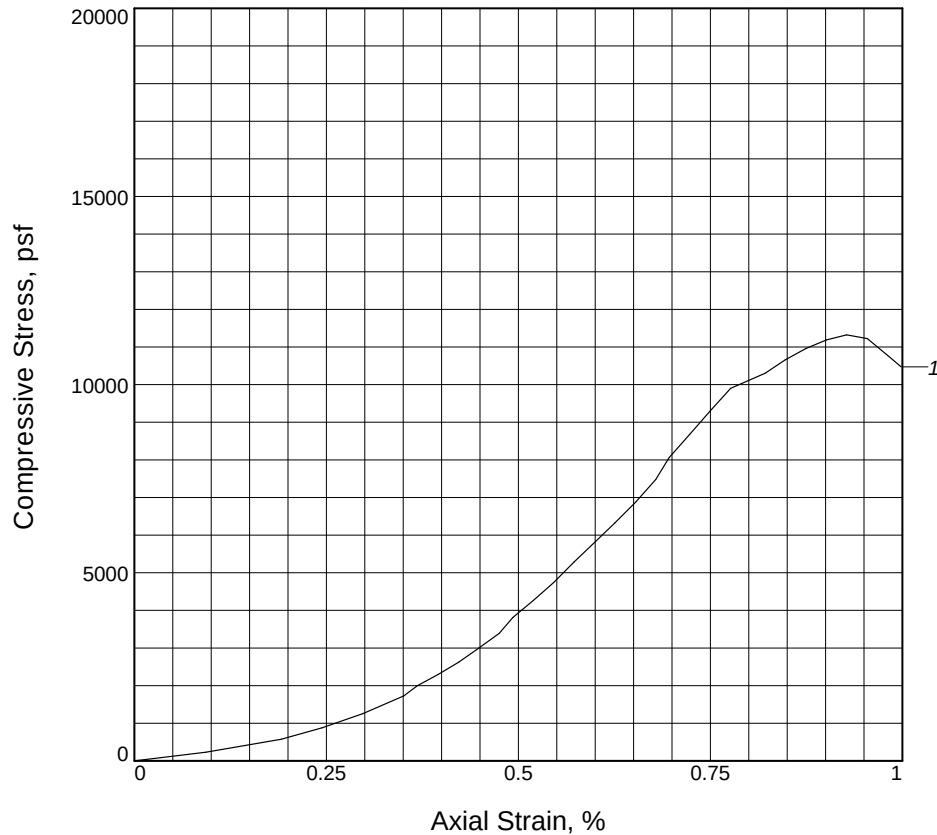
Depth: 57.1-57.6 ft

Figure _____



Tested By: SS Checked By: CC

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	11322			
Undrained shear strength, psf	5661			
Failure strain, %	0.9			
Strain rate, %/min.	0.04			
Water content, %	16.1			
Wet density, pcf	137.7			
Dry density, pcf	118.5			
Saturation, %	98.9			
Void ratio	0.4482			
Specimen diameter, in.	1.99			
Specimen height, in.	3.86			
Height/diameter ratio	1.94			

Description: Shale

LL =	PL =	PI =	Assumed GS= 2.75	Type: Intact
------	------	------	------------------	--------------

Project No.: 4691.12652.01

Date Sampled: 12/4/24

Remarks:
Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-1

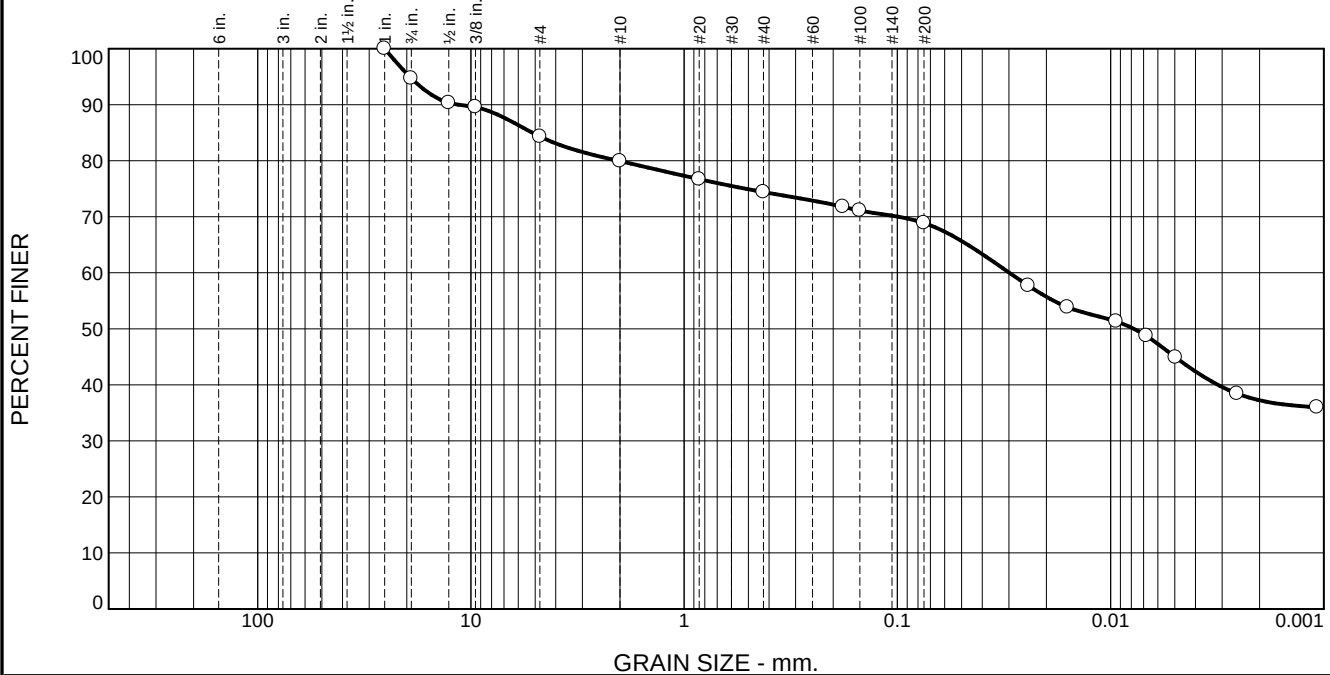
Sample Number: 38301 **Depth:** 63.6-64.4 ft

Figure _____



Tested By: SS **Checked By:** CC

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	5.3	10.4	4.4	5.5	5.5	23.9	45.0

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
1	100.0		
.75	94.7		
.5	90.3		
.375	89.6		
#4	84.3		
#10	79.9		
#20	76.7		
#40	74.4		
#80	71.8		
#100	71.1		
#200	68.9		
0.0243 mm.	57.7		
0.0159 mm.	53.9		
0.0094 mm.	51.3		
0.0068 mm.	48.7		
0.0050 mm.	44.9		
0.0026 mm.	38.4		
0.0011 mm.	36.0		

* (no specification provided)

Material Description
Gravelly Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)
PL= 18 LL= 58 PI= 40

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(26)

Coefficients
D₉₀= 11.5176 D₈₅= 5.1558 D₆₀= 0.0298
D₅₀= 0.0078 D₃₀= D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/20/2024
Tested By: SS
Checked By: CC
Title: Laboratory Supervisor

Location: B-2
Sample Number: 38302 Depth: 2.5-4.0 ft

Date Sampled: 12/04/2024

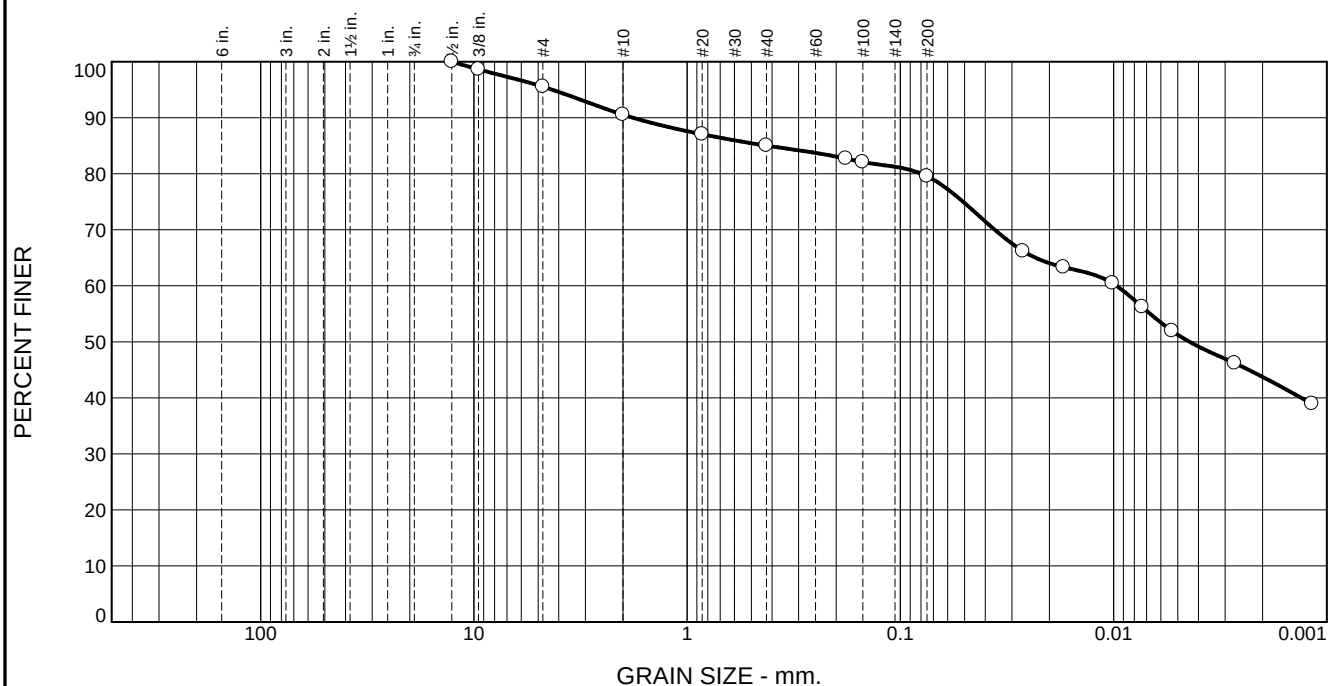


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	4.5	5.0	5.5	5.4	28.4	51.2

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.5	100.0		
.375	98.6		
#4	95.5		
#10	90.5		
#20	87.0		
#40	85.0		
#80	82.7		
#100	82.1		
#200	79.6		
0.0266 mm.	66.2		
0.0172 mm.	63.3		
0.0101 mm.	60.4		
0.0073 mm.	56.2		
0.0053 mm.	51.9		
0.0027 mm.	46.2		
0.0012 mm.	38.9		

* (no specification provided)

Material Description

Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)

PL= 18 LL= 58 PI= 40

Classification

USCS (D 2487)= CH AASHTO (M 145)= A-7-6(33)

Coefficients

D₉₀= 1.7977 D₈₅= 0.4237 D₆₀= 0.0097
D₅₀= 0.0044 D₃₀= C_u=
D₁₀= C_c=

Remarks

Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/20/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: B-2

Sample Number: 38304

Depth: 10.0-11.5 ft

Date Sampled: 12/05/2024

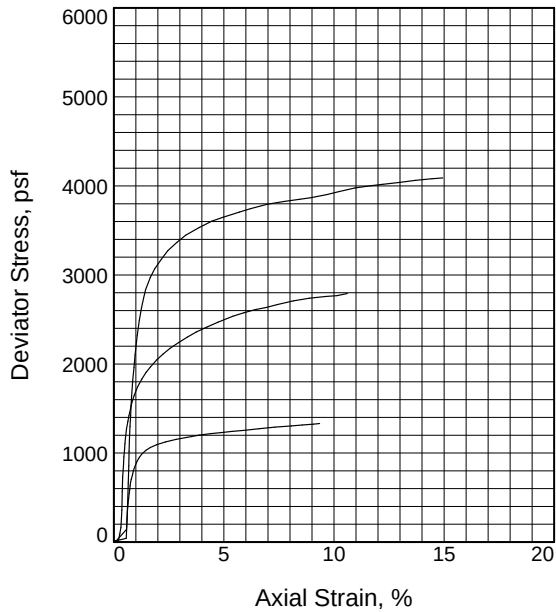
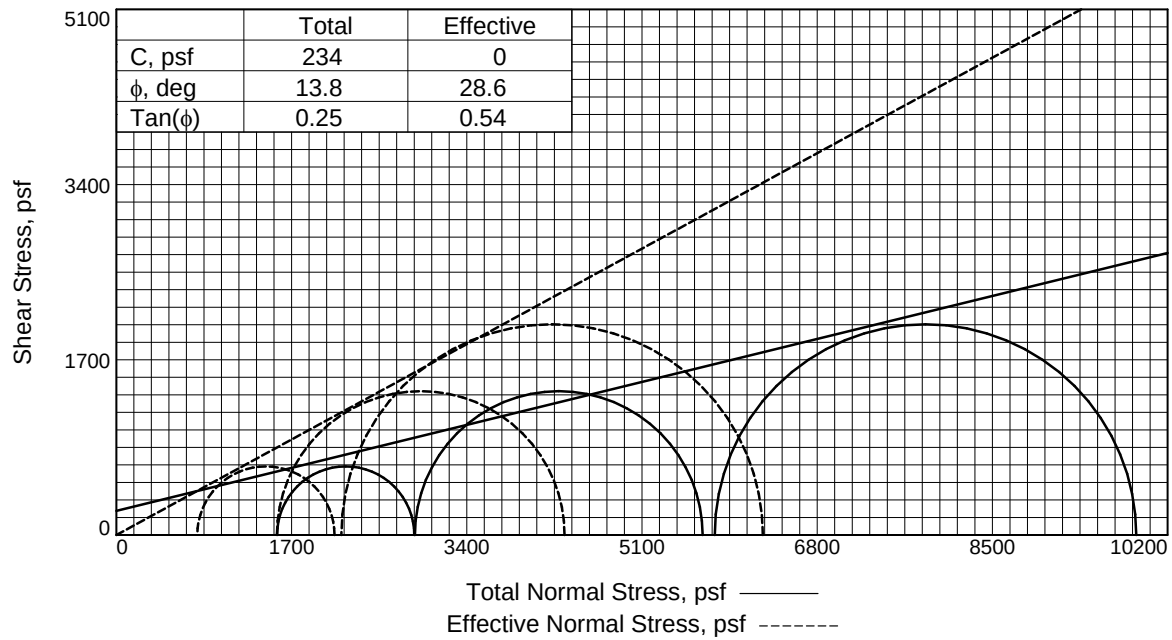


Client: Jackola Engineering

Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure



Sample No.		1	2	3
Initial	Water Content, %	17.8	17.7	17.8
	Dry Density, pcf	101.3	103.2	98.1
	Saturation, %	74.6	78.0	68.7
	Void Ratio	0.6325	0.6028	0.6869
	Diameter, in.	2.87	2.86	2.84
	Height, in.	5.98	5.97	5.97
At Test	Water Content, %	28.4	24.8	26.3
	Dry Density, pcf	94.4	99.8	97.5
	Saturation, %	100.0	100.0	100.0
	Void Ratio	0.7524	0.6580	0.6962
	Diameter, in.	2.93	2.89	2.85
	Height, in.	6.12	6.04	5.99
Strain rate, %/min.		0.0066	0.0066	0.0066
Back Pressure, psi		47.53	48.70	47.40
Cell Pressure, psi		58.36	68.82	87.72
Fail. Stress, psf		1331	2791	4089
Total Pore Pr., psf		7619	8352	10448
Ult. Stress, psf				
Total Pore Pr., psf				
$\bar{\sigma}_1$ Failure, psf		2116	4349	6273
$\bar{\sigma}_3$ Failure, psf		785	1558	2183

Type of Test:

CU with Pore Pressures

Sample Type: Intact

Description: Fat CLAY

Assumed Specific Gravity= 2.65

Remarks: Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-2

Sample Number: 38306

Depth: 20.0-22.0 ft

Proj. No.: 4691.12652.01

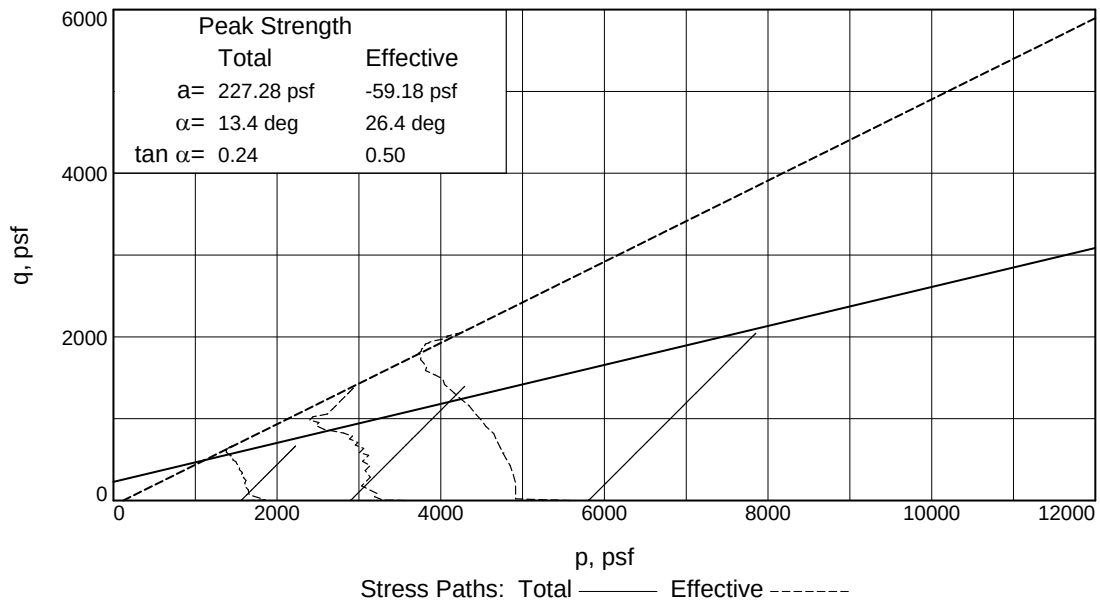
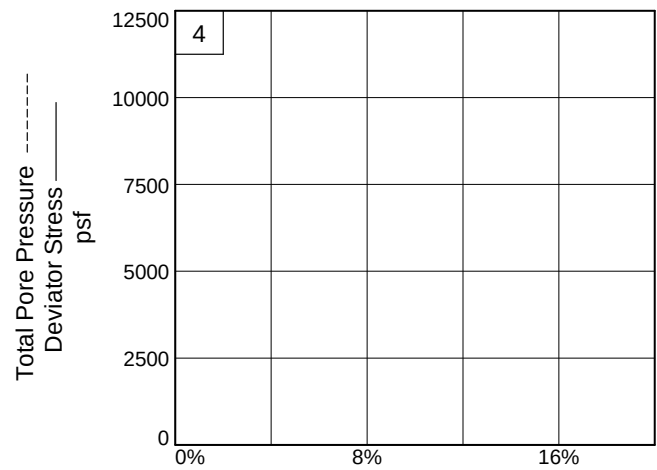
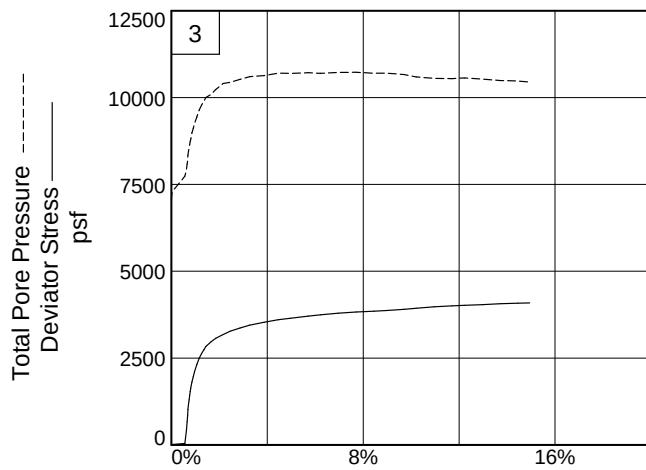
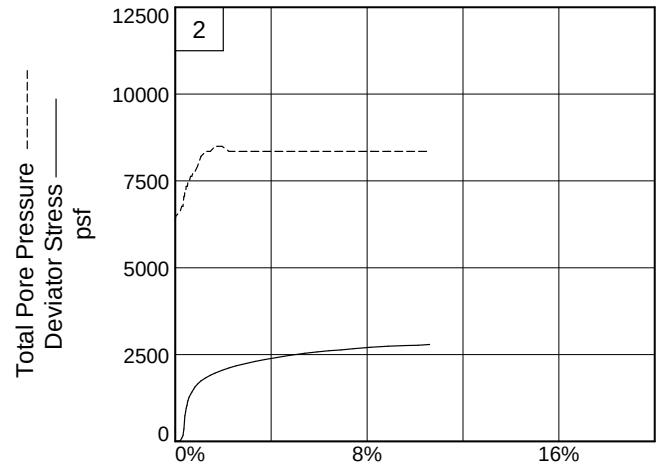
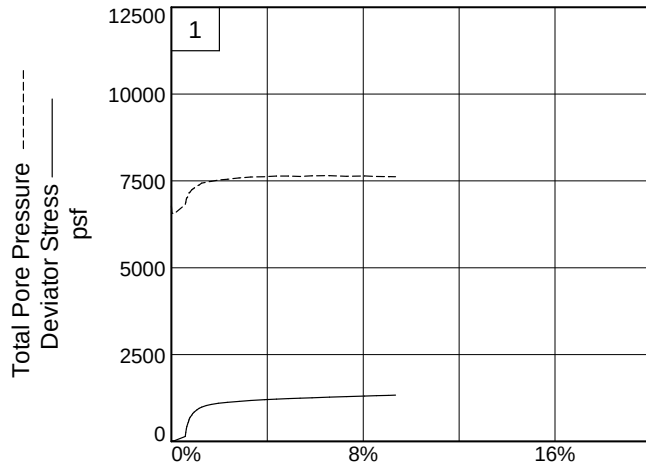
Date Sampled: 12/5/24



Figure _____

Tested By: CC

Checked By: DR



Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-2 **Depth:** 20.0-22.0 ft

Sample Number: 38306

Project No.: 4691.12652.01

Figure _____

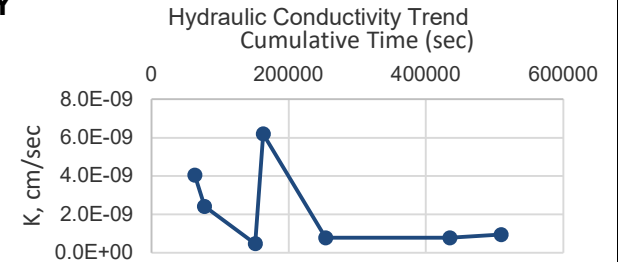
DOWL HKM

Tested By: CC **Checked By:** DR



222 N. 32nd Street, Suite 700
Billings, Montana 59101

FLEXIBLE-WALL HYDRAULIC CONDUCTIVITY ASTM D5084 (Method C)



Project No.: 4691.12652.01

Client: Jackola Engineering

Sample ID: 38306

Project: Bar Island Dam In Tact

Tested by: CC

Sample Description: Fat CLAY

Date Tested: 1/3/2024

Burette Area: 0.880 cm²

Sample Location: B-2 20.0-22.0 ft

Assumed SG: 2.65

Notes:

Effective Consolidation Stress 19.5 psi

Chamber 69.0 psi

Hydraulic Gradient -13.8

Upper 51 psi

System Constant, C 0.440 (cm²)

Lower 48 psi

Confining Stress 19.5 psi

Mass	Initial 1223.5	Final 1286.7	g
Moisture Content	17.7%	24.9%	
Dry Density	103.3	101.1	pcf
Diameter	7.26	7.26	cm
Length	15.16	15.34	cm
Area	41.45	41.45	cm ²

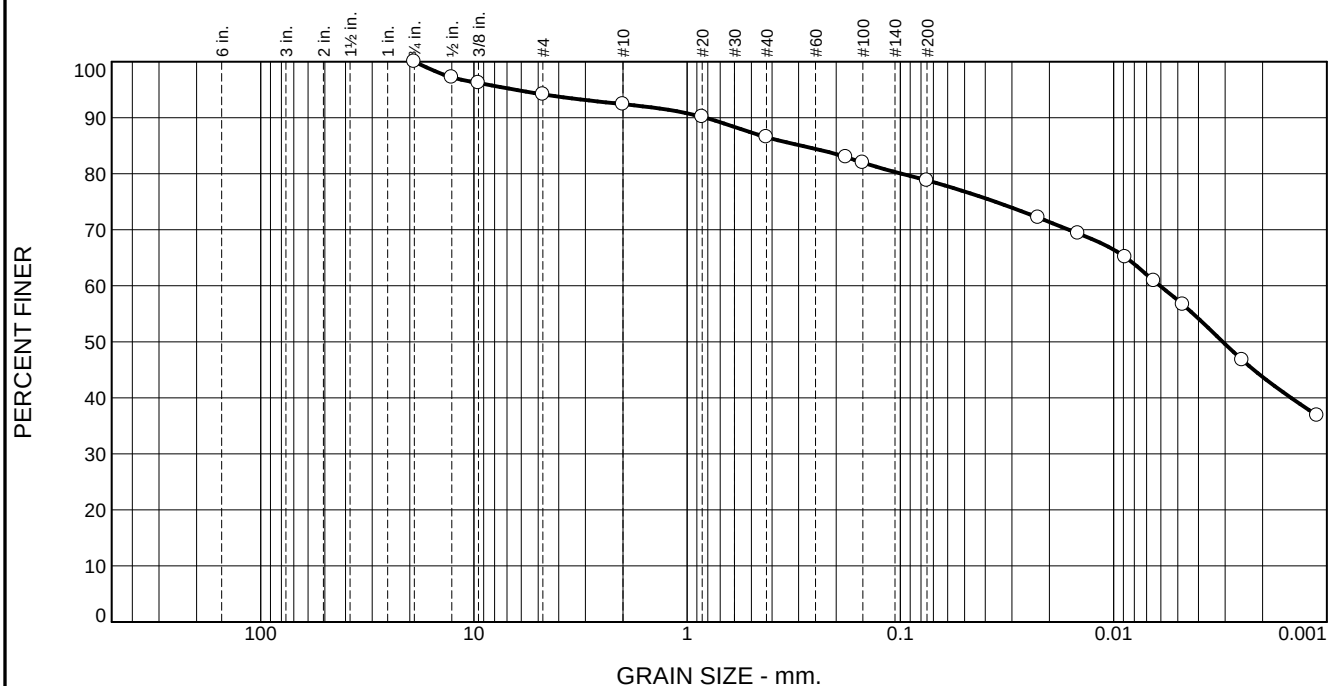
Date	Time (s)	Temperature		Burette Readings		Incremental Flow			Head Difference ¹ h (t)	Preliminary ² kA/L (cm ² /sec)	Final ³ k ₂₀ (x10 ⁻¹⁰) (cm/sec)	ASTM D5084 Method C k ₂₀ (cm/sec)
		°C	R _T	Lower V _U (t) (cc)	Upper V _L (t) (cc)	Inflow (lower) (cc)	Outflow (upper) (cc)	Ratio In/Out				
1/7/2025	0	21.2	0.975	25.00	0.00				-223.9			
1/7/2025	54360	21.2	0.975	24.00	1.50	-1.00	-1.50	0.7	-222.6	4.5E-08	1.6E-08	1.6E-08
1/7/2025	63360	21.2	0.975	23.90	1.50	-0.10	0.00	#DIV/0!	-222.6	1.1E-08	4.0E-09	4.0E-09
1/7/2025	77580	21.2	0.975	23.90	1.60	0.00	-0.10	0.0	-222.5	6.7E-09	2.4E-09	2.4E-09
1/7/2025	151140	21.2	0.975	23.90	1.70	0.00	-0.10	0.0	-222.5	1.3E-09	4.7E-10	4.7E-10
1/7/2025	162900	21.2	0.975	23.70	1.70	-0.20	0.00	#DIV/0!	-222.4	1.7E-08	6.2E-09	6.2E-09
1/7/2025	253620	21.2	0.975	23.60	1.80	-0.10	-0.10	1.0	-222.3	2.2E-09	7.8E-10	7.8E-10
1/7/2025	434700	21.2	0.975	23.40	2.00	-0.20	-0.20	1.0	-222.1	2.2E-09	7.8E-10	7.8E-10
1/7/2025	509400	21.2	0.975	23.30	2.10	-0.10	-0.10	1.0	-222.0	2.6E-09	9.5E-10	9.5E-10

Permeant liquid: Tap Water

k_{avg} = 8.4E-10 cm/s

Reviewed by: C. Cunningham

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	5.8	1.8	5.9	7.7	21.4	57.4

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.75	100.0		
.5	97.2		
.375	96.2		
#4	94.2		
#10	92.4		
#20	90.1		
#40	86.5		
#80	83.0		
#100	82.0		
#200	78.8		
0.0226 mm.	72.2		
0.0147 mm.	69.3		
0.0088 mm.	65.1		
0.0065 mm.	60.9		
0.0047 mm.	56.7		
0.0025 mm.	46.8		
0.0011 mm.	36.8		

* (no specification provided)

Material Description

Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)

PL= 18 LL= 63 PI= 45

Classification

USCS (D 2487)= CH AASHTO (M 145)= A-7-6(36)

Coefficients

D₉₀= 0.8225 D₈₅= 0.2913 D₆₀= 0.0061
D₅₀= 0.0031 D₃₀= C_u=
D₁₀= C_c=

Remarks

Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/20/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: B-2

Sample Number: 38308

Depth: 25.0-26.5 ft

Date Sampled: 12/05/2024



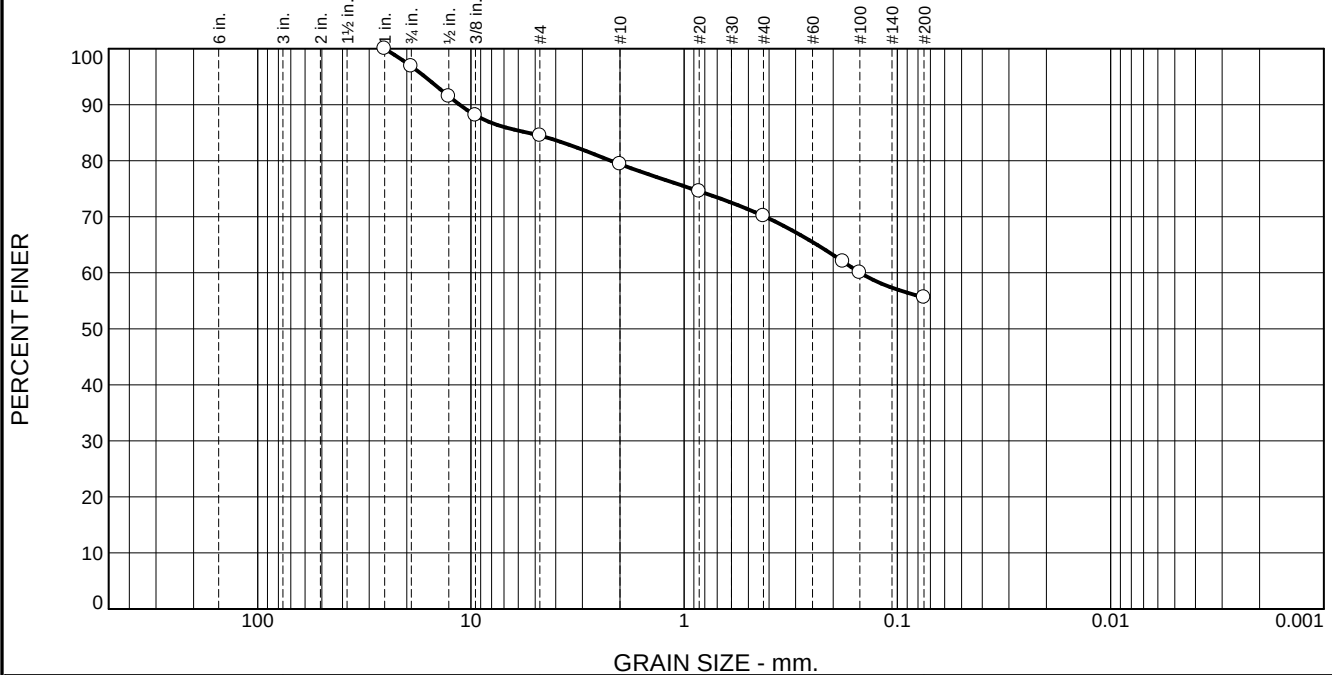
Client: Jackola Engineering

Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	3.1	12.4	5.1	9.3	14.5	55.6	

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
1	100.0		
.75	96.9		
.5	91.5		
.375	88.1		
#4	84.5		
#10	79.4		
#20	74.5		
#40	70.1		
#80	62.0		
#100	60.0		
#200	55.6		

* (no specification provided)

Material Description
Sandy Fat CLAY with Gravel

Atterberg Limits (ASTM D 4318)
PL= 21 LL= 59 PI= 38

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(18)

Coefficients
D₉₀= 11.3003 D₈₅= 5.4149 D₆₀= 0.1495
D₅₀= D₃₀= D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/17/2024
Tested By: SS
Checked By: CC
Title: Lab Supervisor

Location: B-3
Sample Number: 38309 Depth: 2.5-4.0 ft

Date Sampled: 12/05/2024

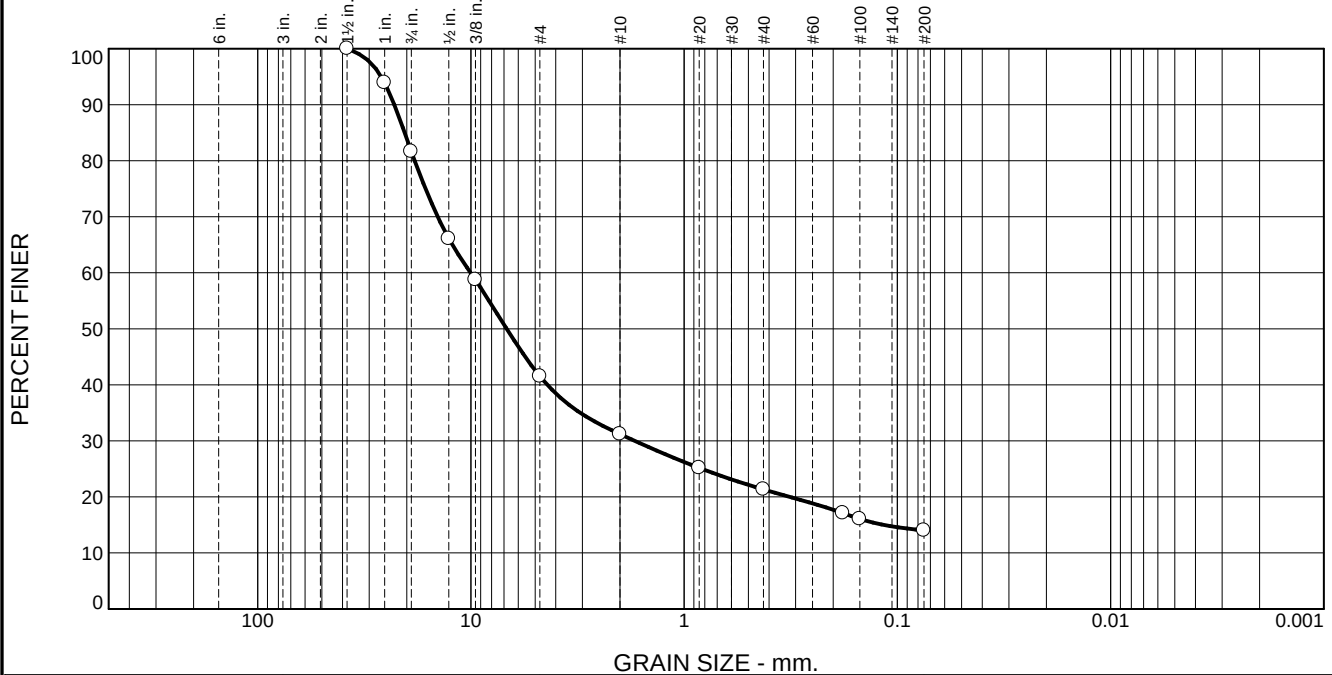


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	18.3	40.2	10.3	9.9	7.3	14.0	

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
1.5	100.0		
1	93.9		
.75	81.7		
.5	66.1		
.375	58.7		
#4	41.5		
#10	31.2		
#20	25.2		
#40	21.3		
#80	17.1		
#100	16.1		
#200	14.0		

* (no specification provided)

Material Description
Clayey GRAVEL with Sand

Atterberg Limits (ASTM D 4318)
PL= 15 LL= 43 PI= 28

Classification
USCS (D 2487)= GC AASHTO (M 145)= A-2-7(0)

Coefficients
D₉₀= 22.8794 D₈₅= 20.4704 D₆₀= 10.0355
D₅₀= 6.7972 D₃₀= 1.7056 D₁₅= 0.1162
D₁₀= Cu= Cc=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/17/2024
Tested By: SS
Checked By: CC
Title: Lab Supervisor

Location: B-3
Sample Number: 38311 Depth: 6.0-7.5 ft

Date Sampled: 12/05/2024

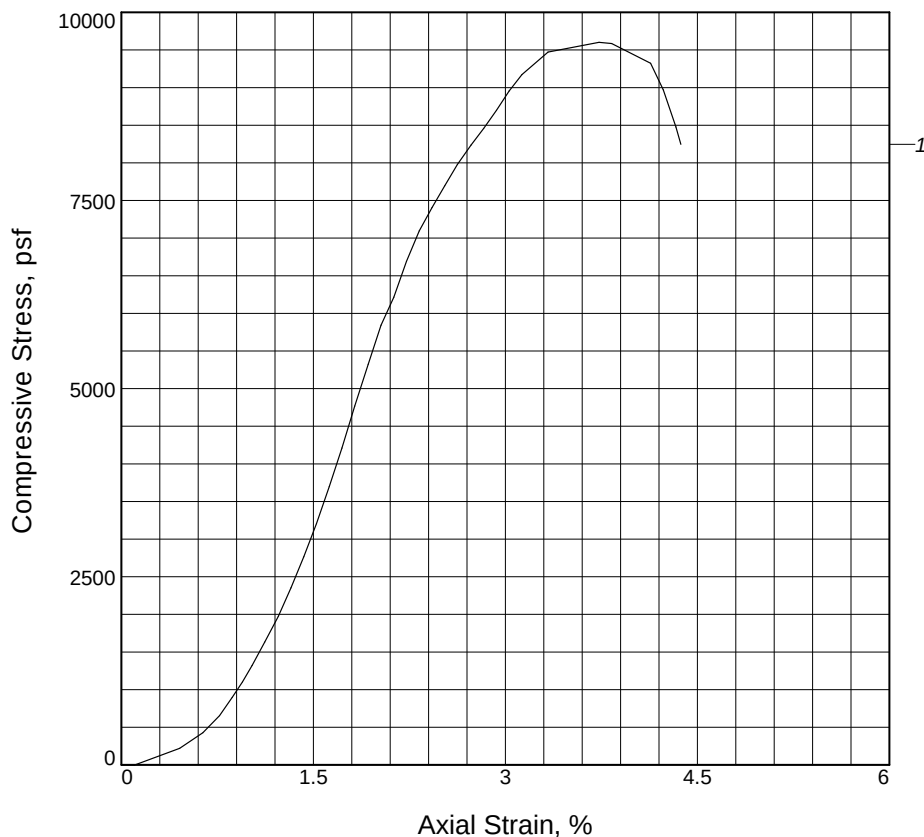


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	9602			
Undrained shear strength, psf	4801			
Failure strain, %	3.7			
Strain rate, %/min.	0.03			
Water content, %	16.3			
Wet density, pcf	134.3			
Dry density, pcf	115.5			
Saturation, %	99.9			
Void ratio	0.4322			
Specimen diameter, in.	1.96			
Specimen height, in.	4.13			
Height/diameter ratio	2.11			

Description: Shale

LL = **PL =** **PI =** **Assumed GS=** 2.65 **Type:** Intact

Project No.: 4691.12652.01

Date Sampled: 12/5/24

Remarks:

Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-3

Sample Number: 38313

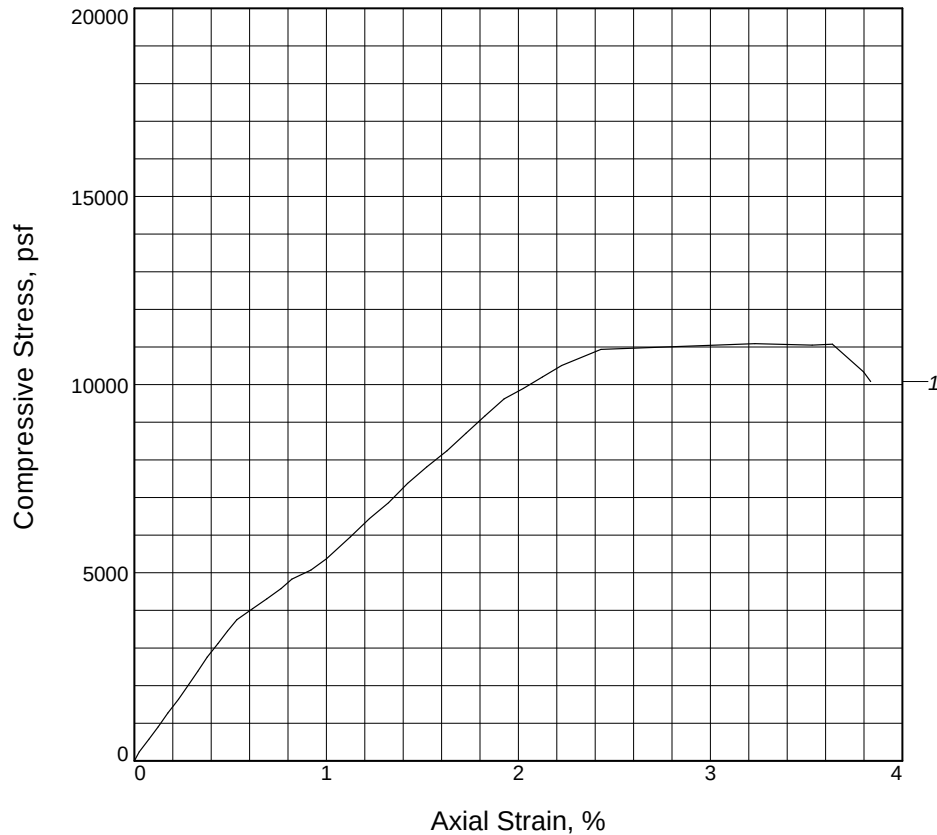
Depth: 10.4-11.3 ft

Figure _____



Tested By: SS **Checked By:** CC

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	11089			
Undrained shear strength, psf	5544			
Failure strain, %	3.2			
Strain rate, %/min.	0.03			
Water content, %	16.1			
Wet density, pcf	134.3			
Dry density, pcf	115.7			
Saturation, %	99.3			
Void ratio	0.4302			
Specimen diameter, in.	1.98			
Specimen height, in.	4.02			
Height/diameter ratio	2.03			

Description: Shale

LL =	PL =	PI =	Assumed GS= 2.65	Type: Intact
------	------	------	------------------	--------------

Project No.: 4691.12652.01

Date Sampled: 12/5/24

Remarks:
Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-3

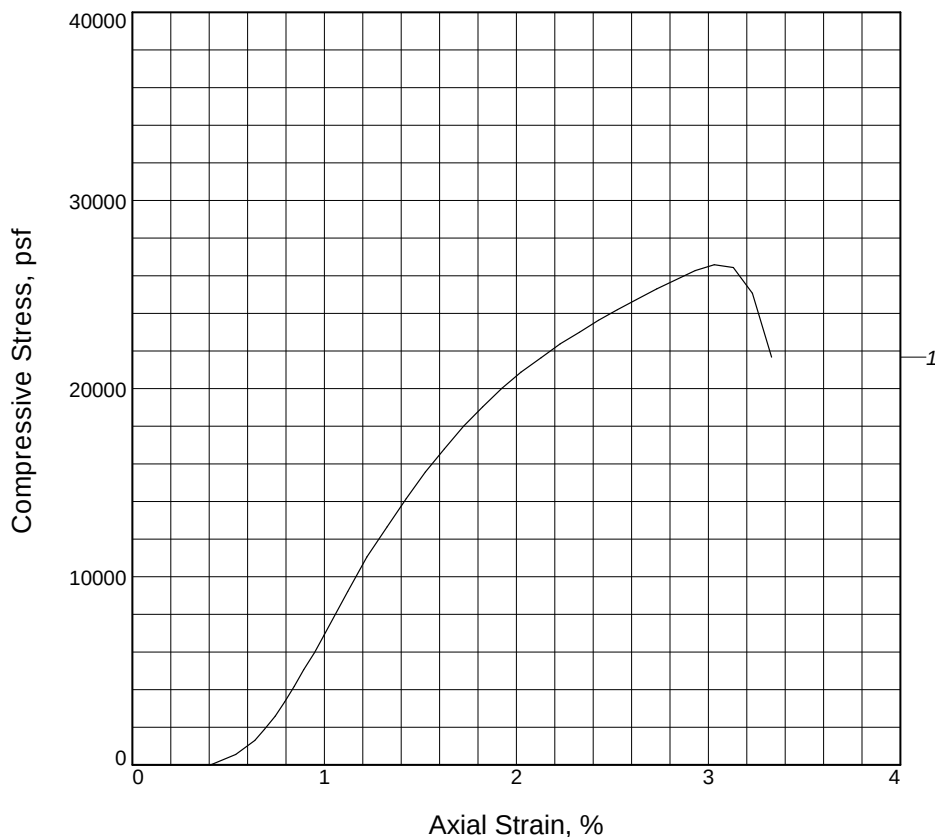
Sample Number: 38317 **Depth:** 16.5-17.2 ft

Figure _____



Tested By: SS **Checked By:** CC

UNCONFINED COMPRESSION TEST



Sample No.	1			
Unconfined strength, psf	26586			
Undrained shear strength, psf	13293			
Failure strain, %	3.0			
Strain rate, %/min.	0.03			
Water content, %	15.1			
Wet density, pcf	138.2			
Dry density, pcf	120.1			
Saturation, %	96.5			
Void ratio	0.4289			
Specimen diameter, in.	1.98			
Specimen height, in.	3.99			
Height/diameter ratio	2.02			

Description: Shale

LL = **PL =** **PI =** **Assumed GS=** 2.75 **Type:** Intact

Project No.: 4691.12652.01

Date Sampled: 12/5/24

Remarks:

Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: B-3

Sample Number: 38321

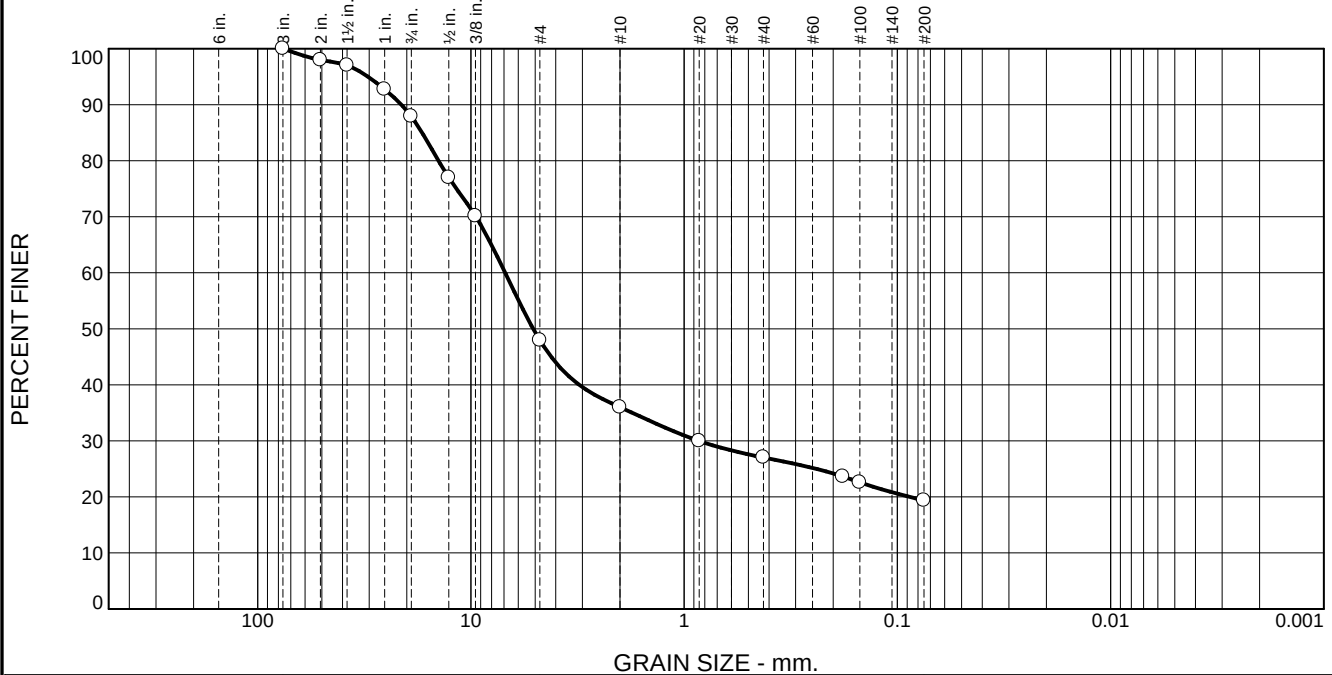
Depth: 24.0-24.7 ft

Figure _____



Tested By: SS **Checked By:** CC

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	12.1	40.0	11.9	9.0	7.6	19.4	

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
3	100.0		
2	98.0		
1.5	97.0		
1	92.7		
.75	87.9		
.5	77.0		
.375	70.1		
#4	47.9		
#10	36.0		
#20	30.0		
#40	27.0		
#80	23.6		
#100	22.6		
#200	19.4		

* (no specification provided)

Material Description
Clayey GRAVEL with Sand

Atterberg Limits (ASTM D 4318)
PL= 15 LL= 46 PI= 31

Classification
USCS (D 2487)= GC AASHTO (M 145)= A-2-7(1)

Coefficients
D₉₀= 21.1926 D₈₅= 16.9517 D₆₀= 6.9093
D₅₀= 5.1076 D₃₀= 0.8537 D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/17/2024
Tested By: SS
Checked By: CC
Title: Lab Supervisor

Location: TP-1
Sample Number: 38323 Depth: 5.0-10.0 ft

Date Sampled: 12/04/2024

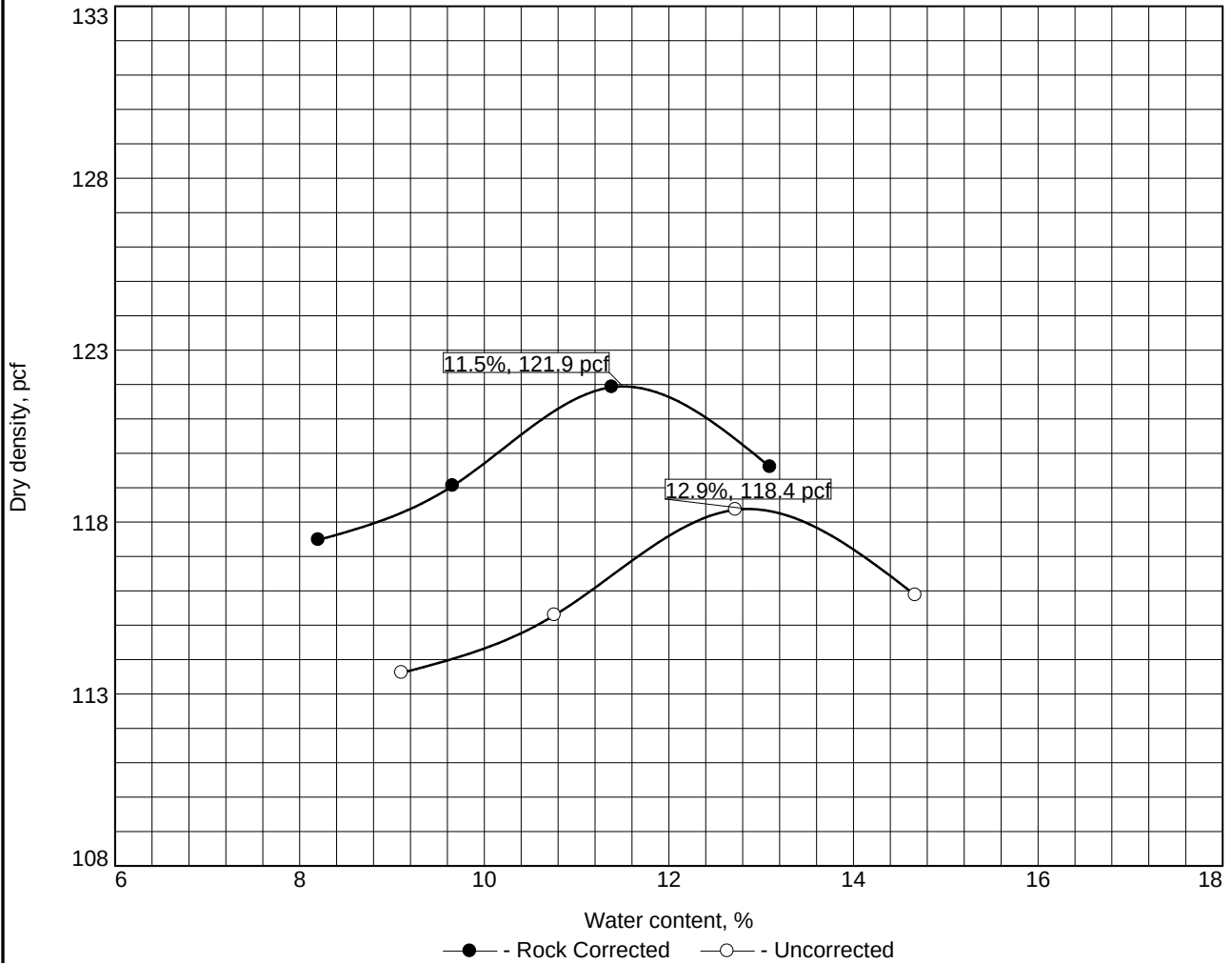


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

COMPACTION TEST REPORT



Test specification: ASTM D 698-12 Method C Standard
 ASTM D4718-15 Oversize Corr. Applied to Each Test Point

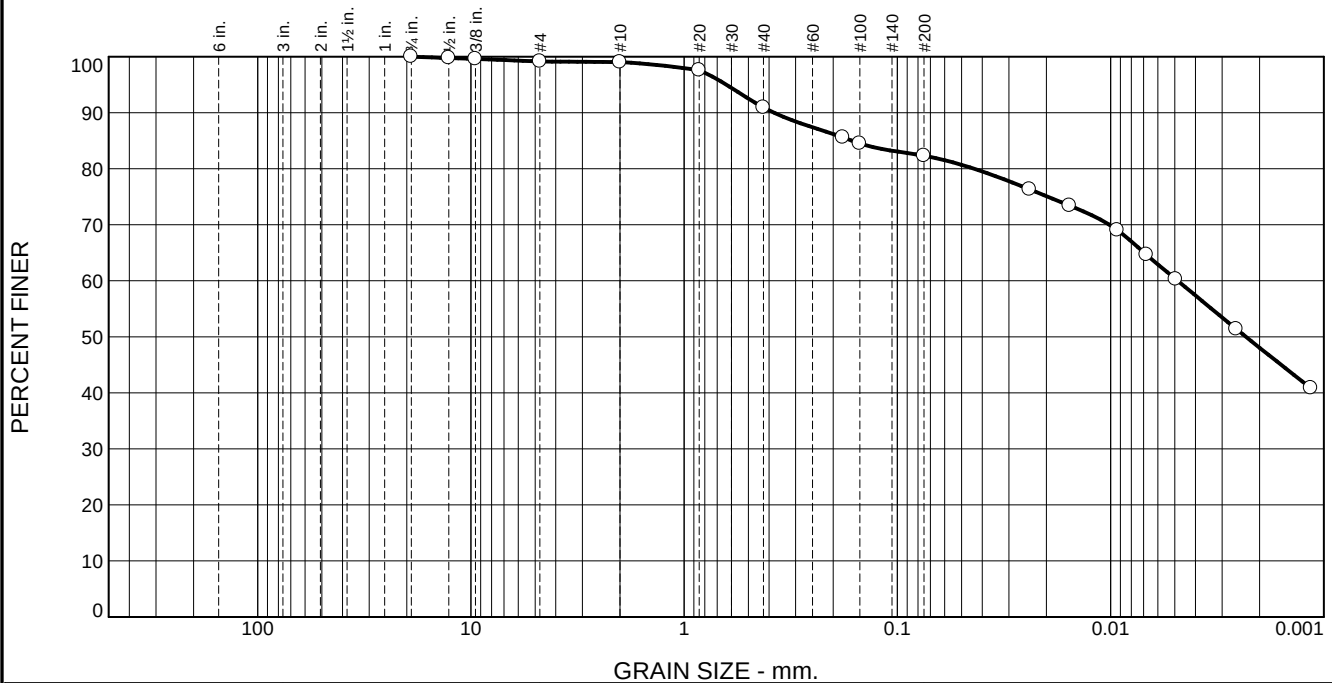
Elev/ Depth	Classification		Nat. Moist.	Sp.G.	LL	PI	% > 3/4 in.	% < No.200
	USCS	AASHTO						
5.0-10.0 ft	GC	A-2-7(1)	6.0	2.500	46	31	12.1	19.4

ROCK CORRECTED TEST RESULTS		UNCORRECTED	MATERIAL DESCRIPTION
Maximum dry density = 121.9 pcf		118.4 pcf	Clayey GRAVEL with Sand
Optimum moisture = 11.5 %		12.9 %	
Project No. 4691.12652.01 Client: Jackola Engineering Project: Bar Island Dam Modifications			Remarks: Sampled by DOWL
○ Location: TP-1 Sample Number: 38323			
			Figure

Figure

Tested By: SS Checked By: CC

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	0.9	0.1	8.1	8.6	21.9	60.4

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
.75	100.0		
.5	99.8		
.375	99.6		
#4	99.1		
#10	99.0		
#20	97.6		
#40	90.9		
#80	85.6		
#100	84.5		
#200	82.3		
0.0240 mm.	76.3		
0.0156 mm.	73.4		
0.0093 mm.	69.0		
0.0068 mm.	64.7		
0.0049 mm.	60.3		
0.0026 mm.	51.4		
0.0011 mm.	40.8		

* (no specification provided)

Material Description
Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)
PL= 20 LL= 55 PI= 35

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(30)

Coefficients
D₉₀= 0.3793 D₈₅= 0.1634 D₆₀= 0.0048
D₅₀= 0.0023 D₃₀= D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 **Date Tested:** 12/30/2024

Tested By: SS

Checked By: CC

Title: Laboratory Supervisor

Location: TP-2
Sample Number: 38322

Depth: 5.0-10.0 ft

Date Sampled: 12/04/2024



Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

DISPERSIVE CHARACTERISTICS OF CLAYEY SOILS

Double Hydrometer, Crumb & Pinhole

ASTM D6572/D4221/D4647

Project: 4691.12652.01

Lab No: 38322

Project Name: Bar Island Dam Mod.

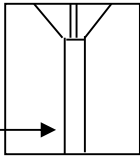
Date Sampled/By: 12/4/24 DD

Sample Location: TP-2 5.0-10.0

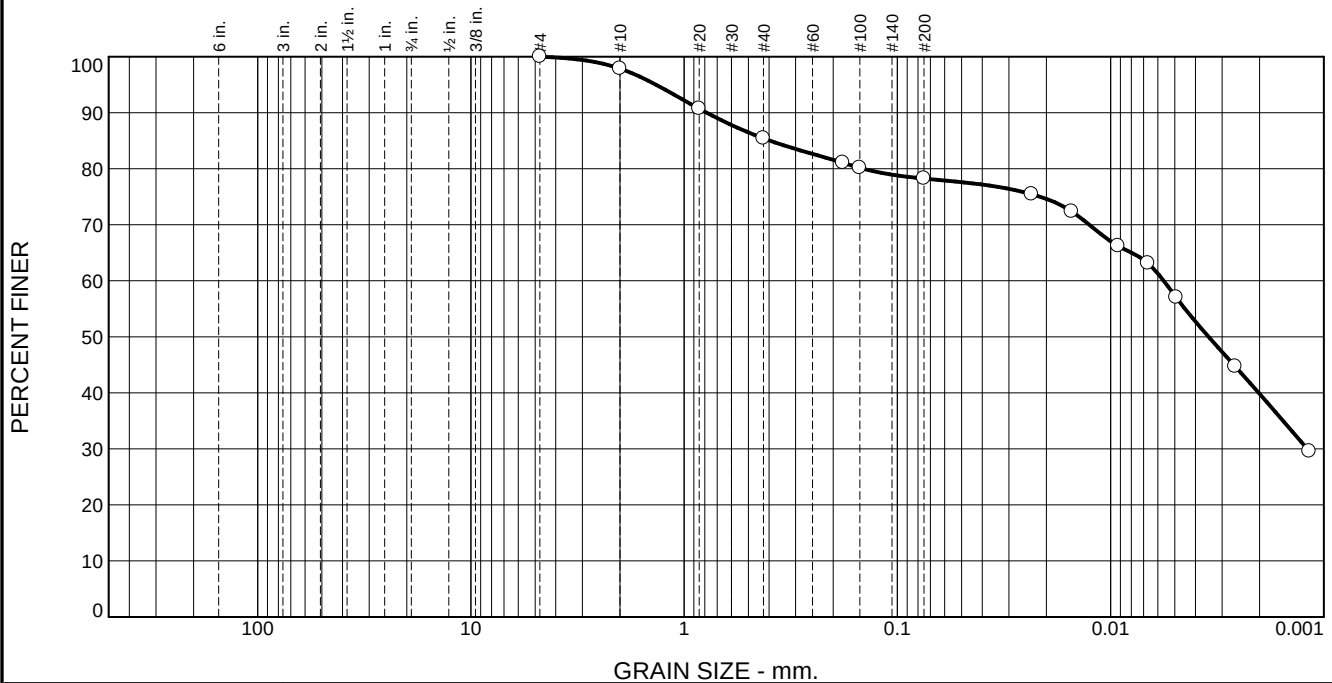
Date Test/By: 1/7/25 SS

Other:

Date Check/By: 1/7/25 CC

PINHOLE DISPERSION		GRADATION		ATTERBERG LIMITS	
Final Hole Diameter 3.0 mm  Specimen after Test		.75	100.0	PL	20
		.5	99.8	LL	55
		.375	99.6	PI	35
		#4	99.1		
		#10	99.0		
		#20	97.6		
		#40	90.9		
		#80	85.6		
		#100	84.5		
		#200	82.3		
		0.0240 mm.	76.3		
		0.0156 mm.	73.4		
		0.0093 mm.	69.0		
		0.0068 mm.	64.7		
		0.0049 mm.	60.3		
		0.0026 mm.	51.4		
		0.0011 mm.	40.8		
SOIL SPECIMAN DATA Nat. Moisture 19.5 Curing Time 24 Hours % Compaction 95% Test Method A				CLASSIFICATION USCS CH AASHTO A-7-6(30)	
DISPERSIVE CLASSIFICATION D1				DESCRIPTION Fat CLAY with sand	
CRUMB DISPERSION Type of Test ☒ Natural ☐ Remolded		PHOTOGRAPH OF CRUMB TEST AFTER 6 HOURS 			
Nat. Moisture 19.5 Curing Time 24 Hours					
DISPERSIVE CLASSIFICATION ☐ Non Dispersive ☐ Dispersive ☒ Intermediate ☐ Highly Dispersive					

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	0.0	2.1	12.5	7.2	20.8	57.4

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
#4	100.0		
#10	97.9		
#20	90.7		
#40	85.4		
#80	81.1		
#100	80.2		
#200	78.2		
0.0234 mm.	75.4		
0.0152 mm.	72.4		
0.0092 mm.	66.2		
0.0067 mm.	63.1		
0.0049 mm.	57.0		
0.0026 mm.	44.7		
0.0012 mm.	29.6		

* (no specification provided)

Material Description
Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)
PL= 19 LL= 56 PI= 37

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(29)

Coefficients
D₉₀= 0.7841 D₈₅= 0.3958 D₆₀= 0.0056
D₅₀= 0.0035 D₃₀= 0.0012 D₁₅=
D₁₀= C_u= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 Date Tested: 12/23/2024
Tested By: SS
Checked By: CC
Title: Laboratory Supervisor

Location: TP-3
Sample Number: 38324 Depth: 5.0-10.0 ft

Date Sampled: 12/04/2024

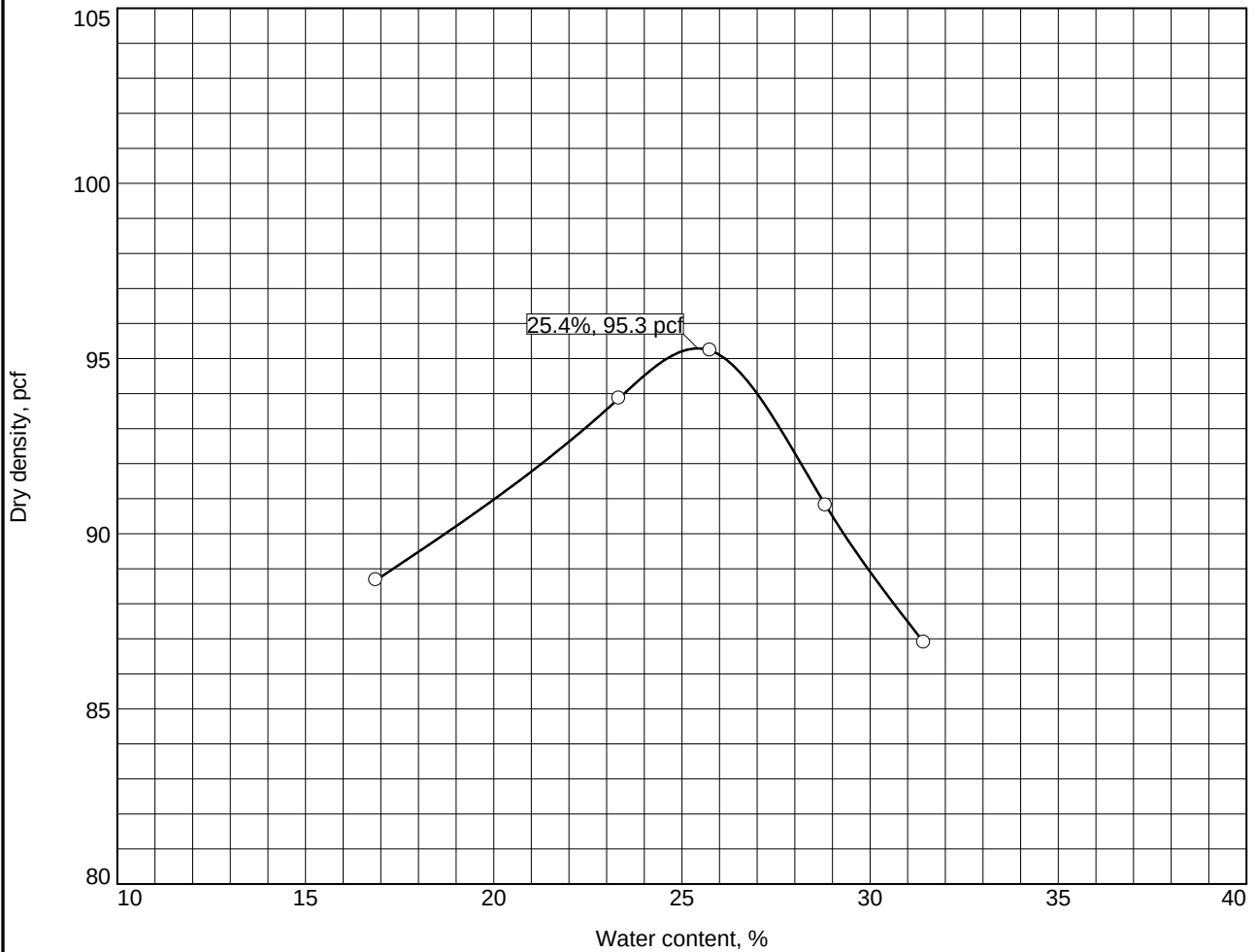


Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure

COMPACTION TEST REPORT



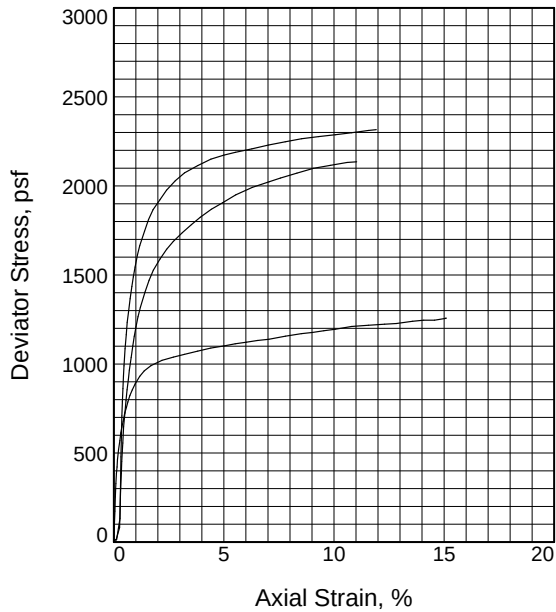
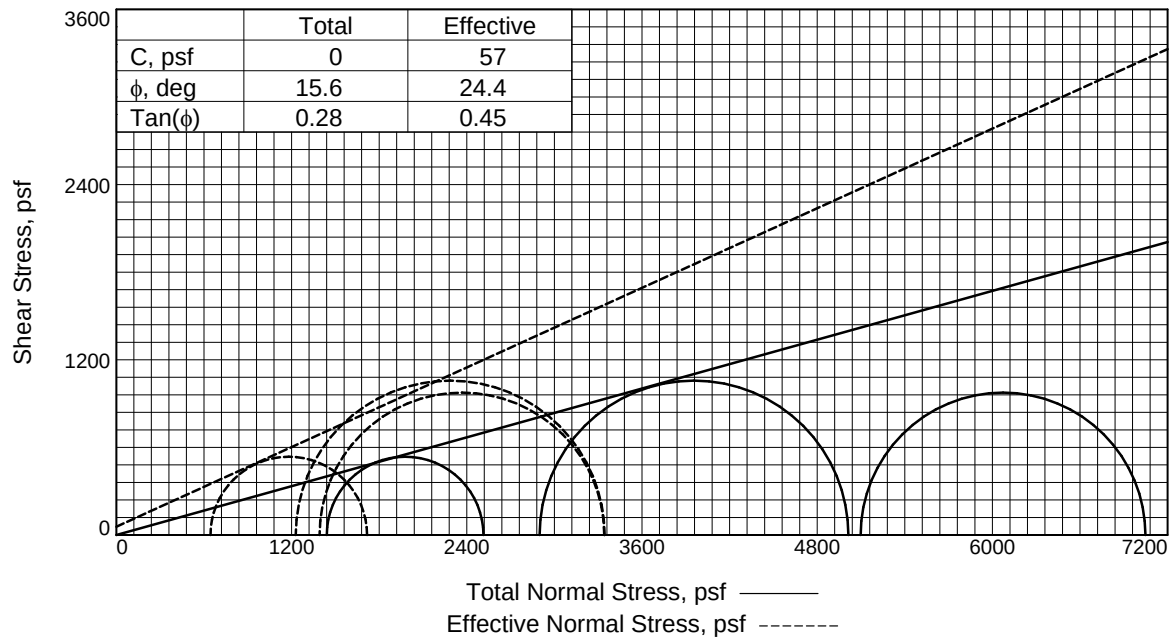
Test specification: ASTM D 698-12 Method A Standard

Elev/ Depth	Classification		Nat. Moist.	Sp.G.	LL	PI	% > #4	% < No.200
	USCS	AASHTO						
5.0-10.0 ft	CH	A-7-6(29)	16.8		56	37	0.0	78.2

TEST RESULTS		MATERIAL DESCRIPTION
Maximum dry density = 95.3 pcf		Fat CLAY with Sand
Optimum moisture = 25.4 %		
Project No. 4691.12652.01 Client: Jackola Engineering Project: Bar Island Dam Modifications		Remarks: Sampled by DOWL
Location: TP-3 Sample Number: 38324		
		
		Figure

Figure

Tested By: SS Checked By: CC



Sample No.		1	2	3
Initial	Water Content, %	26.1	25.9	25.9
	Dry Density, pcf	89.4	92.1	91.3
	Saturation, %	81.5	86.4	84.6
	Void Ratio	0.8495	0.7955	0.8116
	Diameter, in.	2.88	2.88	2.88
	Height, in.	6.02	5.85	5.91
At Test	Water Content, %	31.0	29.1	29.3
	Dry Density, pcf	90.8	93.4	93.2
	Saturation, %	100.0	100.0	100.0
	Void Ratio	0.8228	0.7707	0.7759
	Diameter, in.	2.87	2.87	2.86
	Height, in.	5.99	5.82	5.87
Strain rate, %/min.		0.0010	0.0010	0.0010
Back Pressure, psi		43.33	43.59	47.72
Cell Pressure, psi		53.34	63.72	83.12
Fail. Stress, psf		1073	2115	1950
Total Pore Pr., psf		7036	7947	10578
Ult. Stress, psf				
Total Pore Pr., psf				
$\bar{\sigma}_1$ Failure, psf		1717	3343	3342
$\bar{\sigma}_3$ Failure, psf		645	1229	1392

Type of Test:

CU with Pore Pressures

Sample Type: Remolded

Description: Fat CLAY with Sand

LL= 56 PL= 19 PI= 37

Assumed Specific Gravity= 2.65

Remarks: Sampled by DOWL

Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: TP-3

Sample Number: 38324

Depth: 5.0-10.0 ft

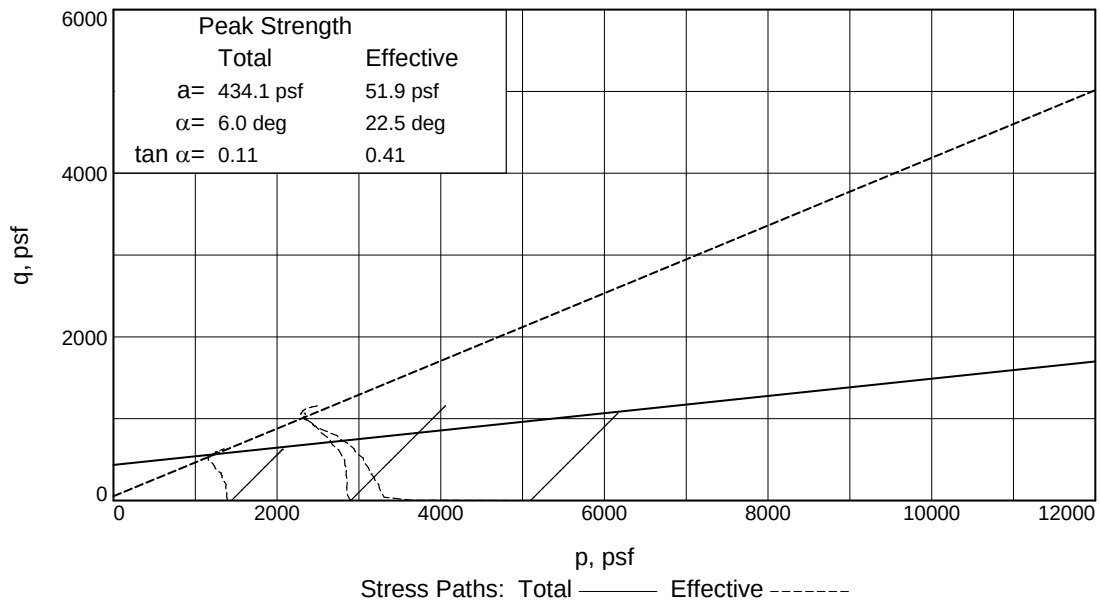
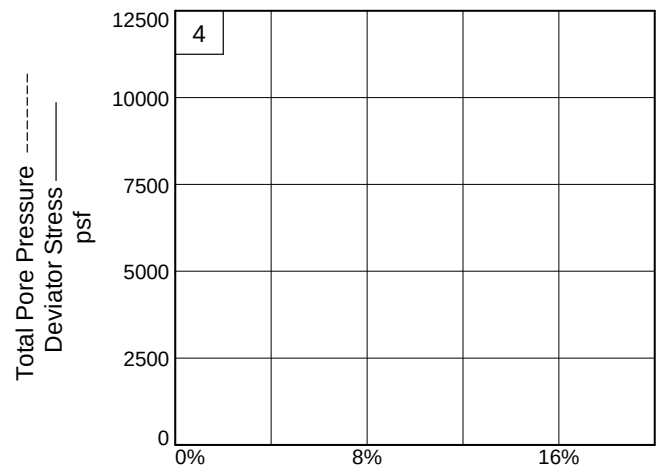
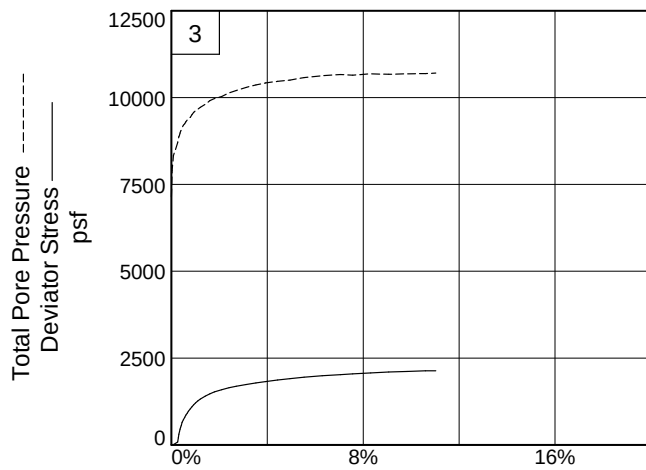
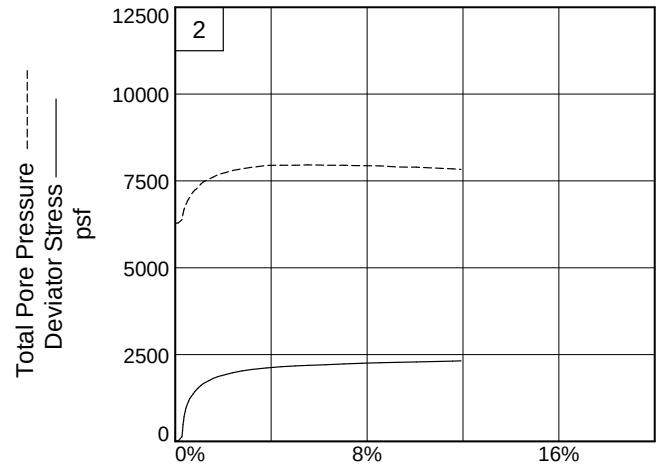
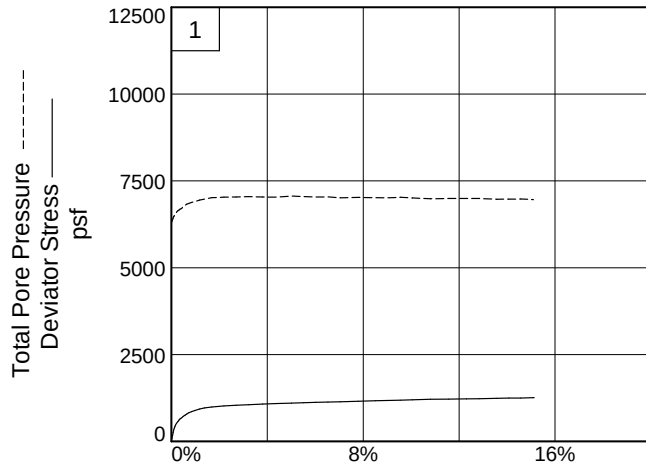
Proj. No.: 4691.12652.01

Date Sampled: 12/04/2024



Figure _____

Tested By: CC **Checked By:** DR



Client: Jackola Engineering

Project: Bar Island Dam Modifications

Location: TP-3 **Depth:** 5.0-10.0 ft

Sample Number: 38324

Project No.: 4691.12652.01

Figure _____

DOWL HKM

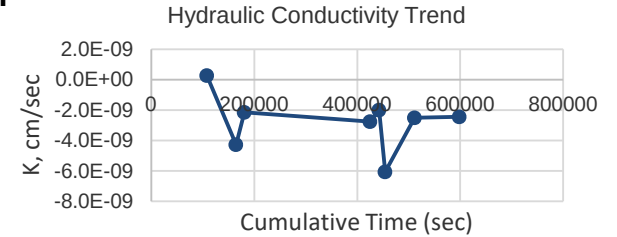
Tested By: CC

Checked By: DR



222 N. 32nd Street, Suite 700
Billings, Montana 59101

FLEXIBLE-WALL HYDRAULIC CONDUCTIVITY ASTM D5084 (Method C)



Project No.: 4691.12652.01

Client: Jackola Engineering

Sample ID: 38324

Project: Bar Island Dam Modifications

In Tact

Tested by: CC

Sample Description: Fat CLAY with sand

Date Tested: 1/22/2025

Burette Area: 0.880 cm²

Sample Location: TP-3 5.0-10.0 ft

Assumed SG: 2.65

Notes: _____

Effective Consolidation Stress 18.5 psi

Chamber 63.0 psi

Hydraulic Gradient -14.2

Upper 46 psi

System Constant, C 0.440 (cm²)

Lower 43 psi

Confining Stress 18.5 psi

	Initial	Final	
Mass	<u>1160.7</u>	<u>1160.7</u>	g
Moisture Content	<u>25.9%</u>	<u>25.9%</u>	
Dry Density	<u>92.1</u>	<u>92.2</u>	pcf
Diameter	<u>2.88</u>	<u>7.32</u>	cm
Length	<u>5.85</u>	<u>14.86</u>	cm
Area	<u>42.03</u>	<u>42.03</u>	cm ²

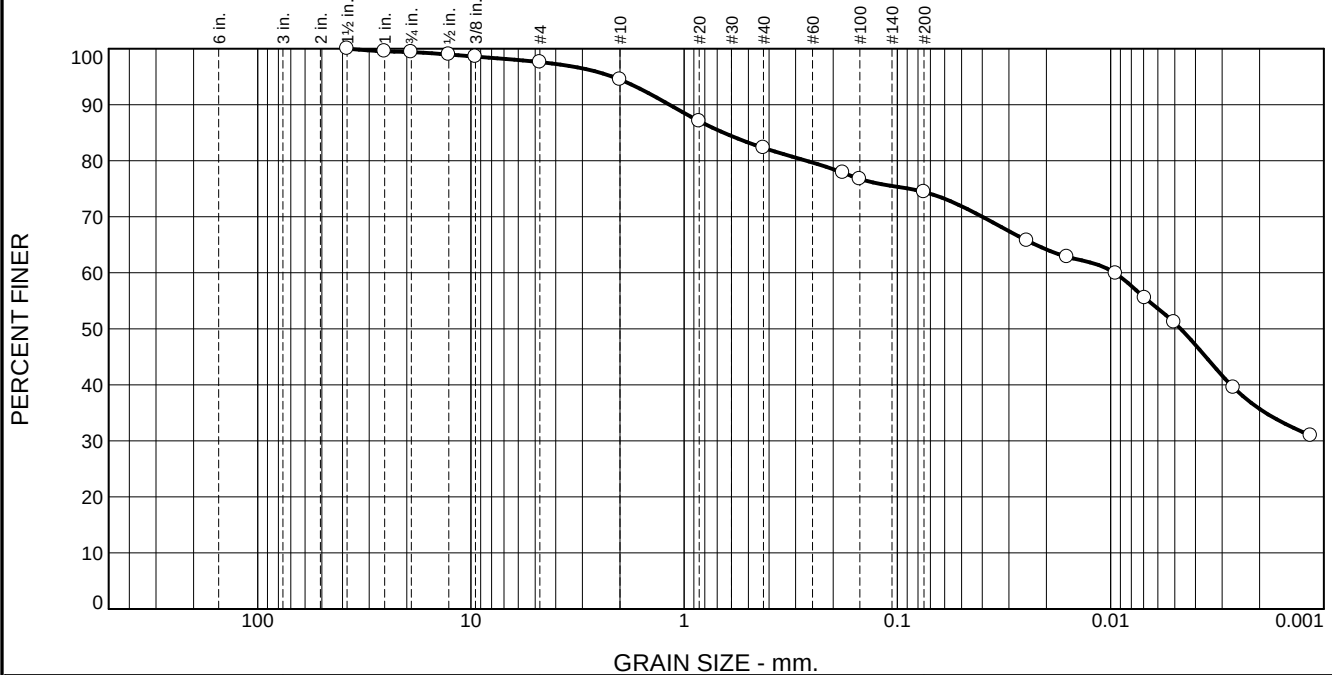
Date	Time (s)	Temperature		Burette Readings		Incremental Flow			Head Difference ¹ h (t)	Preliminary ² kA/L (cm ² /sec)	Final ³ k ₂₀ (x10 ⁻⁷) (cm/sec)	ASTM D5084 Method C k ₂₀ (cm/sec)
		°C	R _T	Lower V _U (t) (cc)	Upper V _L (t) (cc)	Inflow (lower) (cc)	Outflow (upper) (cc)	Ratio In/Out				
1/23/2025	<u>79140</u>	<u>21.2</u>	0.975	<u>9.10</u>	<u>16.00</u>				-208.1			
1/23/2025	<u>99900</u>	<u>21.2</u>	0.975	<u>9.00</u>	<u>15.50</u>	-0.10	0.50	-0.2	-208.3	-1.9E-08	-6.6E-09	-6.6E-09
1/23/2025	<u>108180</u>	<u>21.2</u>	0.975	<u>8.90</u>	<u>15.40</u>	-0.10	0.10	-1.0	-208.3	7.7E-10	2.6E-10	2.6E-10
1/24/2025	<u>164700</u>	<u>21.2</u>	0.975	<u>8.80</u>	<u>14.60</u>	-0.10	0.80	-0.1	-208.6	-1.2E-08	-4.3E-09	-4.3E-09
1/24/2025	<u>180840</u>	<u>21.2</u>	0.975	<u>8.80</u>	<u>14.50</u>	0.00	0.10	0.0	-208.6	-6.3E-09	-2.2E-09	-2.2E-09
1/27/2025	<u>425040</u>	<u>21.2</u>	0.975	<u>9.40</u>	<u>13.20</u>	0.60	1.30	0.5	-209.6	-8.0E-09	-2.8E-09	-2.8E-09
1/27/2025	<u>442320</u>	<u>21.2</u>	0.975	<u>9.40</u>	<u>13.10</u>	0.00	0.10	0.0	-209.6	-5.8E-09	-2.0E-09	-2.0E-09
1/27/2025	<u>454140</u>	<u>21.2</u>	0.975	<u>9.50</u>	<u>13.00</u>	0.10	0.10	1.0	-209.7	-1.8E-08	-6.1E-09	-6.1E-09
1/28/2025	<u>511260</u>	<u>21.2</u>	0.975	<u>9.70</u>	<u>12.80</u>	0.20	0.20	1.0	-209.9	-7.3E-09	-2.5E-09	-2.5E-09
1/29/2025	<u>598140</u>	<u>21.2</u>	0.975	<u>9.90</u>	<u>12.40</u>	0.20	0.40	0.5	-210.2	-7.1E-09	-2.4E-09	-2.4E-09

Permeant liquid: Tap Water

kavg = -3.0E-09 cm/s

Reviewed by: C. Cunningham

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.6	1.8	3.1	12.2	7.9	23.4	51.0

TEST RESULTS			
Opening Size	Percent Finer	Spec.* (Percent)	Pass? (X=Fail)
1.5	100.0		
1	99.5		
.75	99.4		
.5	98.9		
.375	98.6		
#4	97.6		
#10	94.5		
#20	87.1		
#40	82.3		
#80	77.9		
#100	76.8		
#200	74.4		
0.0247 mm.	65.8		
0.0160 mm.	62.8		
0.0095 mm.	59.9		
0.0069 mm.	55.5		
0.0050 mm.	51.2		
0.0027 mm.	39.5		
0.0012 mm.	30.9		

* (no specification provided)

Material Description
Fat CLAY with Sand

Atterberg Limits (ASTM D 4318)
PL= 18 LL= 58 PI= 40

Classification
USCS (D 2487)= CH AASHTO (M 145)= A-7-6(29)

Coefficients
D₉₀= 1.1727 D₈₅= 0.6531 D₆₀= 0.0096
D₅₀= 0.0047 D₃₀= C_u= D₁₅= C_c=

Remarks
Sampled by DOWL

Date Received: 12/06/2024 **Date Tested:** 12/23/2024
Tested By: SS
Checked By: CC
Title: Laboratory Supervisor

Location: TP-4
Sample Number: 38325 **Depth:** 5.0-10.0 ft

Date Sampled: 12/04/2024.



Client: Jackola Engineering
Project: Bar Island Dam Modifications

Project No: 4691.12652.01

Figure



LABORATORY ANALYTICAL REPORT

Prepared by Billings, MT Branch

Client: DOWL
Project: 4691.12652.01 Bar Island Dam

Report Date: 12/20/24

Lab ID: B24120799-001
Client Sample ID: 38288 B-1 15.0-20.0

Collection Date: 12/04/24 12:00
DateReceived: 12/10/24
Matrix: Soil

Analyses	Result	Units	Qualifiers	RL	MCL/ QCL	Method	Analysis Date / By
SULFATE BY MT DOT METHOD 532							
Sulfate	16100	mg/kg		2		E300.0	12/17/24 19:13 / spb
WATER EXTRACTABLE CONSTITUENTS							
Conductivity, 1:2	4.6	mmhos/cm		0.1		ASA10-3	12/19/24 15:22 / srm
MT DOT 232-16							
pH	7.5	s.u.		0.1		MTDOT 232-1	12/19/24 15:21 / srm
SATURATED PASTE EXTRACT							
Calcium, sat. paste	24.6	meq/L		0.05		SW6010B	12/19/24 18:05 / enb
Magnesium, sat. paste	22.0	meq/L		0.08		SW6010B	12/19/24 18:05 / enb
Potassium, sat. paste	1.36	meq/L		0.08		SW6010B	12/19/24 18:05 / enb
Sodium, sat. paste	66.8	meq/L		0.04		SW6010B	12/19/24 18:05 / enb
RESISTIVITY OF SOIL							
Resistivity	217	ohm-cm		1		A2510 B	12/19/24 15:22 / srm

Lab ID: B24120799-002
Client Sample ID: 38322 TP-2 5.0-10.0

Collection Date: 12/04/24 12:00
DateReceived: 12/10/24
Matrix: Soil

Analyses	Result	Units	Qualifiers	RL	MCL/ QCL	Method	Analysis Date / By
SULFATE BY MT DOT METHOD 532							
Sulfate	25600	mg/kg		2		E300.0	12/17/24 20:21 / spb
WATER EXTRACTABLE CONSTITUENTS							
Conductivity, 1:2	5.9	mmhos/cm		0.1		ASA10-3	12/19/24 15:22 / srm
MT DOT 232-16							
pH	6.5	s.u.		0.1		MTDOT 232-1	12/19/24 15:21 / srm
SATURATED PASTE EXTRACT							
Calcium, sat. paste	22.8	meq/L		0.05		SW6010B	12/19/24 18:06 / enb
Magnesium, sat. paste	31.0	meq/L		0.08		SW6010B	12/19/24 18:06 / enb
Potassium, sat. paste	1.28	meq/L		0.08		SW6010B	12/19/24 18:06 / enb
Sodium, sat. paste	104	meq/L		0.04		SW6010B	12/19/24 18:06 / enb
RESISTIVITY OF SOIL							
Resistivity	168	ohm-cm		1		A2510 B	12/19/24 15:22 / srm

Report RL - Analyte Reporting Limit
Definitions: QCL - Quality Control Limit

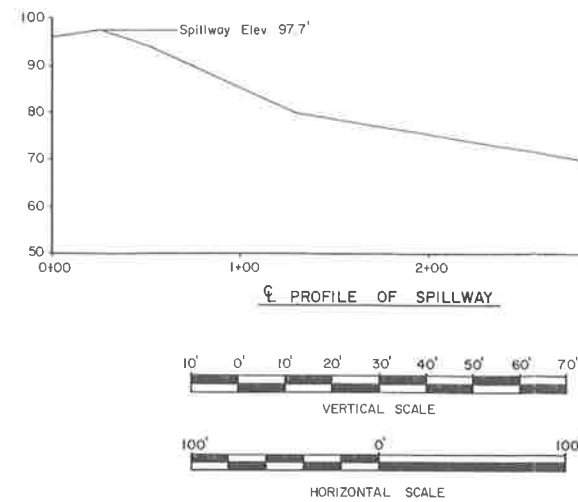
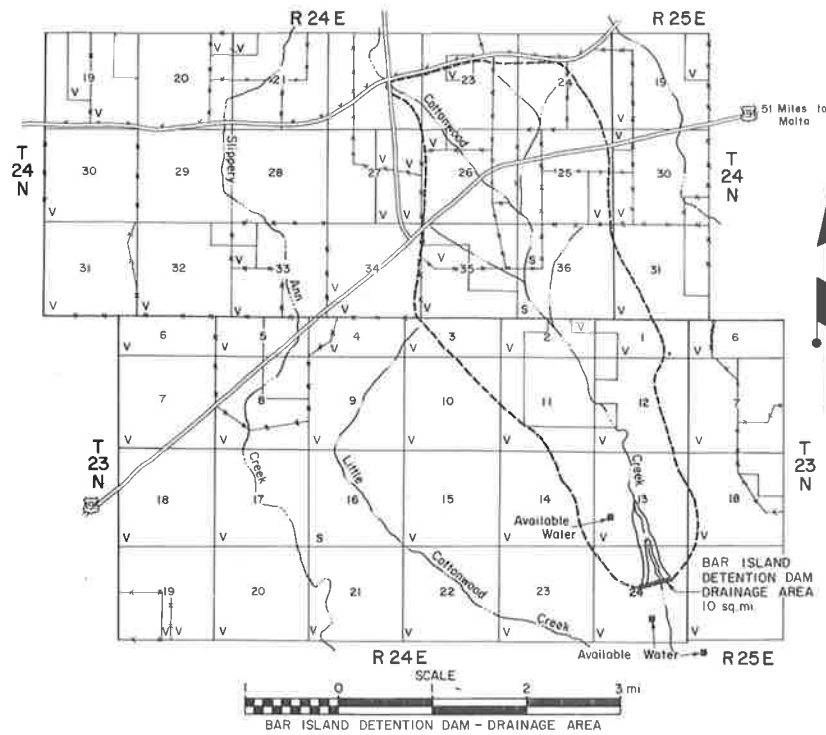
MCL - Maximum Contaminant Level
ND - Not detected at the Reporting Limit (RL)

APPENDIX B

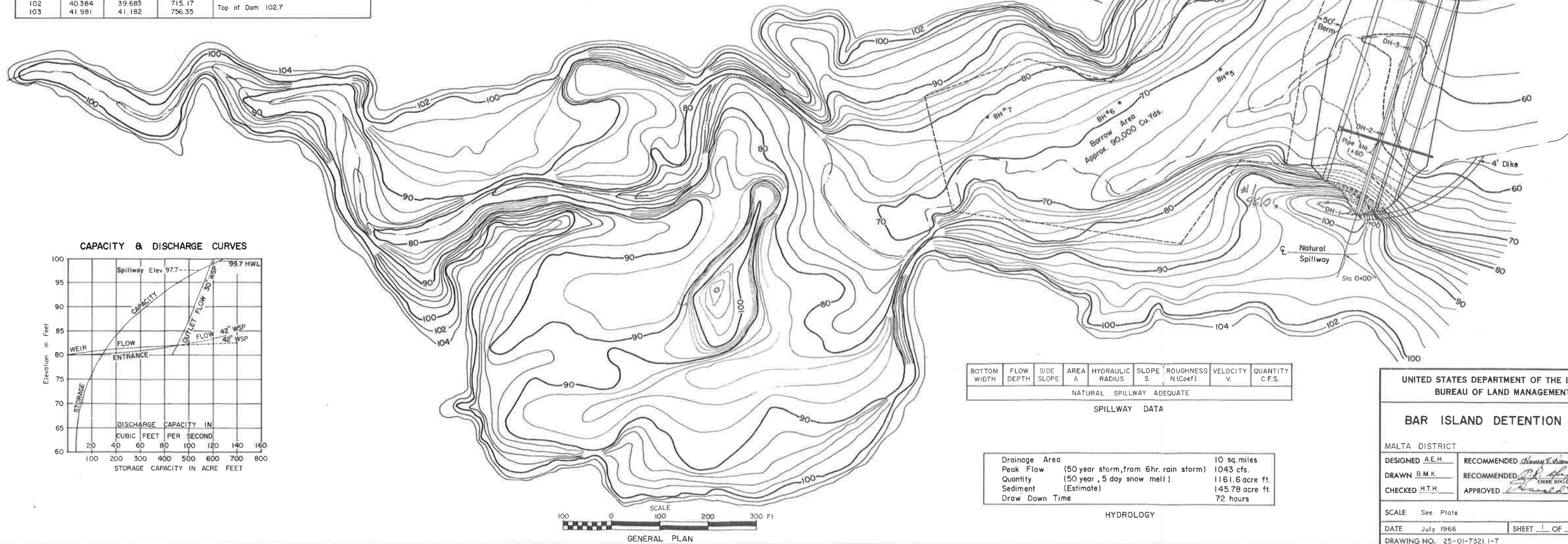
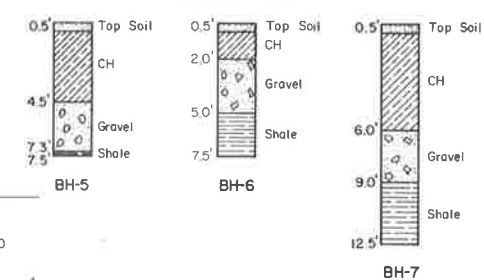
ORIGINAL DRAWINGS

RESERVOIR CAPACITY TABLE

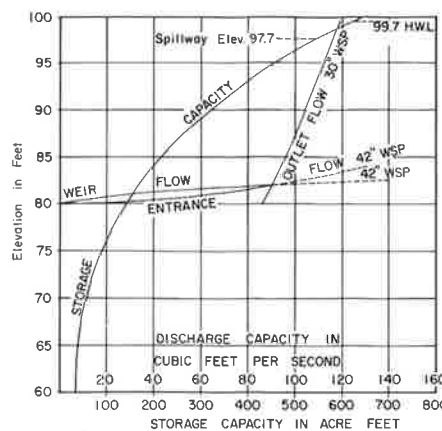
CONTOUR ELEV.	SURFACE ACRES	CAPACITY ACRE FEET	CUM. CAP. ACRE FEET	REMARKS
59	.004		35.000	Add 35 acre feet for Excavation and Channel.
60	.148	.076	35.076	
61	.531	.339	35.422	
62	.990	.760	36.18	
63	1.342	1.166	37.34	
64	1.775	1.558	38.90	
65	2.232	2.004	40.90	
66	2.559	2.395	43.30	
67	3.226	2.893	46.20	
68	4.013	3.620	49.81	
69	4.755	4.384	54.20	Entrance Lip 80.0
70	5.399	5.077	59.27	
71	6.043	5.721	65.00	
72	6.687	6.365	71.36	
73	7.281	6.984	78.34	
74	7.785	7.533	85.88	
75	8.311	8.048	93.92	
76	8.265	8.788	102.71	
77	10.209	9.737	112.45	
78	10.811	10.510	122.96	
79	11.335	11.073	134.03	Spillway 97.7
80	12.166	11.751	145.78	
81	12.945	12.555	158.34	
82	13.757	13.351	171.69	
83	14.409	14.083	185.77	
84	15.311	14.860	200.63	
85	16.330	15.820	216.45	
86	17.905	17.117	233.57	
87	19.716	18.810	252.38	
88	21.147	20.432	272.81	H.W.L. 99.7
89	22.778	21.962	294.77	
90	23.694	23.236	318.01	
91	25.257	24.475	342.48	
92	27.170	26.214	368.70	
93	29.705	28.437	397.13	
94	31.873	30.789	427.92	
95	33.424	32.648	460.57	
96	34.137	33.780	494.35	
97	34.911	34.524	528.88	Top of Dam 102.7
98	35.368	35.140	564.02	
99	36.620	35.994	600.01	
100	37.681	37.150	637.16	
101	38.982	38.331	675.49	
102	40.384	39.683	715.17	
103	41.981	41.182	756.35	



BORROW AREA



CAPACITY & DISCHARGE CURVES



BOTTOM WIDTH	FLOW DEPTH	SIDE SLOPE	AREA A	HYDRAULIC RADIUS	SLOPE S	ROUGHNESS N (Coef)	VELOCITY V	QUANTITY C.F.S.
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NATURAL SPILLWAY ADEQUATE

SPILLWAY DATA

Drainage Area	10 sq. miles
Peak Flow (50 year storm, from 6 hr. rain storm)	1043 cfs.
Quantity (50 year, 5 day snow melt)	1161.6 acre ft.
Sediment (Estimate)	145.78 acre ft.
Draw Down Time	72 hours

HYDROLOGY

UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

BAR ISLAND DETENTION DAM

MALTA DISTRICT MONTANA

DESIGNED A.E.H. RECOMMENDED Henry F. Hammerstein

DRAWN B.M.K. RECOMMENDED J. H. H. H.

CHECKED H.T.H. APPROVED

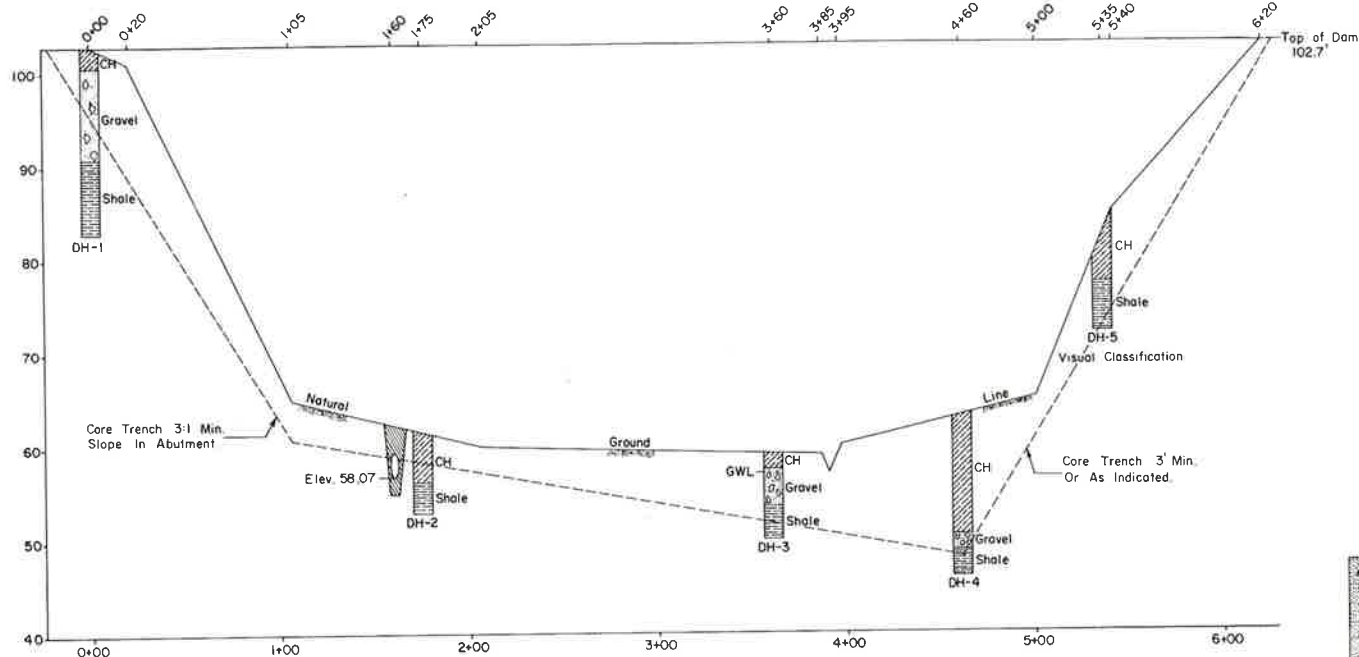
SCALE See Plate

DATE July 1966 SHEET 1 OF 2

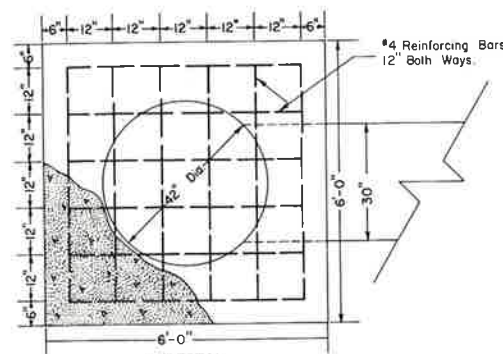
DRAWING NO. 25-01-7321.1-7



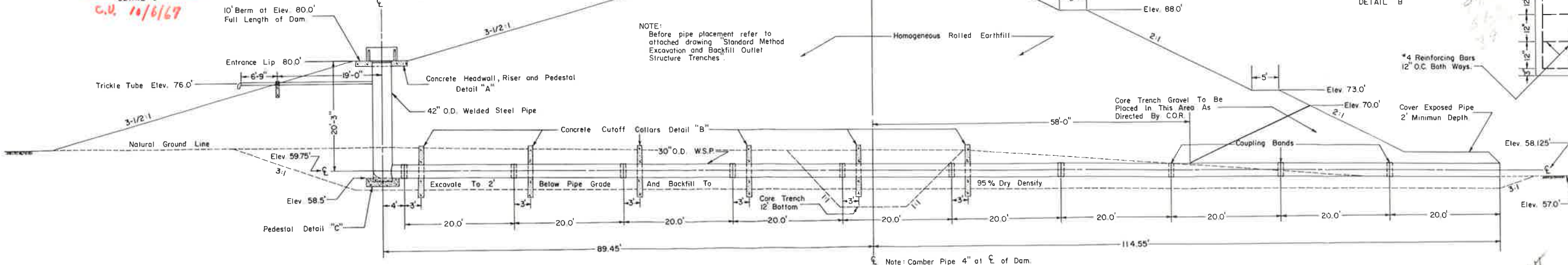
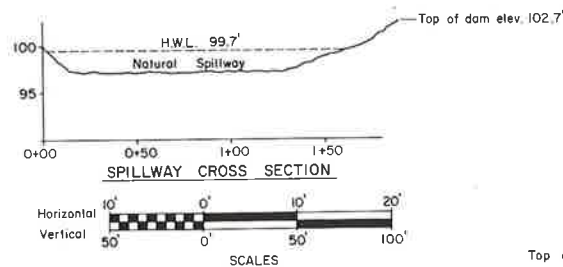
NOTE: Camber Top of Dam 6"



PROFILE OF DAM



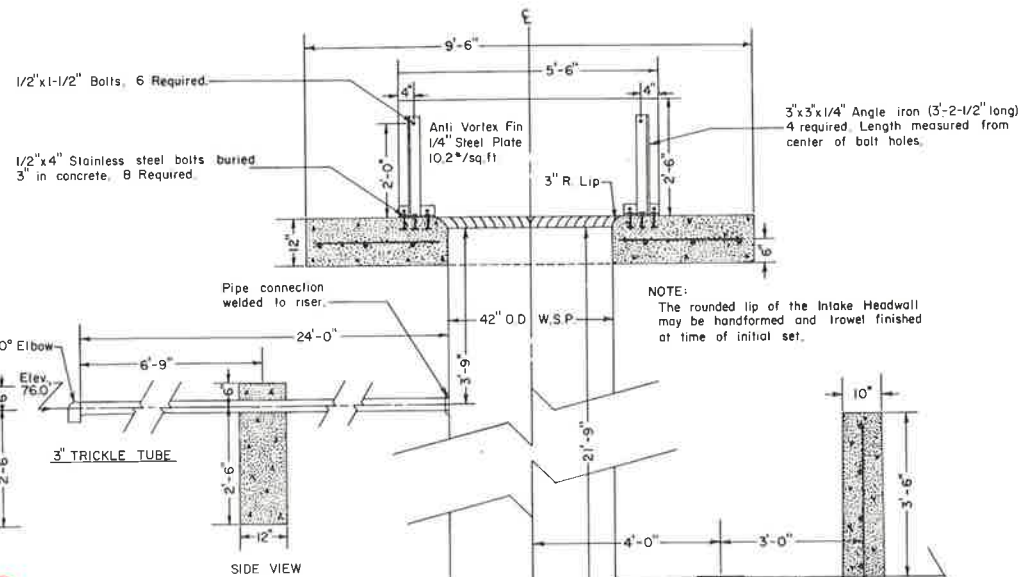
PEDESTAL TOP VIEW
2.0 Cu. Yds. Reinforced Concrete
DETAIL "C"



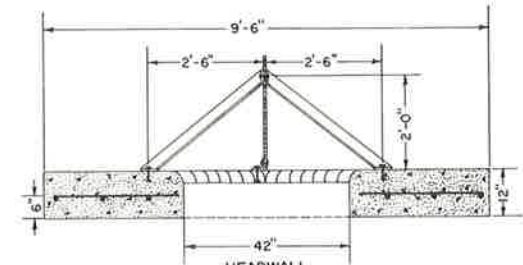
CROSS SECTION of DAM at STATION 1+60
SCALE

BILL OF MATERIALS

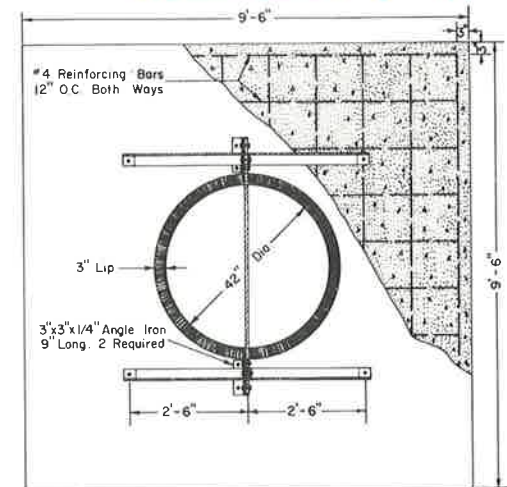
- 10 - 20' Sections of 30" outside diameter, .625" wall thickness, welded steel pipe with coal tar paint protective coating on exterior surface.
- 1 - Manhole connection as detailed, shop fabricated from 42" outside diameter, .625" wall thickness and 30" outside diameter, .625" wall thickness, welded steel pipe with coal tar paint protective coating on exterior surfaces.
- 10 - Style No. 38 dresser coupling or equal for 30" outside diameter steel pipe. Middle ring of coupling shall not be less than .625" thickness and not less than 10" in length. Gaskets shall be suitable for cold water service.
- 23.2 - Cu. Yds. reinforced concrete.
- 1 - 1/4" steel plate, 10.2" sq. ft. (5'-6" x 2'-6") as detailed.
- 2 - 9" Long, 3"x3"x1/4" Angle iron, 4.9"/ft.
- 4 - 3'-2 1/2" Long, 3"x3"x1/4" Angle iron, 4.9"/ft.
- 8 - 1/2"x4" Stainless steel bolts with nuts.
- 6 - 1/2"x1-1/2" Bolts with nuts.
- 24 - Of 3" black iron pipe with a 90° elbow.



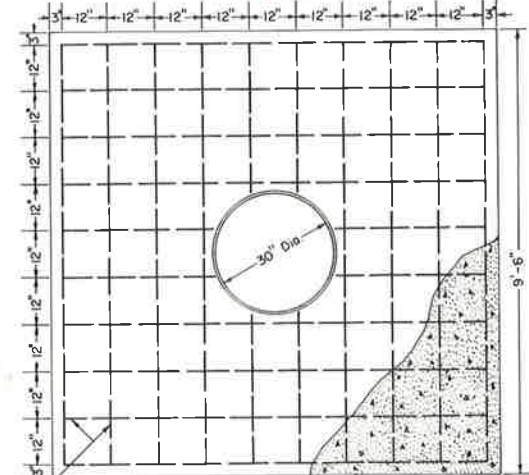
FRONT VIEW
0.2 Cu. Yds. Concrete
C.U. 10/6/67



HEADWALL FRONT VIEW
3.0 Cu. Yds. Reinforced Concrete C.U. 10/6/67



HEADWALL TOP VIEW
3.0 Cu. Yds. Reinforced Concrete
C.U. 10/6/67



CUTOFF COLLAR FRONT VIEW
3.0 Cu. Yds. Reinforced Concrete C.U. 10/6/67

UNITED STATES DEPARTMENT OF THE INTERIOR BUREAU OF LAND MANAGEMENT	
BAR ISLAND DETENTION DAM	
MALTA DISTRICT	MONTANA
DESIGNED A.E.H.	RECOMMENDED <i>Henry T. Hammermark</i>
DRAWN B.M.K.	RECOMMENDED <i>A.P. George</i>
CHECKED H.T.H.	APPROVED <i>Harold Lytle</i>
SCALE See Plate	
DATE July 1966	SHEET 2 OF 2
DRAWING NO. 25-01-7321.1-7	